Network Layer

UNIT 3

Presented By

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Network Layer

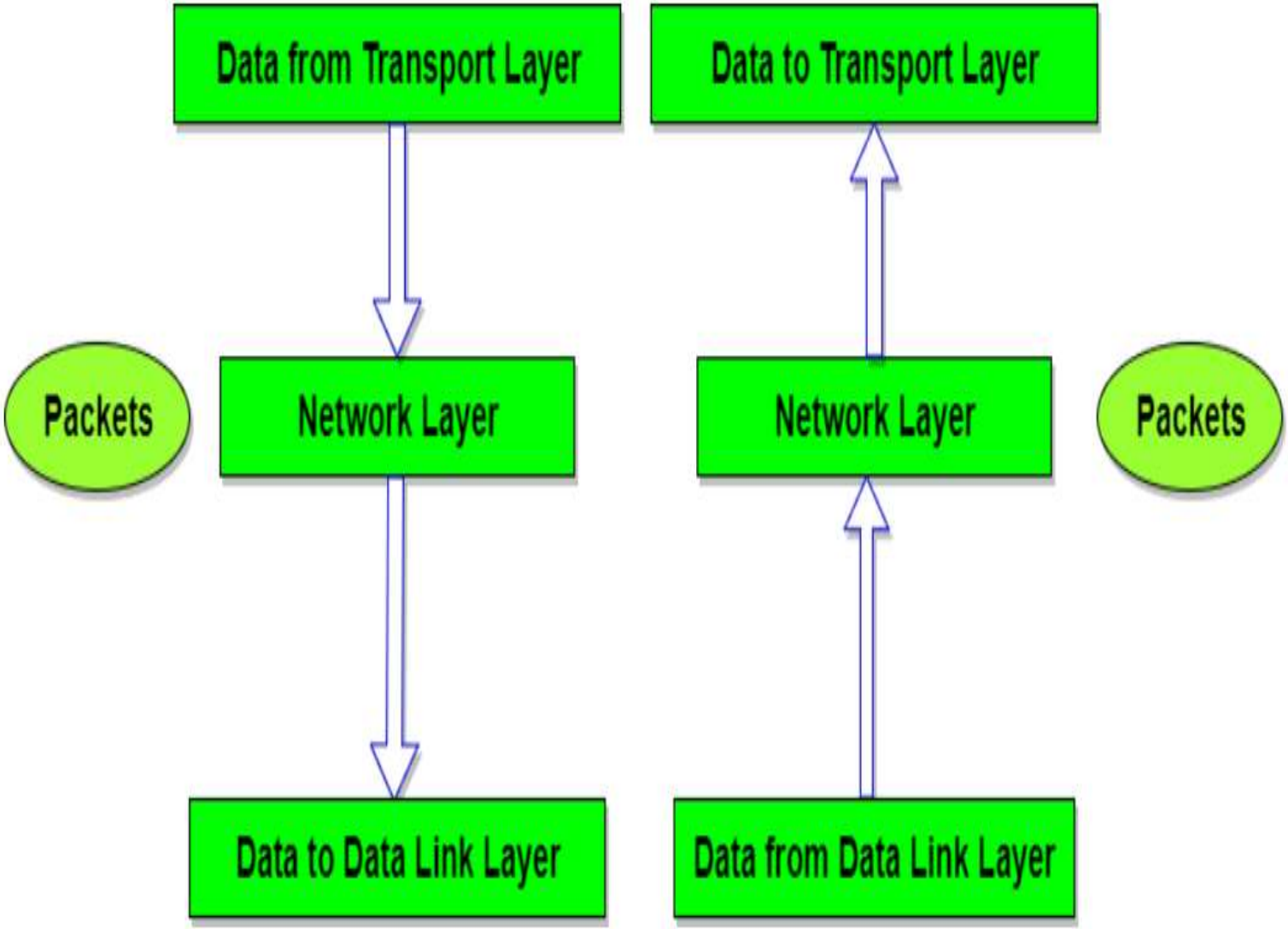
- The Network Layer is the **third layer** of the OSI model.
- It handles the **service requests from the transport layer** and further **forwards the service request to the data link layer**.
- The network layer translates the **logical addresses into physical addresses**
- It determines the **route from the source to the destination** and also manages the traffic problems such as switching, routing and controls the congestion of data packets.
- The main **role of the network layer is to move the packets from sending host to the receiving host (Source to Destination delivery).**

Functions of Network Layer

- **Routing:** When a packet reaches **the router's input link, the router will move the packets to the router's output link**. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.
- **Logical Addressing:** The data link layer implements the physical addressing and network layer implements the logical addressing. Logical addressing is also used to **distinguish between source and destination system.** The network layer **adds a header to the packet which includes the logical addresses of both the sender and the receiver.**

Functions of Network Layer

- **Internetworking:** This is the main role of the **network layer that it provides the logical connection between different types of networks**.
- **Fragmentation:** The fragmentation is a process of **breaking the packets into the smallest individual data units** that travel through different networks.


Data from Transport Layer
Data to Transport Layer
Packets
Network Layer
Network Layer
Packets
Data to Data Link Layer
Data from Data Link Layer

Features

- Main responsibility of Network layer is **to carry the data packets from the source to the destination without changing or using it.**
- **If the packets are too large for delivery, they are fragmented** i.e., broken down into smaller packets.
- It decides **the root to be taken by the packets to travel from the source to the destination among the multiple roots available in a network** (also called as routing).
- The **source and destination addresses are added to the data packets inside the network layer.**

Network Layer Services

- **Guaranteed delivery:** This layer provides the service which **guarantees that the packet will arrive at its destination.**
- **Guaranteed delivery with bounded delay:** This service guarantees that the **packet will be delivered within a specified host-to-host delay bound.**
- **In-Order packets:** This service ensures that the packet **arrives at the destination in the order in which they are sent.**

Network Layer Services

- **Guaranteed max jitter:** This service ensures that the amount of time taken **between two successive transmissions at the sender is equal to the time between their receipt at the destination. (**If a sender sends packets at **regular intervals**, the receiver also gets them at **the same regular intervals** — no variation in spacing. So, the **delay difference between packets (jitter)** is controlled or eliminated.)
- **Security services:** The network layer provides **security by using a session key between the source and destination host**. The network layer in the **source host encrypts the payloads of datagrams being sent to the destination host**. The network layer in the **destination host would then decrypt the payload**. In such a way, **the network layer maintains the data integrity and source authentication services.**

Other services in Network Layer

- **Error Control:** Although it can be implemented in the network layer, but it is usually not preferred because the data packet in a network layer maybe fragmented at each router, **which makes error checking inefficient in the network layer**.
- **Congestion Control:** Congestion occurs when the number of **datagrams sent by source is beyond the capacity of network or routers.** This is another issue in the network layer protocol. If congestion continues, sometimes **a situation may arrive where the system collapses and no datagrams are delivered**. Although congestion control is indirectly implemented in network layer, but still there is a lack of congestion control in the network layer.

Other services in Network Layer

Flow Control:

- It regulates the amount of data a source can send without overloading the receiver.
- If the source produces a data at a very faster rate than the receiver can consume it, the receiver will be overloaded with data.
- To control the flow of data, the receiver should send a feedback to the sender to inform the latter that it is overloaded with data.
- There is a lack of flow control in the design of the network layer. It does not directly provide any flow control.
- The datagrams are sent by the sender when they are ready, without any attention to the readiness of the receiver.

Advantages of Network Layer services

- Packetization service **in network layer provides an ease of transportation of the data packets.**
- Packetization also **eliminates single points of failure in data communication systems.**
- Routers present in the network layer **reduce network traffic by creating collision and broadcast domains.**
- With the help of Forwarding, **data packets are transferred from one place to another in the network.**

Disadvantages of Network Layer services

- There is a **lack of flow control in the design of the network layer**.
- Congestion occurs sometimes due to the presence of too many datagrams in a network which are beyond the capacity of network or the routers. Due to this, **some routers may drop some of the datagrams and some important piece of information maybe lost.**
- Although indirectly error control is present in network layer, **but there is a lack of proper error control mechanisms as due to presence of fragmented data packets, error control becomes difficult to implement**.

Packet switching

- **Packet switching** is a method of transferring the data to a network in form of packets.
- In order to transfer the file fast and efficient manner over the network and minimize the transmission latency, the data is broken into small pieces of variable length, called **Packet**.
- At the destination, all these small-parts (packets) has to be reassembled, belonging to the same file. A packet composes of payload and various control information. No pre-setup or reservation of resources is needed.

Packet switching

- Packet Switching uses **Store and Forward** technique while switching the packets; while forwarding the packet each hop first store that packet then forward. This technique is very beneficial because packets may get discarded at any hop due to some reason.
- Packet-Switched networks were designed to overcome the *weaknesses* of Circuit-Switched networks since circuit-switched networks were not very effective for small messages.

Advantages of Packet Switching

- More efficient in terms of bandwidth, since the concept of reserving circuit is not there.
- Minimal transmission latency.
- More reliable as destination can detect the missing packet.
- More fault tolerant because packets may follow different path in case any link is down, Unlike Circuit Switching.
- Cost effective and comparatively cheaper to implement.

Disadvantages of Packet Switching

- Packet Switching don’t give packets in order, whereas Circuit Switching provides ordered delivery of packets because all the packets follow the same path.
- Since the packets are unordered, we need to provide sequence numbers to each packet.
- Complexity is more at each node because of the facility to follow multiple path.
- Transmission delay is more because of rerouting.
- Packet Switching is beneficial only for small messages, but for bursty data (large messages) Circuit Switching is better.

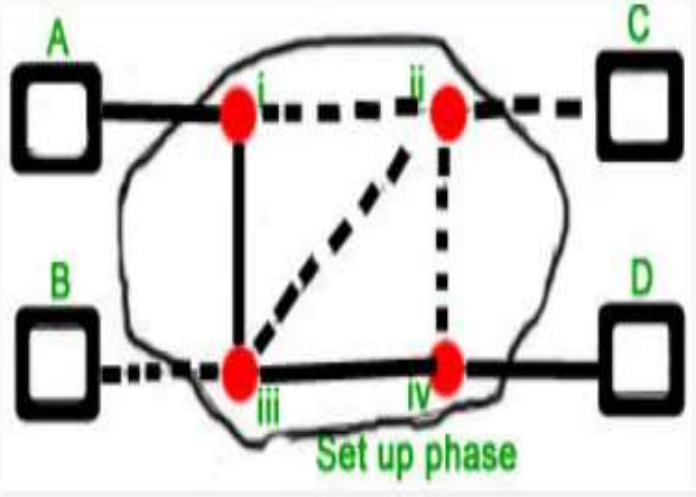
Modes in packet switching

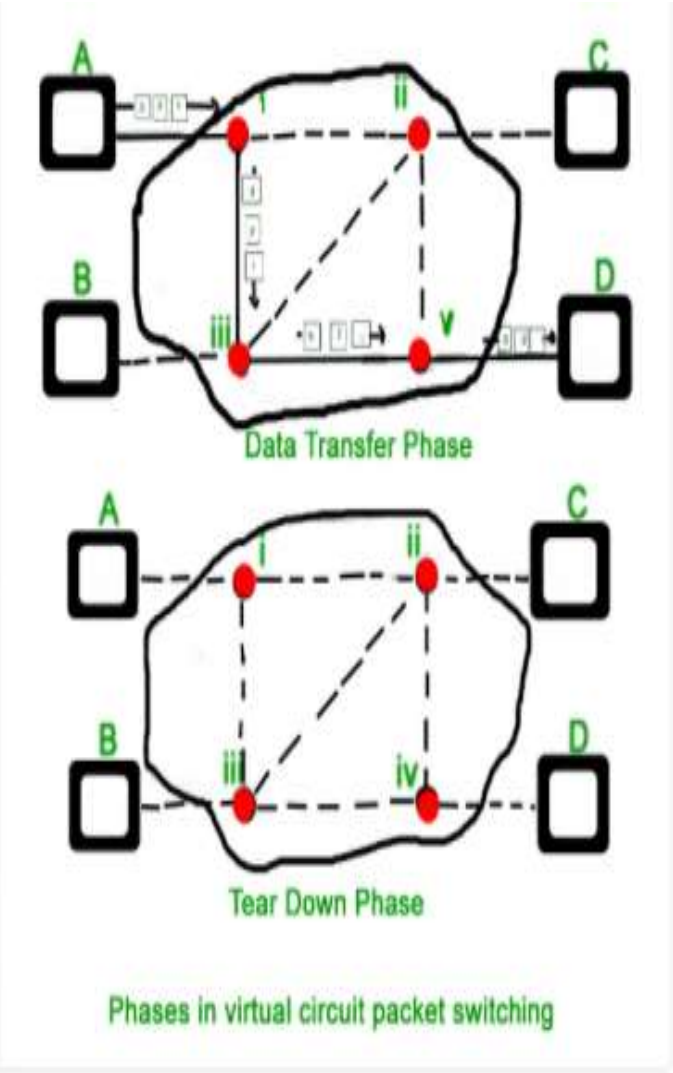
- Connection oriented packet switching
- Connection less packet switching

Connection oriented packet switching

- Before starting the transmission, it establishes a logical path or virtual connection using signaling protocol, between sender and receiver and all packets belongs to this flow will follow this predefined route.
- Virtual Circuit ID is provided by switches/routers to uniquely identify this virtual connection.
- Data is divided into small units and all these small units are appended with help of sequence number. Overall, three phases takes place here- Setup, data transfer and tear down phase.

Connection oriented packet switching





Connection oriented packet switching

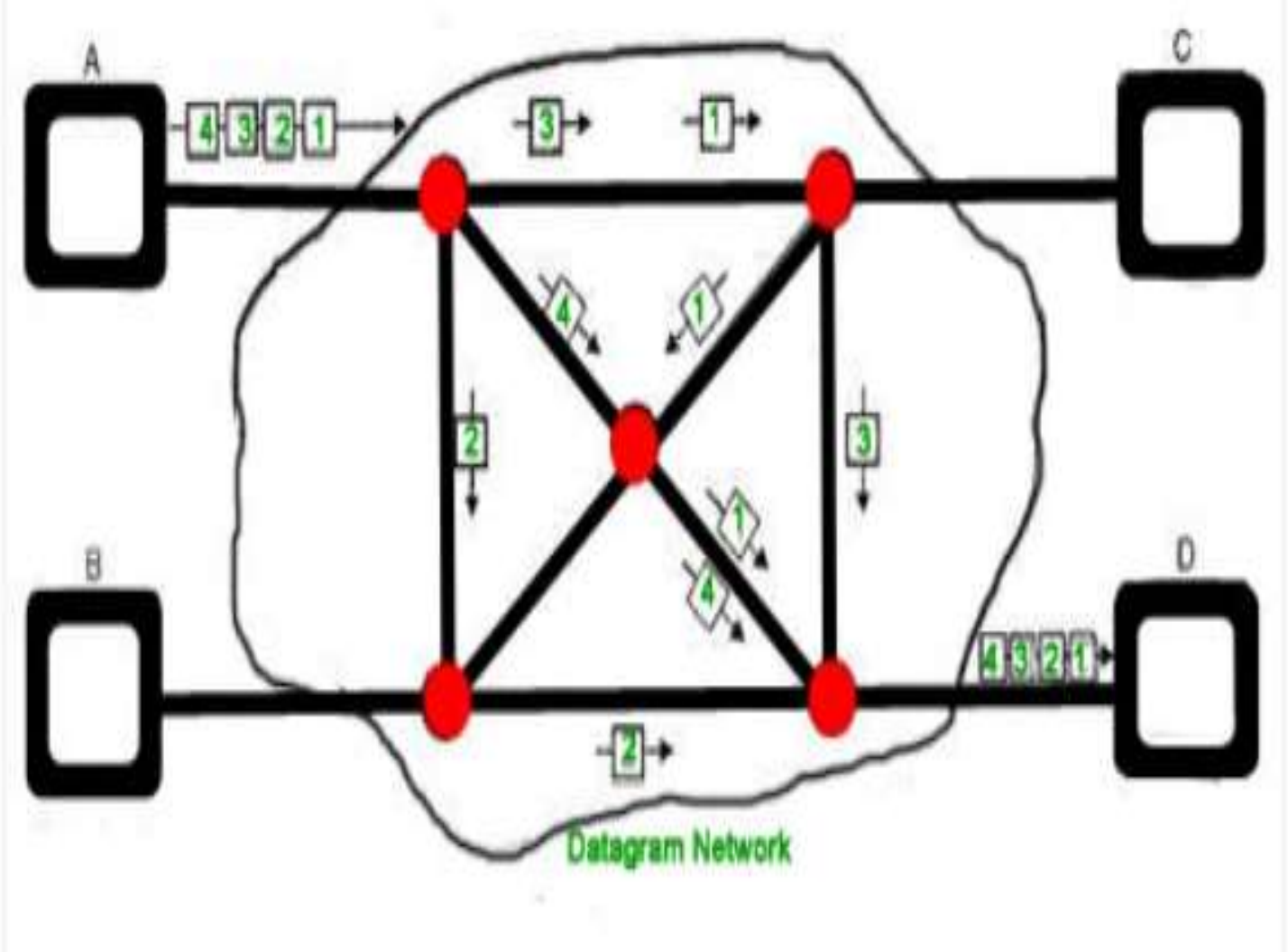
- All address information is only transferred during **setup phase**. Once the route to destination is discovered, entry is added to switching table of each intermediate node.
- During **data transfer**, packet header (local header) may contain information such as length, timestamp, sequence number etc.

  Connection-oriented switching is very useful in switched WAN.
- Some popular protocols which use Virtual Circuit Switching approach are X.25, Frame-Relay, ATM and MPLS(Multi-Protocol Label Switching).

Connectionless packet switching

- In Connectionless Packet Switching each packet contains all necessary addressing information such as **source address, destination address and port numbers** etc.
- In Datagram Packet Switching, each packet is treated **independently**.
- Packets belonging to one flow may take different routes because routing decisions are made dynamically, so the packets arrived at destination might be out of order.
- It has no connection setup and teardown phase, like Virtual Circuits.
- Packet delivery is not guaranteed in connectionless packet switching, so the reliable delivery must be provided by end systems using additional protocols.

Connectionless packet switching



Delays in packet switching

- Transmission Delay
- Propagation Delay
- Queuing Delay
- Processing Delay

Transmission Delay

- Time taken to put a packet onto link. In other words, it is simply time required to put data bits on the wire/communication medium. It depends on length of packet and bandwidth of network.

- Transmission Delay = Data size / bandwidth = (L/B) second

Propagation Delay

- Time taken by the first bit to travel from sender to receiver end of the link. In other words, it is simply the time required for bits to reach the destination from the start point. Factors on which Propagation delay depends are Distance and propagation speed.
- Propagation delay = distance/transmission speed = d/s

Queuing Delay

- Queuing delay is the time a job waits in a queue until it can be executed.
- It depends on congestion. It is the time difference between when the packet arrived Destination and when the packet data was processed or executed.
- It may be caused by mainly three reasons i.e. originating switches, intermediate switches or call receiver servicing switches.

Average Queuing delay = (N-1)L/(2*R)

where N = no. of packets

L=size of packet

R=bandwidth

Processing Delay

- Processing delay is the time it takes routers to process the packet header.
- Processing of packets helps in detecting bit-level errors that occur during transmission of a packet to the destination. Processing delays in high-speed routers are typically on the order of microseconds or less.
- In simple words, it is just the time taken to process packets.

Processing Delay

- Total time or End-to-End time

= Transmission delay + Propagation delay+ Queuing delay + Processing delay

For M hops and N packets –

- Total delay

= M*(Transmission delay + propagation delay)+
(M-1)*(Processing delay + Queuing delay) +
(N-1)*(Transmission delay)

For N connecting link in the circuit –

- Transmission delay = N*L/R
- Propagation delay = N*(d/s)

Problem

How much time will it take to send a packet of size L bits from A to B in given setup if Bandwidth is R bps, propagation speed is t meter/sec and distance b/w any two points is d meters (ignore processing and queuing delay) ?

A---R1---R2---B

Ans:

N = no. of links = no. of hops = no. of routers +1 = 3

File size = L bits

Bandwidth = R bps

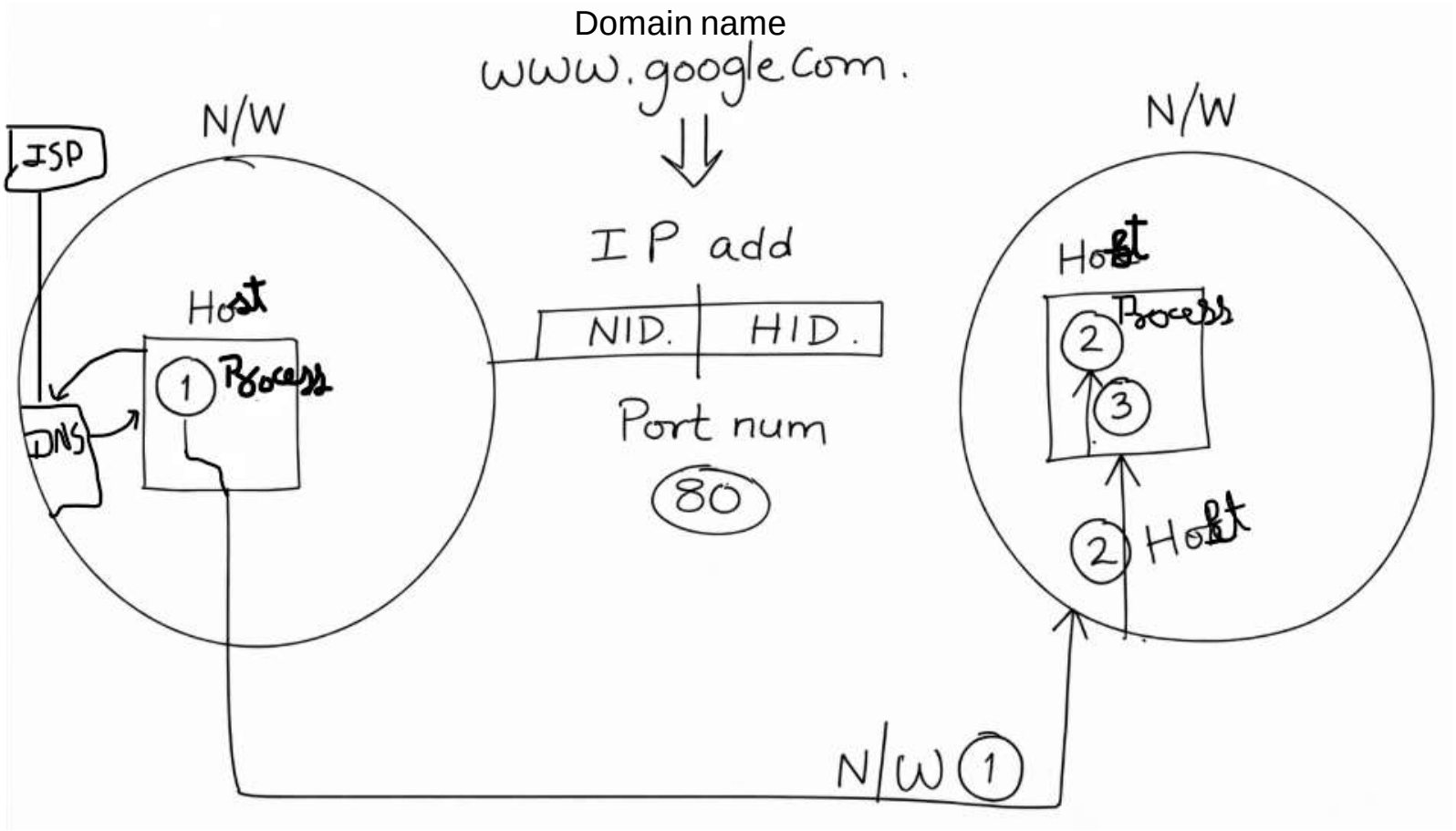
Propagation speed = t meter/sec

Distance = d meters

Transmission delay = (N*L)/R = (3*L)/R sec

Propagation delay = N*(d/t) = (3*d)/t sec

Total time = 3*(L/R + d/t) sec


Domain name
www.google.com.
IP add
NID.
HID.
Port num
80
N/W
ISP
DNS
Host
1
Process
N/W
Host
2
Process
3
2
Host
N/W
1

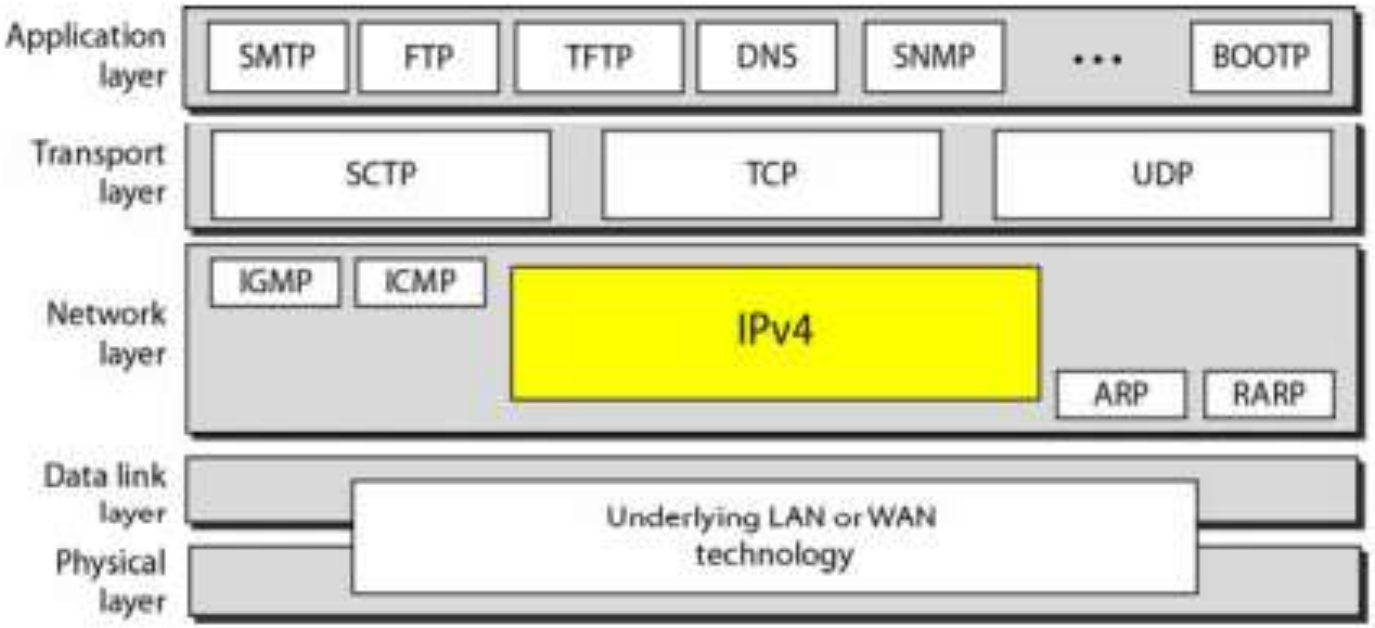
IPV4 address

- An IP address (*internet protocol address*) is a numerical representation that uniquely identifies a specific interface on the network.
- Addresses in **IPv4 are 32-bits** long. This allows for a maximum of 4,294,967,296 (2^{32}) unique addresses.
- Addresses in **IPv6 are 128-bits**, which allows for 3.4 x 10^{38} (2^{128}) unique addresses.

IPV4 Address

- IP addresses are binary numbers but are typically expressed in decimal form (IPv4) or hexadecimal form (IPv6) to make reading and using them easier for humans.
- IP stands for **Internet Protocol** and describes a set of standards and **requirements for creating and transmitting data packets**, or datagrams, across networks.
- The Internet Protocol (IP) is part of the Internet layer of the Internet protocol suite. In the OSI model, IP would be considered part of the network layer.
- IP is traditionally used in conjunction with a higher-level protocol, most notably TCP. The IP standard is governed by RFC 791 (Request for Comments 791).

Position of IPv4 in TCP/IP protocol suite



Based on
Data Communications and Networking, 4th Edition by Behrouz A. Forouzan, McGraw-Hill Companies, Inc., 2007

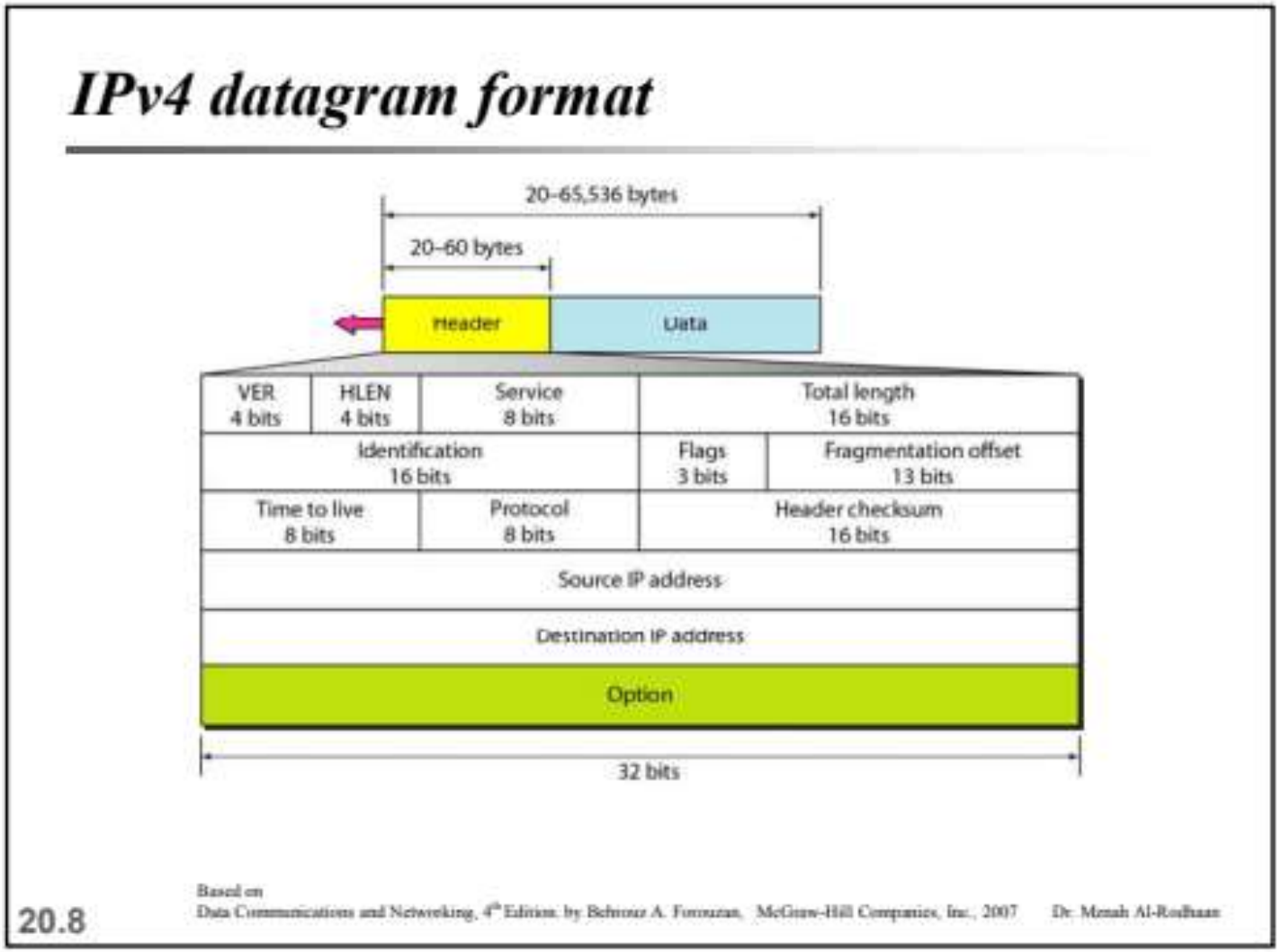
Dr. Menah Al-Rodhaan

IPV4 Address

- IPv4 addresses are actually **32-bit** binary numbers, consisting of the two sub addresses (identifiers) mentioned above which, respectively, **identify the network and the host to the network**, with an imaginary boundary separating the two.
- An IP address is, as such, generally shown as **4 octets** of numbers from 0-255 represented in decimal form instead of binary form.
- An IPv4 address is typically expressed in **dotted-decimal notation**, with every eight bits (octet) represented by a number from **one to 255**, each separated by a dot.
- An example IPv4 address would look like this:

  192.168.17.12

IPV4 PACKET FORMAT



- **VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4
- **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.
- **Type of service:** *Priority,* Low Delay, High Throughput, Reliability, Shortest path (cost)(8 bits)
- **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
- **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
- **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)
- **Fragment Offset:** Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes. Indicates the **position of a fragment** within the original IP packet.
- **Time to live:** Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination. (0-255)
- **Protocol:** Name of the protocol to which the data is to be passed (8 bits)
- **Header Checksum:** 16 bits header checksum for checking errors in the datagram header
- **Source IP address:** 32 bits IP address of the sender
- **Destination IP address:** 32 bits IP address of the receiver
- **Option:** Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not. (0-40 bytes)

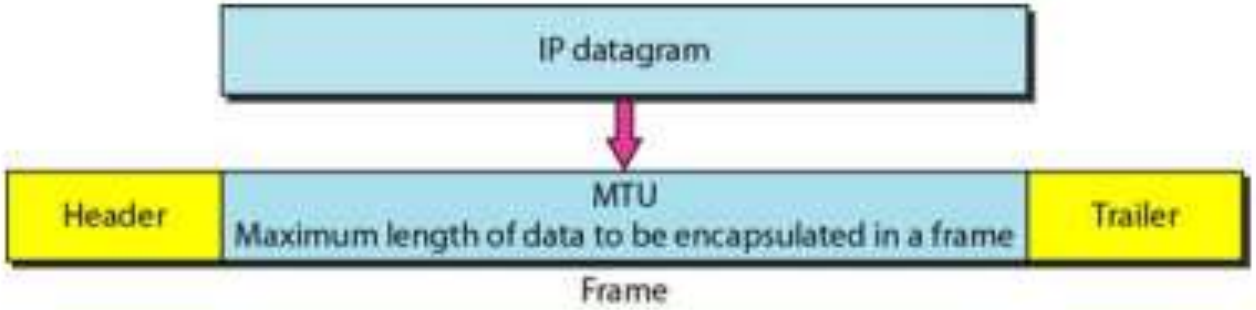
Fragmentation and Reassembly

- Fragmentation and reassembly is a method for transmission of messages larger than networks Maximum Transmission Unit (MTU).

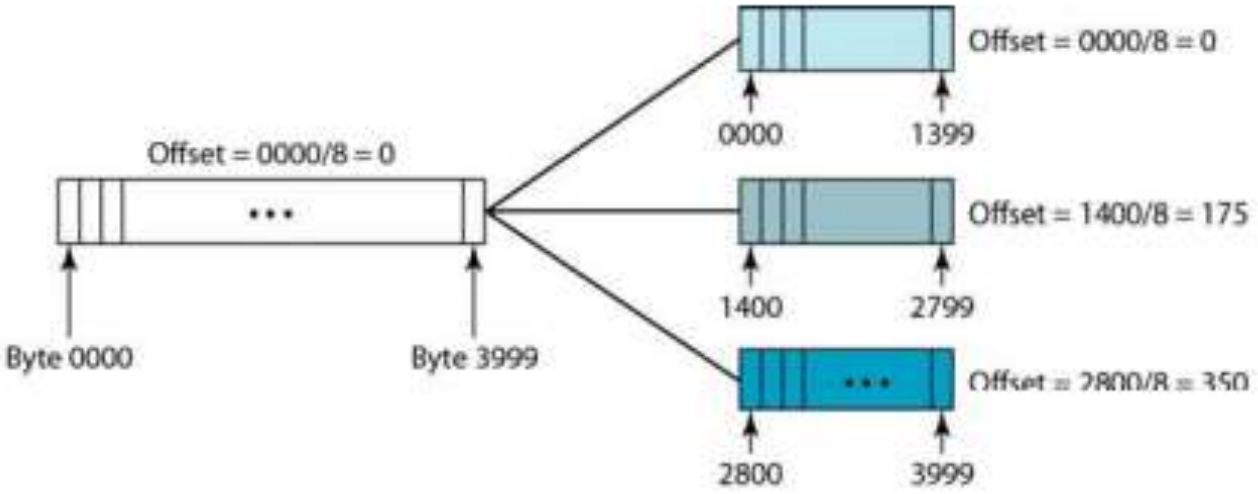
- Messages are fragmented into small pieces by the sender and reassembled by the receiver.

Fragmentation and Reassembly

Maximum transfer unit (MTU)



Fragmentation and Reassembly



Global Address

- The term IP address is used to mean a logical address in the network layer and is used in the Source address and Destination address field of the IP packets.

Logical Address

- Logical address are necessary for universal communications to identify each host in the internet for delivery of a packet from host to host.
- All IPv4 addresses are 32 bits long (equivalently 4 bytes).
- Within an IP address it encodes its network number and host number.
- The physical (network) addresses will change from hop to hop, but the logical addresses remains the same.

Network id and Host id

- Network ID is the portion of an IP address that identifies the TCP/IP network on which a host resides.
- The network ID portion of an IP address uniquely identifies the host's network on an internetwork, while the host ID portion of the IP address identifies the host within its network.
- Together, the host ID and network ID, which make up the entire IP address of a host, uniquely identify the host on a TCP/IP internetwork.



Format of an IP Address

Address Space

- An address space is the total number of addresses used by the protocol.
- For an N bits address, 2^N bits address space can be used, because each bit can have two different values (0 or 1).

Notations

- Two types of notations in IPv4:

1. **Binary Notation**: The binary notation in IPv4 is a 32 bit address or 4 bytes addres.

   Ex: 01000000 00100000 00010000 00000111

2. **Dotted- Decimal Notation**: To make 32 bit form shorter and easier to read, Internet addresses are usually written in Decimal form with decimal points separating the bytes called as Dotted decimal notation. Each number varies from 0 to 255.

   Example:

   01000000 00100000 00010000 00000111

   64 32 16 7

Classful Addressing

- IPv4 addresses were divided into 5 categories.
  - Class A
  - Class B
  - Class C
  - Class D
  - Class E
- This allocation has come to be called classful addressing

Classful Addressing

- IPv4 address of 4 bytes defines 3 fields.
  - Class Type
  - Network ID(Netid)
  - HostID

Class A



- Class A addresses were designed for large organizations with a large number of attached hosts or routers.
- In a Class A network, the first eight bits, or the first dotted decimal, is the network part of the address, with the remaining part of the address being the host part of the address. There are 128 possible Class A networks.

0.0.0.0 to 127.0.0.0

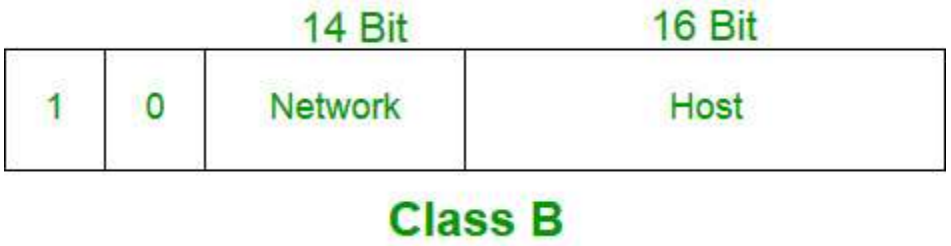
- Total number of addresses/connections in class A: 2^31
- Total Networks= 2^7 -2
- However, any address that begins with 127.0.0.1 is considered a loopback address and IP address 0.0.0.0 is reserved for broadcasting

Class A



- There are $2^{24} - 2$ HostID in Class A.
- When all Host ID bits are 0, it represents the Network ID for the network.
- When all Host ID bits are 1, it represents the Broadcast Address.

Class B



- Class B addresses were designed for midsize organizations with tens of thousands of attached hosts or routers.
- In a Class B network, 2 bytes for class type and netid and 2 bytes for hosted.
- All Class B networks have their first bit set to 1 and the second bit set to 0.
- In dotted decimal notation, that makes 128.0.0.0 to 191.255.0.0 as Class B networks.
- There are (2^{14}) 16,384 possible Class B networks.
- Total Number of Hosts = $2^{16} - 2$
- Example for a Class B IP address:

135.168.24.14

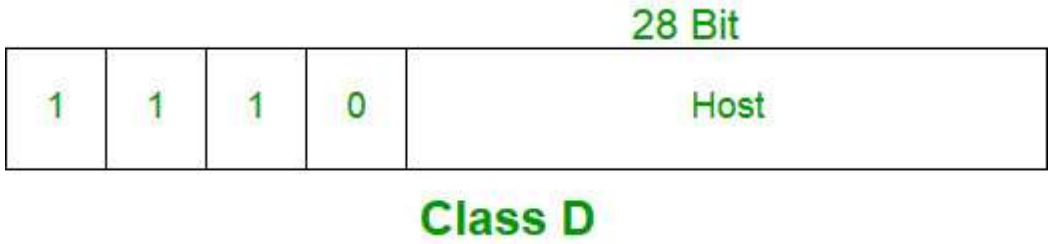
Class C



- Class C addresses were designed for small organizations with a small number of attached hosts or routers.
- It uses 3 bytes for class type and netid and 1 byte for hostid.
- In a Class C network, the first two bits are set to 1, and the third bit is set to 0.
- That makes the first 24 bits of the address the network address and the remainder as the host address.
- Total number of connections in class C is 2^{29}
- Total number of networks available in class C is 2^{21}
- Total number of hosts that can be configured in every network in class C is $2^{8} - 2$
- Class C network addresses range from 192.0.0.0 to 223.255.255.0. There are over 2 million possible Class C networks.

192.168.178.1

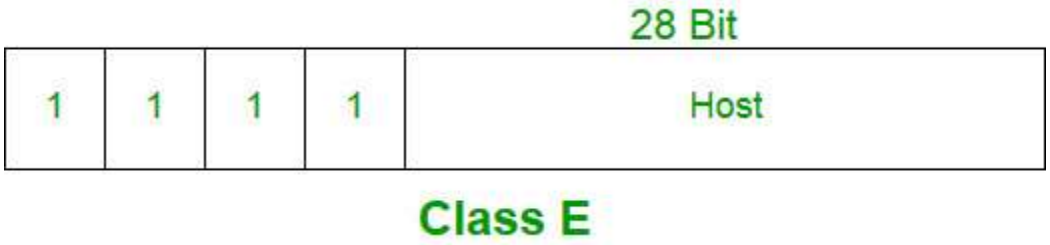
Class D



- Class D addresses are used for multicasting applications.
- Class D addresses have their first three bits set to “1” and their fourth bit set to “0”.
- Class D addresses are 32-bit network addresses, meaning that all the values within the range of 224.0.0.0 – 239.255.255.255 are used to uniquely identify multicast groups.
- There are no host addresses within the Class D address space, since all the hosts within a group share the group’s IP address for receiver purposes.

227.16.6.176

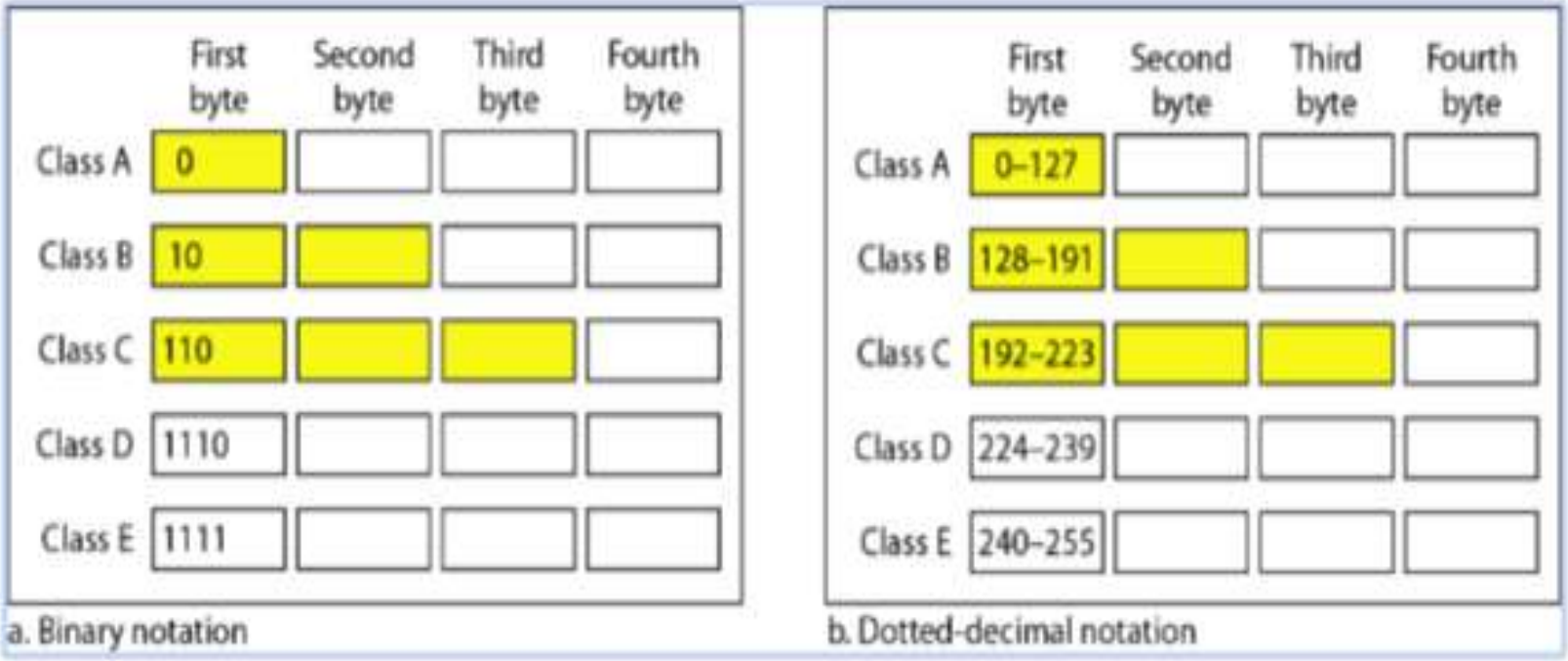
Class E



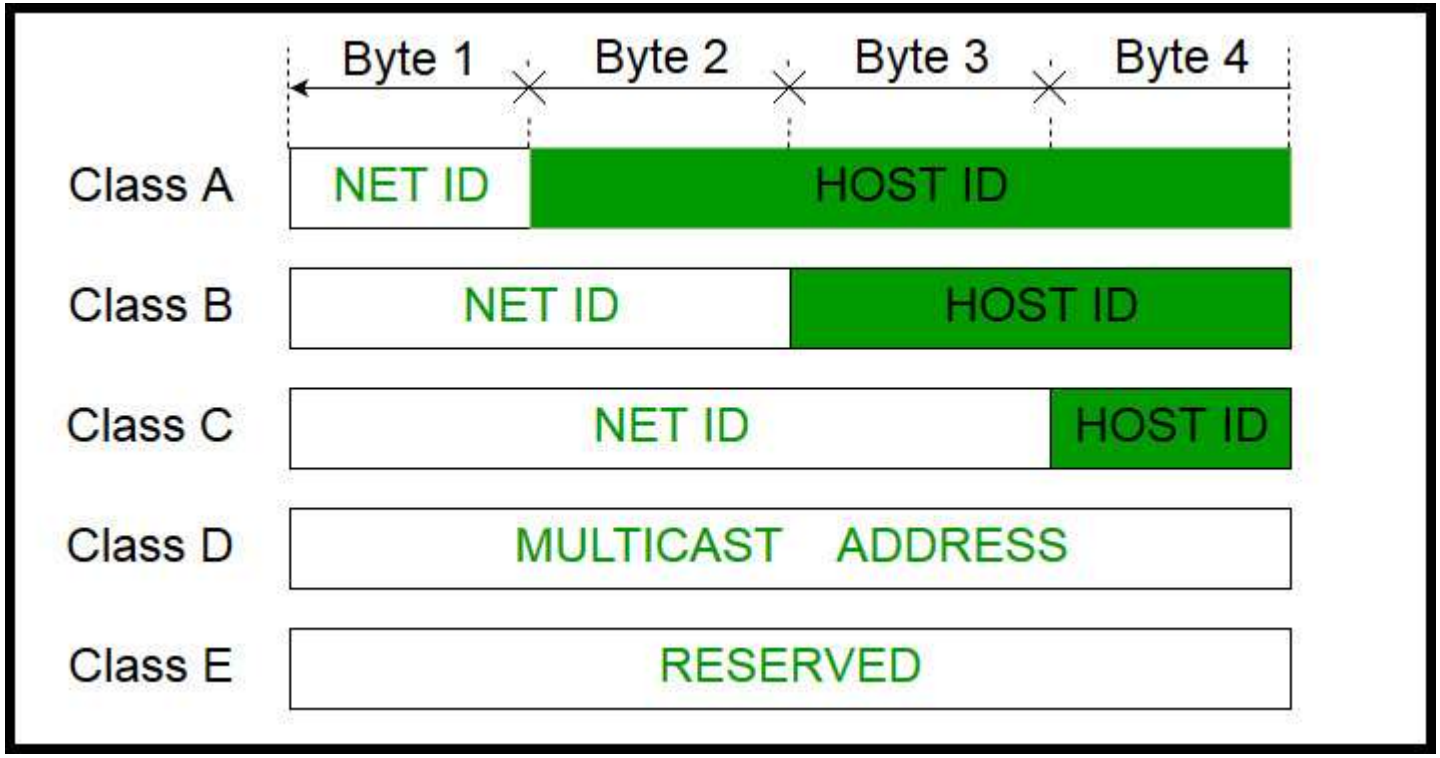
- Class E addresses are reserved for future use, in which addresses beginning with 1111.
- Class E networks are defined by having the first four network address bits as 1.
- That encompasses addresses from 240.0.0.0 to 255.255.255.255. While this class is reserved, its usage was never defined.
- As a result, most network implementations discard these addresses as illegal or undefined.

243.164.89.28

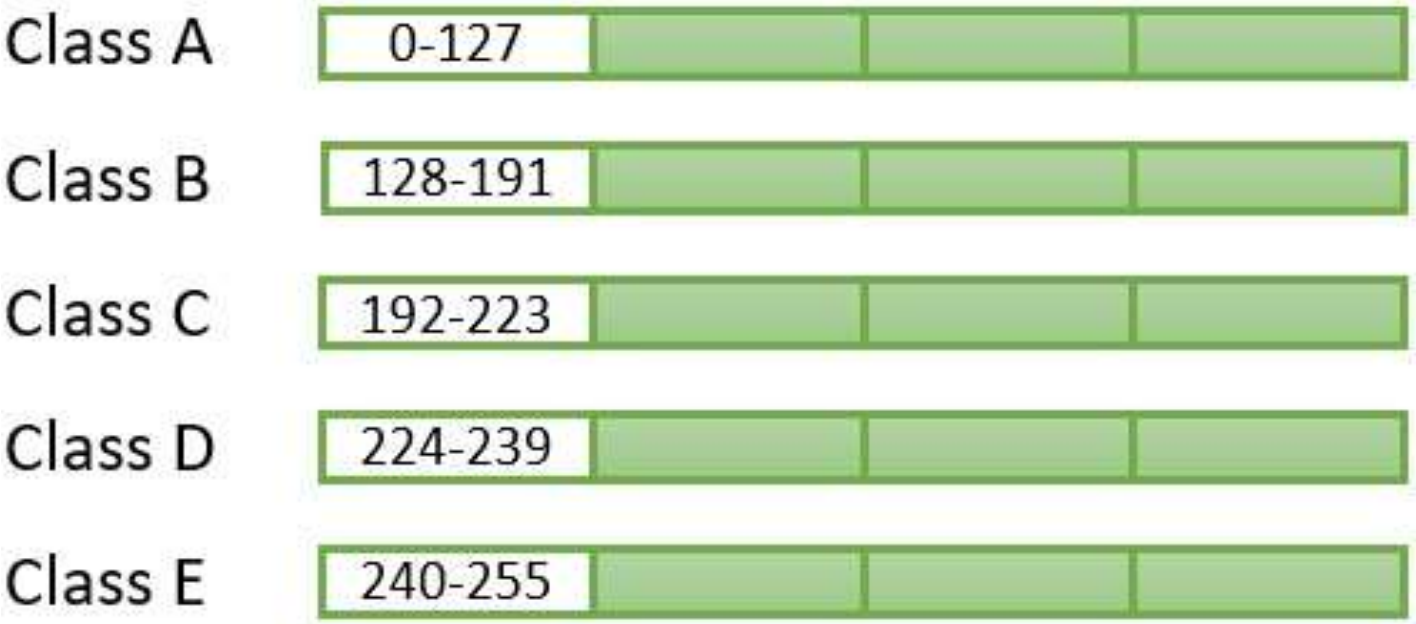
Classful Addressing



Classful Addressing



Classful Addressing



Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0
D	224.0.0.0 to 239.255.255.255	Reserved for Multicasting
E	240.0.0.0 to 254.255.255.255	Experimental

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback testing.

Class	Leading bits	Size of *network number* bit field	Size of *rest* bit field	Number of networks	Addresses per network	Total addresses in class	Start address	End address
Class A	0	8	24	128 (2^7)	16,777,216 (2^{24})	2,147,483,648 (2^{31})	0.0.0.0	127.255.255.255
Class B	10	16	16	16,384 (2^{14})	65,536 (2^{16})	1,073,741,824 (2^{30})	128.0.0.0	191.255.255.255
Class C	110	24	8	2,097,152 (2^{21})	256 (2^8)	536,870,912 (2^{29})	192.0.0.0	223.255.255.255
Class D (multicast)	1110	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	224.0.0.0	239.255.255.255
Class E (reserved)	1111	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	240.0.0.0	255.255.255.255

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

Find the error, if any, in the following IPv4 addresses.

a. 111.56.045.78

b. 221.34.7.8.20

c. 75.45.301.14

d. 11100010.23.14.67

Solution

a. There must be no leading zero (045).

b. There can be no more than four numbers.

c. Each number needs to be less than or equal to 255.

d. A mixture of binary notation and dotted-decimal notation is not allowed.

Find the class of each address.

a. 0**0000001** 00001011 00001011 11101111

b. 11**000001** 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111

Solution

a. The first bit is 0. This is a class A address.

b. The first 2 bits are 1; the third bit is 0. This is a class C address.

c. The first byte is 14; the class is A.

d. The first byte is 252; the class is E.

Disadvantages of Classful Addressing

- If we consider class A, the number of addresses in each block is **more than enough** for almost any organization. So, it results in wastage of addresses.
- Same is the case with class B, probably an organization receiving block from class B would not require that much of addresses. So, it also results in **wastage of addresses**.
- A block in class C may be **too small** to **fulfil the addresses requirement** of an organization.
- Each address in class D defines a group of hosts. Hosts need to **multicast** the address. So, the addresses are wasted here too.
- Addresses of class E are **reserved for the future purpose** which is also wastage of addresses.
- The main issue here is; we are **not assigning** addresses according to **user requirements**. We directly assign a **block of a fixed size** which has a **fixed number of addresses** which leads to wastage of address.

Subnetting and Supernetting

- To overcome the flaws of classful addressing, these two solutions were introduced to compensate for the wastage of addresses.

Subnetting

- As class blocks of A & B are too large for any organization. So, they can **divide** their large network in the **smaller subnetwork** and **share** them with other organizations. This whole concept is **subnetting**.

Supernetting

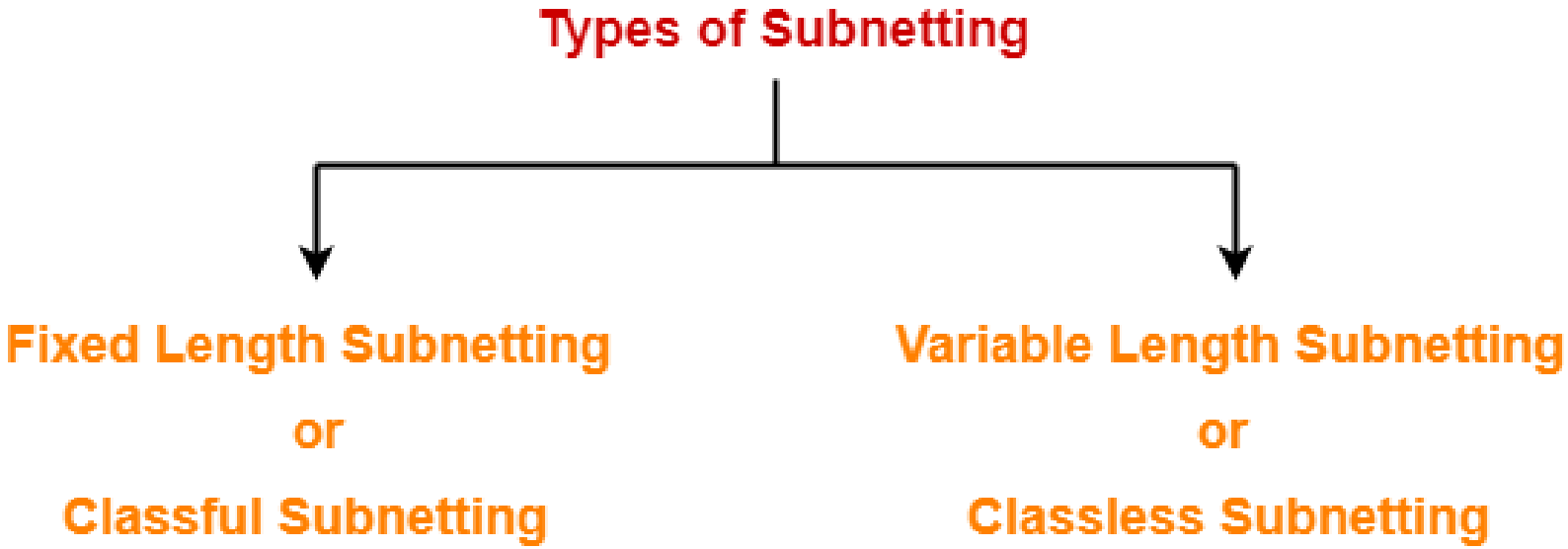
- As the blocks in class A and B were almost consumed so, new organizations consider class C. But, the block of class C is too small then the requirement of the organization. In this case, the solution which came out is supernetting which grants to **join the blocks of class C** to **form a larger block** which satisfies the address requirement of the organization.

Reason for Subnetting

- Maintenance of a very big network like class A and B is very difficult for network administrator.
- Having all the computer from different departments in a company on the same network is less secure from a company perspective.
- So, if an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets) or, in rare cases, share part of the addresses with neighbours.

Subnetting

- A **subnet**, or subnetwork, is a segmented piece of a larger network.
- More specifically, **subnets** are a logical partition of an IP network into multiple, smaller network segments.
- Organizations will use a **subnet** to subdivide large networks into smaller, more efficient subnetworks.
- When a bigger network is divided into smaller networks, in order to maintain security, then that is known as Subnetting. so, maintenance is easier for smaller networks.


Types of Subnetting
Fixed Length Subnetting
or
Classful Subnetting
Variable Length Subnetting
or
Classless Subnetting

Fixed Length Subnet Mask (FLSM)

- All subnets are created using the **same subnet mask**.
  Each subnet has **equal number of hosts**.
- **Features:**
- Simple to design and manage
- Wastes IP addresses if different subnet sizes are required
- Same number of usable IPs in every subnet
- **Example:**
  A Class C network 192.168.1.0/24 divided into /26
- Host bits: $32 - 26 = 6$, Total addresses per subnet: $2^6 = 64$
- Borrowed bits from host part: $26 - 24 = 2$ *bits borrowed*
- Number of subnets: $2^2 = 4$ *subnets*
- Subnet 1: 192.168.1.0/26
- Subnet 2: 192.168.1.64/26
- Subnet 3: 192.168.1.128/26
- Subnet 4: 192.168.1.192/26

Variable Length Subnet Mask (VLSM)

- Subnets are created using **different subnet masks** based on requirement.
- **Features:**
- Efficient IP address utilization
- Supports networks of different sizes
- Slightly more complex to design
- **Example:**
  From 192.168.1.0/24
- Sales department → /26 (62 hosts)
- HR department → /27 (30 hosts)
- Admin → /28 (14 hosts)
- Each subnet size matches the actual need.

Private Address

- Within the address space, certain networks are reserved for private networks.
- Packets from these networks are not routed across the public internet. This provides a way for private networks to use internal IP addresses without interfering with other networks. The private networks are

10.0.0.1 - 10.255.255.255

172.16.0.0 - 172.32.255.255

192.168.0.0 - 192.168.255.255

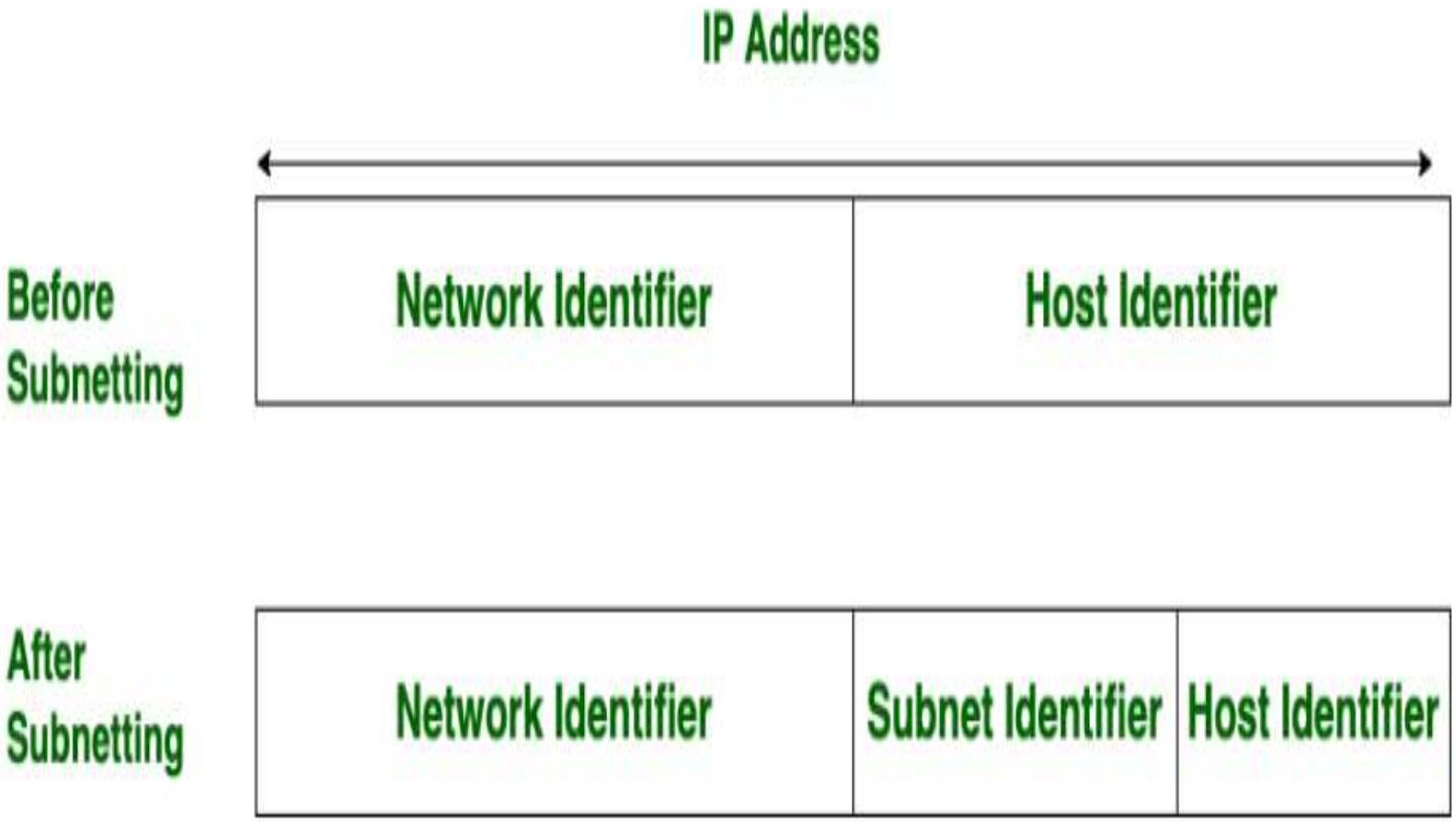
Special addresses

Certain IPv4 addresses are set aside for specific uses:

127.0.0.0	Loopback address (the host's own interface)
224.0.0.0	IP Multicast
255.255.255.255	Broadcast (sent to all interfaces on network)

Subnetting and Super netting

- Subnetting is the procedure to divide the network into sub-networks or small networks.
- Super netting is the procedure of combine the small networks into larger space.
- In subnetting, Network addresses's bits are increased. on the other hand, in subnetting, Host addresses's bits are increased.
- Subnetting is implemented via Variable-length subnet masking, While supernetting is implemented via Classless interdomain routing.


IP Address
Before Subnetting
Network Identifier
Host Identifier
After Subnetting
Network Identifier
Subnet Identifier
Host Identifier

S.NO	SUBNETTING	SUPERNETTING
1.	Subnetting is the procedure to divide the network into sub-networks.	While supernetting is the procedure of combine the small networks.
2.	In subnetting, Network addresses's bits are increased.	While in subnetting, Host addresses's bits are increased.
3.	In subnetting, The mask bits are moved towards right.	While In supernetting, The mask bits are moved towards left.
4.	Subnetting is implemented via Variable-length subnet masking.	While supernetting is implemented via Classless interdomain routing.
5.	In subnetting, Address depletion is reduced or removed.	While It is used for simplify routing process.

What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0 ?

Solution:

- Convert the given destination address into binary format.

200.45.34.56 = 11001000 00101101 00100010 00111000

- Convert the given subnet mask into binary format.

255.255.240.0 = 11111111 11111111 11110000 00000000

- Do the AND operation using destination address and subnet mask address.

200.45.34.56 = 11001000 00101101 00100010 00111000

255.255.240.0 = 11111111 11111111 11110000 00000000

11001000 00101101 00100000 00000000

subnetwork address is 200.45.32.0

Find the subnetwork address for the following:

S.No	IP address	Mask
A	140.11.36.22	255.255.255.0

Solution:

IP address	140.11.36.22
Mask	255.255.255.0
	140.11.36.0

The network having IP address 200.1.2.0 (Class C). Divide this network into two subnets.

1st Subnet

IP Address of the subnet / Subnet id = 200.1.2.0

Direct Broadcast Address = 200.1.2.01111111 = 200.1.2.127

Total number of IP Addresses = 2^7 = 128

Range of IP Addresses = [200.1.2.0, 200.1.2.127]

Total number of hosts that can be configured = 128 − 2 = 126

Range of Allocated IP Addresses = [200.1.2.1, 200.1.2.126]

2nd Subnet

IP Address of the subnet / Subnet id = 200.1.2.128

Direct Broadcast Address = 200.1.2.11111111 = 200.1.2.255

Total number of IP Addresses = 2^7 = 128

Range of IP Addresses = [200.1.2.128, 200.1.2.255]

Total number of hosts that can be configured = 128 − 2 = 126

Range of Allocated IP Addresses = [200.1.2.129, 200.1.2.224]

4 subnets

1st Subnet
IP Address of the subnet / Subnet id = 200.1.2.0
Direct Broadcast Address = 200.1.2.00111111 = 200.1.2.63
Total number of IP Addresses = 2^6 = 64
Range of IP Addresses = [200.1.2.0, 200.1.2.63]
Total number of hosts that can be configured = 64 − 2 = 62
Range of Allocated IP Addresses = [200.1.2.1, 200.1.2.62]

2nd Subnet
IP Address of the subnet / Subnet id = 200.1.2.64
Direct Broadcast Address = 200.1.2.01111111 = 200.1.2.127
Total number of IP Addresses = 2^6 = 64
Range of IP Addresses = [200.1.2.64, 200.1.2.127]
Total number of hosts that can be configured = 64 − 2 = 62
Range of Allocated IP Addresses = [200.1.2.65, 200.1.2.126]

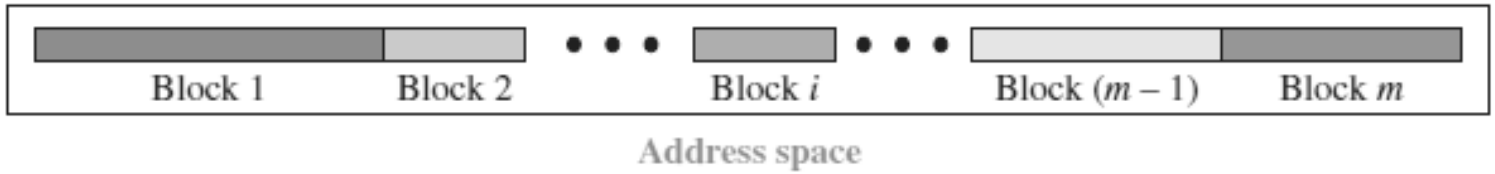
3rd Subnet
IP Address of the subnet / Subnet id = **200.1.2.128**
Direct Broadcast Address = **200.1.2.10111111 = 200.1.2.191**
Total number of IP Addresses = **2^6 = 64**
Range of IP Addresses = **[200.1.2.128, 200.1.2.191]**
Total number of hosts that can be configured = **64 − 2 = 62**
Range of Allocated IP Addresses = **[200.1.2.129, 200.1.2.190]**

4th Subnet
IP Address of the subnet / Subnet id = **200.1.2.192**
Direct Broadcast Address = **200.1.2.11111111 = 200.1.2.255**
Total number of IP Addresses = **2^6 = 64**
Range of IP Addresses = **[200.1.2.192, 200.1.2.255]**
Total number of hosts that can be configured = **64 − 2 = 62**
Range of Allocated IP Addresses = **[200.1.2.193, 200.1.2.254]**

CLASSLESS ADDRESSING

- In classless addressing, variable-length blocks are used that belong to no classes.
- We can have a block of 1 address, 2 addresses, 4 addresses, 128 addresses, and so on.
- In classless addressing, the whole address space is divided into variable length blocks.
- The prefix in an address defines the block (network); the suffix defines the node (device).
- The number of addresses in a block needs to be a power of 2 (2^n).
- An organization can be granted one block of addresses.

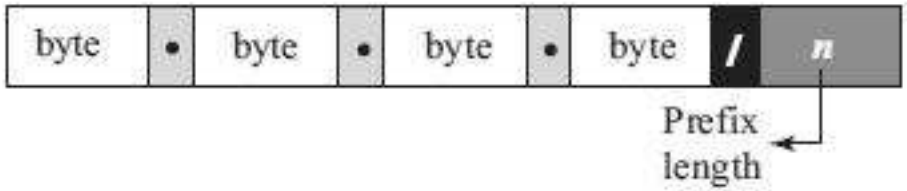
CLASSLESS ADDRESSING



- The prefix length in classless addressing is variable.
- We can have a prefix length that ranges from 0 to 32.
- The size of the network is inversely proportional to the length of the prefix.
- A small prefix means a larger network; a large prefix means a smaller network.
- The idea of classless addressing can be easily applied to classful addressing.
- An address in class A can be thought of as a classless address in which the prefix length is 8.
- An address in class B can be thought of as a classless address in which the prefix is 16, and so on. In other words, classful addressing is a special case of classless addressing.

Notation used in Classless Addressing

- The notation used in classless addressing is informally referred to as slash notation and formally as classless interdomain routing or CIDR.
- For example , 192.168.100.14 ***/24*** represents the IP address 192.168.100.14 and, its subnet mask 255.255.255.0, which has 24 leading 1-bits.



Examples:
12.24.76.8/8
23.14.67.92/12
220.8.24.255/25

Rules for CIDR (Classless Inter-Domain Routing) Blocks

1. CIDR Notation Rule

CIDR uses the format: IP Address / Prefix Length

IP Address → Network address

/n → Number of network bits

Example:

/24 → first 24 bits are network bits remaining bits are host bits

2. Block Size Rule

- Number of addresses in a CIDR block: Block *Size* $= 2^{(32-n)}$
- Where **n = prefix length**
- Example:
- /24 → $2^8 = 256$ addresses ...../26 → $2^6 = 64$ addresses

3. **Contiguous Address Rule (Very Important)**

- All addresses in a CIDR block must be **continuous (consecutive)**.
- Correct: 200.1.2.0 – 200.1.2.63
- Wrong: 200.1.2.0, 200.1.2.10, 200.1.2.50

4. Power of Two Rule

- The number of IP addresses must always be a **power of 2**.
- Allowed sizes: 2, 4, 8, 16, 32, 64, 128, 256 ...

CIDR BLOCK: 200.1.2.64/26

1. Prefix = **/26**

- Network bits = 26 11111111 11111111 11111111 11000000
- Host bits = $32 - 26 = 6$

2. Block Size Rule

- Block *Size* $= 2^{(32-26)} = 2^6 = 64$
- CIDR block contains **64 IP addresses**. (Power of 2 rule satisfied)

3. Range of Addresses (Contiguous Rule)

- CIDR block must be continuous:
- 200.1.2.64 → 200.1.2.127
- (First IP of block: Network Address = 200.1.2.64 (All host bits 0))
- (Broadcast Address = 200.1.2.127) (All host bits = 1)
- All addresses are consecutive.

Address Aggregation

- One of the advantages of the CIDR strategy is **address aggregation** (sometimes called *address summarization* or *route summarization*).
- When blocks of addresses are combined to create a larger block, routing can be done based on the prefix of the larger block.
- ICANN (**Internet Corporation for Assigned Names and Numbers)** assigns a large block of addresses to an ISP.
- Each ISP in turn divides its assigned block into smaller subblocks and grants the subblocks to its customers.

Special Addresses in IPv4

- There are five special addresses that are used for special purposes:
  - *this-host* address,
  - *limited-broadcast* address,
  - *loopback* address,
  - *private* addresses, and
  - *multicast* addresses.

This-host Address

- The only address in the block **0.0.0.0/32** is called the *this-host* address.
- It is used whenever a host needs to send an IP datagram but it does not know its own address to use as the source address. (This happens when a device **knows its MAC address** but **does not know its IP address**.)

Limited-broadcast Address

- The only address in the block **255.255.255.255/32** is called the *limitedbroadcast* address.
- It is used whenever a router or a host needs to send a datagram to all devices in a network.
- The routers in the network, however, block the packet having this address as the destination;the packet cannot travel outside the network.

Loopback Address

- The block **127.0.0.0/8** is called the *loopback* address.
- A packet with one of the addresses in this block as the destination address never leaves the host; it will remain in the host.

Private Addresses

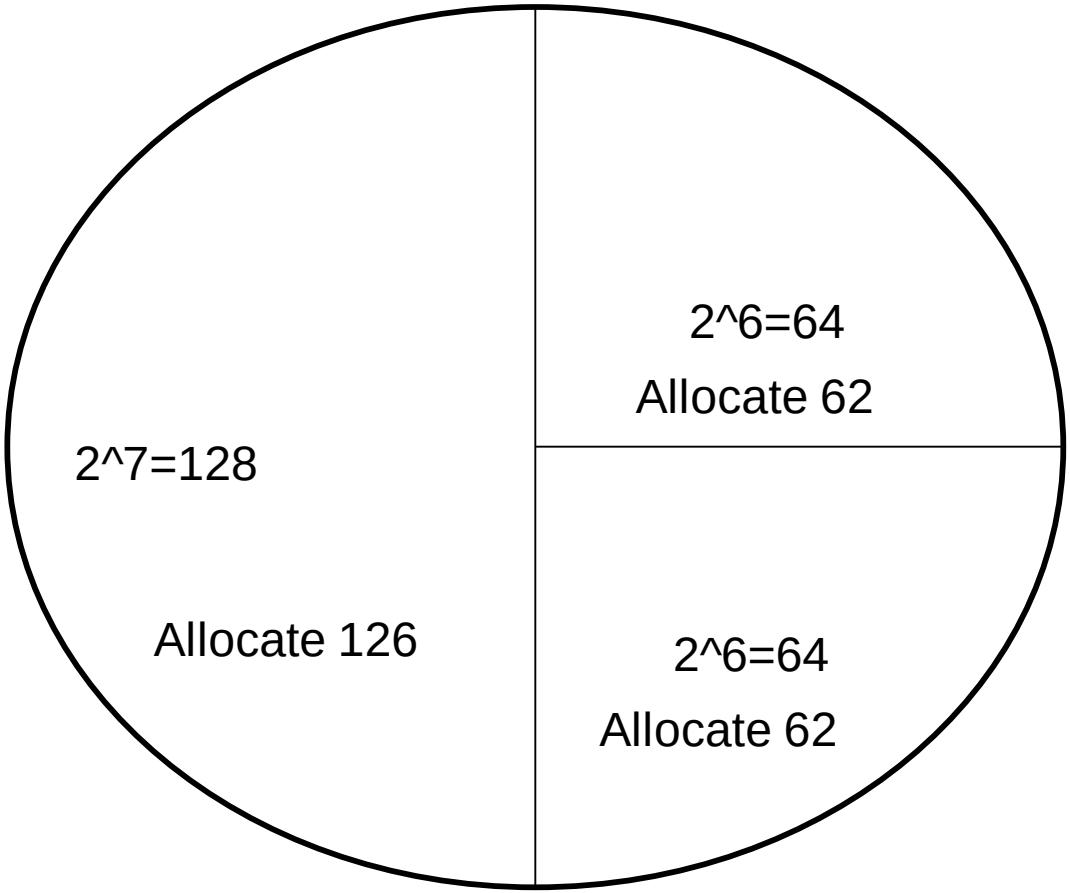
- Four blocks are assigned as private addresses: 10.0.0.0/**8**, 172.16.0.0/**12**,
- 192.168.0.0/**16**, and 169.254.0.0/**16**.

Multicast Addresses

- The block 224.0.0.0/**4** is reserved for multicast addresses.

Variable length subnetting

Consider we have a big single network having IP Address **200.1.2.0**. We want to do subnetting and divide this network into **3 subnets**, such that first contains **126 hosts**, and other two contains **62 hosts each**.



1st Subnet

- IP Address of the subnet / Subnet id = **200.1.2.0**
- Direct Broadcast Address = **200.1.2.01111111 = 200.1.2.127**
- Total number of IP Addresses = **2^7 = 128**
- Range of IP Addresses = **[200.1.2.0, 200.1.2.127]**
- Total number of hosts that can be configured = **128 − 2 = 126**
- Range of Allocated IP Addresses = **[200.1.2.1, 200.1.2.126]**

2nd Subnet

- IP Address of the subnet / Subnet id = **200.1.2.128**
- Direct Broadcast Address = **200.1.2.10111111 = 200.1.2.191**
- Total number of IP Addresses = **2^6 = 64**
- Range of IP Addresses = **[200.1.2.128, 200.1.2.191]**
- Total number of hosts that can be configured = **64 − 2 = 62**
- Range of Allocated IP Addresses = **[200.1.2.129, 200.1.2.190]**

3rd Subnet

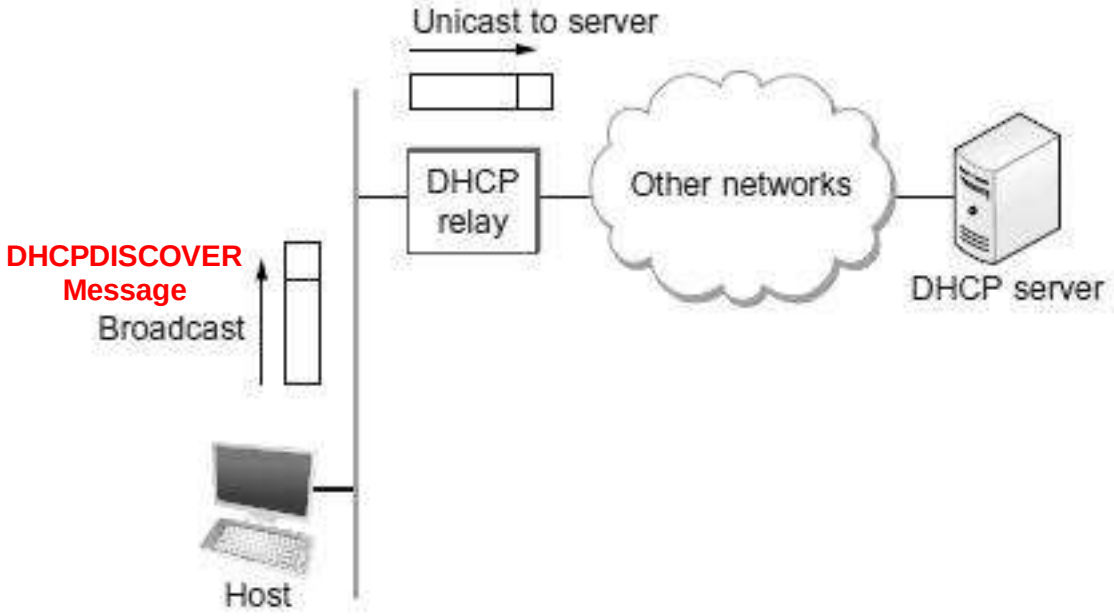
- IP Address of the subnet / Subnet id = **200.1.2.192**
- Direct Broadcast Address = **200.1.2.11111111 = 200.1.2.255**
- Total number of IP Addresses = **2^6 = 64**
- Range of IP Addresses = **[200.1.2.192, 200.1.2.255]**
- Total number of hosts that can be configured = **64 − 2 = 62**
- Range of Allocated IP Addresses = **[200.1.2.193, 200.1.2.254]**

DHCP – DYNAMIC HOST CONFIGURATION PROTOCOL

- The dynamic host configuration protocol is used to simplify the installation and maintenance of networked computers.
- DHCP is derived from an earlier protocol called BOOTP.
- Ethernet addresses are configured into network by manufacturer and they are unique.
- IP addresses must be unique on a given internetwork but also must reflect the structure of the internetwork.
- The main goal of DHCP is to minimize the amount of manual configuration required for a host.
- DHCP works at **Application Layer.**
- Uses **UDP Port 67 (Server)** and **UDP Port 68 (Client)**

- If a new computer is connected to a network, DHCP can provide it with all the necessary information for full system integration into the network.
- DHCP is based on a client/server model.
- DHCP clients send a request to a DHCP server to which the server responds with an IP address
- DHCP server is responsible for providing configuration information to hosts.
- There is at least one DHCP server for an administrative domain.
- The DHCP server can function just as a centralized repository for host configuration information.
- The DHCP server maintains a pool of available addresses that it hands out to hosts on demand.
- Dynamically allots these addresses to nodes for a defined time, then they need to renew it.

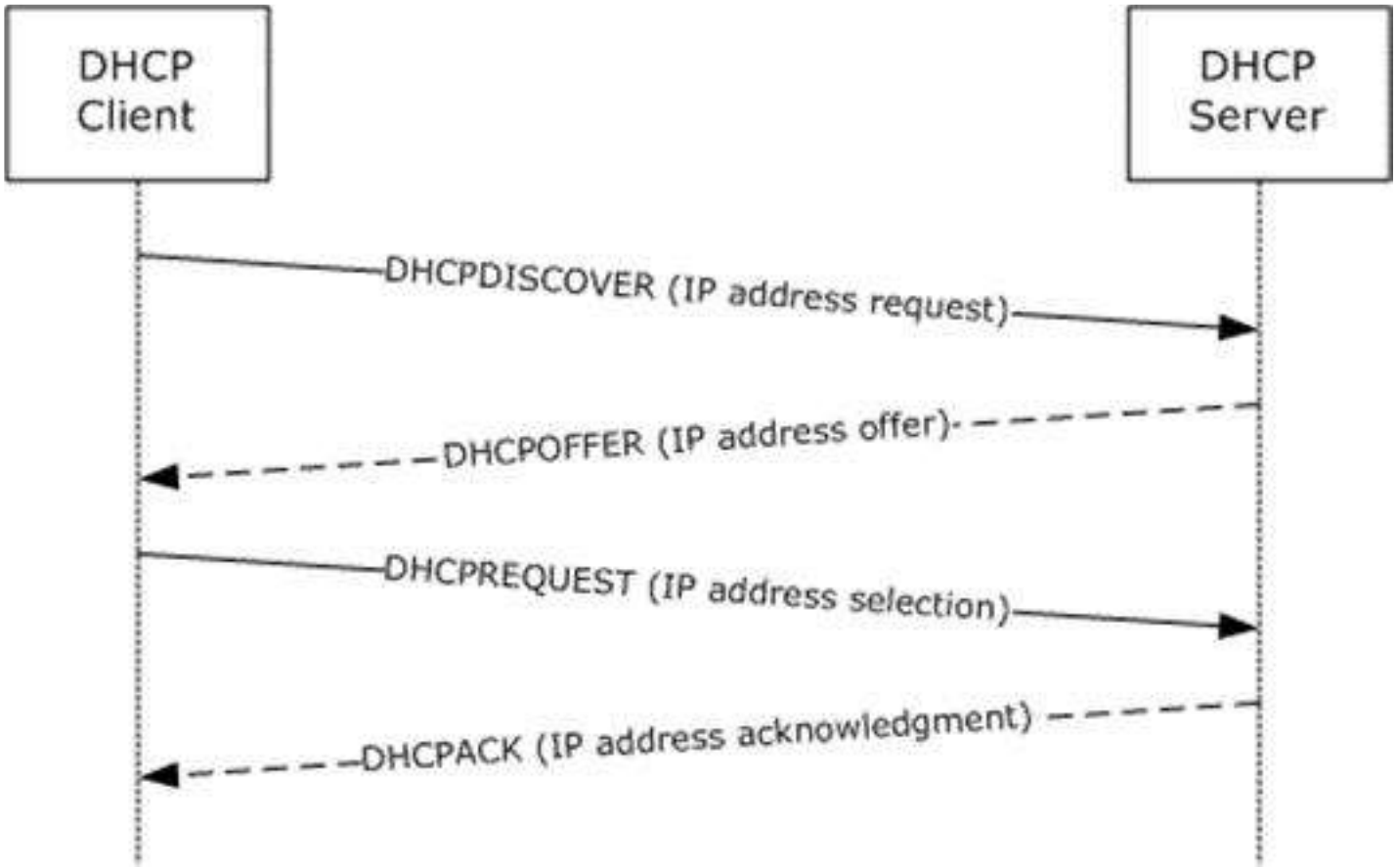
DHCP – DYNAMIC HOST CONFIGURATION PROTOCOL



DHCP – DYNAMIC HOST CONFIGURATION PROTOCOL

- A newly booted or attached host sends a DHCPDISCOVER message to a special IP address (255.255.255.255, which is an IP broadcast address.
- This means it will be received by all hosts and routers on that network.
- DHCP uses the concept of a *relay agent*. There is at least one relay agent on each network.
- DHCP relay agent is configured with the IP address of the DHCP server.
- When a relay agent receives a DHCPDISCOVER message, it unicasts it to the DHCP server and awaits the response, which it will then send back to the requesting client.

DHCP – DYNAMIC HOST CONFIGURATION PROTOCOL



IP Address Allocation Mechanism in DHCP

1. Dynamic Allocation: IP address is assigned **automatically for a limited time** (called *lease time*). After lease expiry, the address returns to the DHCP pool. The client may receive the same or a different IP next time.

Features

- Most commonly used method.
- Efficient use of IP addresses.
- Suitable for large networks.
- Example: Public Wi-Fi networks.

2. Automatic Allocation: DHCP assigns an IP address **permanently** to a device. Once assigned, the same IP is always given to that device. Stored in DHCP server records.

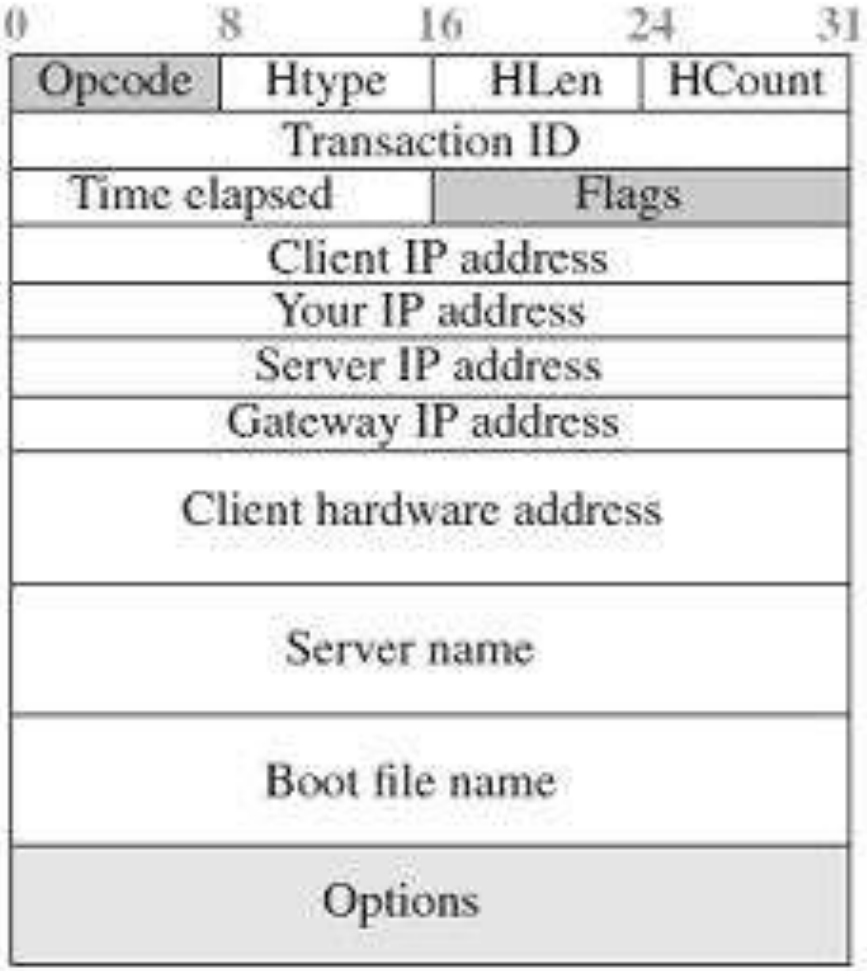
Features

- No lease expiration.
- Useful for frequently used devices.
- Example: Office computers.

3. Manual Allocation (Static Allocation / Reservation): Network administrator manually maps: **MAC Address → Specific IP Address.** DHCP always assigns the predefined IP to that device.

- **Features**
- Combines benefits of static IP and DHCP automation.
- Used for important devices.
- Example: Printers, Servers, Network cameras

DHCP Message Format

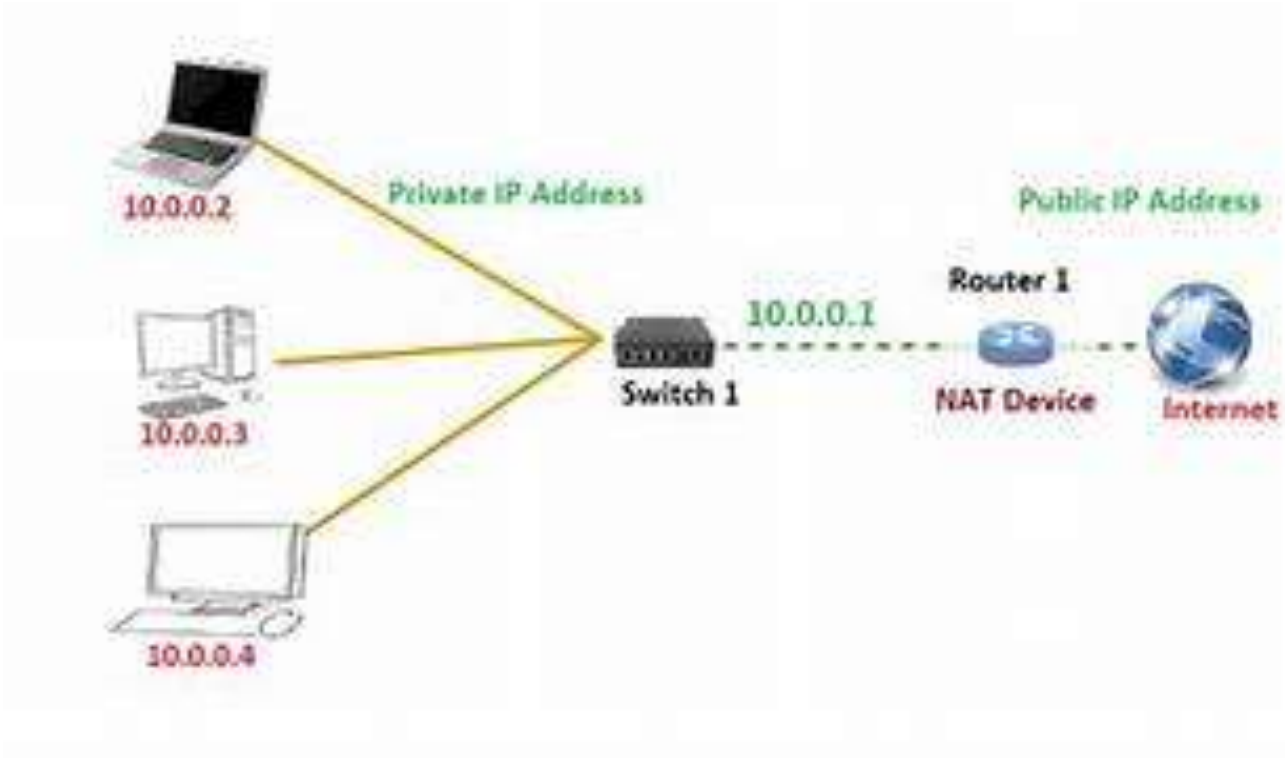


Field	Size	Description
opcode	1 byte	Message type (1 = request, 2 = reply)
htype	1 byte	Hardware type (Ethernet = 1)
hlen	1 byte	Hardware address length (MAC = 6)
hops	1 byte	Used by relay agents
xid	4 bytes	Transaction ID to match requests & replies
secs	2 bytes	Seconds elapsed since request started
flags	2 bytes	Broadcast flag information
ciaddr	4 bytes	Client IP address (if already assigned)
yiaddr	4 bytes	IP address offered to client
siaddr	4 bytes	DHCP server IP address
giaddr	4 bytes	Relay agent (gateway) address
chaddr	16 bytes	Client MAC address
sname	64 bytes	Optional server hostname
file	128 bytes	Boot file name
options	Variable	DHCP configuration parameters (DHCP Message Type (Discover, Offer, Request, ACK) Subnet Mask Default Gateway DNS Server Lease Time Server Identifier)

NETWORK ADDRESS TRANSLATION (NAT)

- A technology that can provide the mapping between the private and universal (external)addresses, and at the same time support virtual private networks is called as **Network Address Translation (NAT).**
- The technology allows a site to use a set of private addresses for internal communication and a set of global Internet addresses (at least one) for communication with the rest of the world.
- The site must have only one connection to the global Internet through a NAT capable router that runs NAT software.

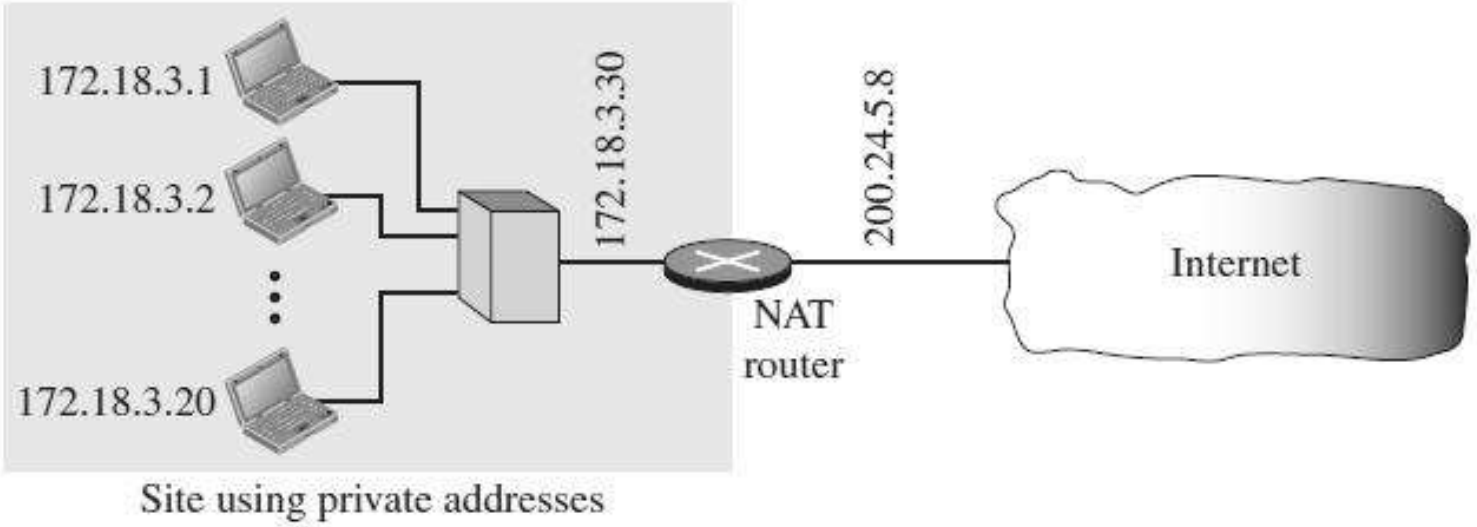
NAT (Example)



Private IP Address Ranges

IPv4	Class A	10.0.0.0 to 10.255.255.255
	Class B	172.16.0.0 to 172.31.255.255
	Class C	192.168.0.0 to 192.168.255.255

NETWORK ADDRESS TRANSLATION (NAT)



Types of NAT

- Two types of NAT exists .

(a) One-to-one translation of IP addresses

(b) One-to-many translation of IP addresses

One-to-one translation of IP addresses

- In **Static NAT**, each **private IP address** is permanently mapped to **one unique public IP address**.

Characteristics

- Fixed mapping (does not change)
- One private IP ↔ One public IP
- Suitable for servers that must be reachable from the Internet
- **Example**

Private IP		Public IP
192.168.1.10	→	203.0.113.10

Uses

- Web servers
- Mail servers
- Devices requiring external access

Advantages

- Predictable communication
- Supports inbound connections

Disadvantage: Requires many public IP addresses

One-to-many translation of IP addresses

- Multiple private IP addresses share **one public IP address** using **port numbers**. This is also called **PAT (Port Address Translation)** or **NAT Overload**.

Characteristics

- Many private IPs → Single public IP
- Uses port numbers to distinguish connections
- Most common NAT type in home/office networks
- **Example**
- Private IP Public IP (same)
  192.168.1.10 → 203.0.113.5:3001
  192.168.1.11 → 203.0.113.5:3002
  192.168.1.12 → 203.0.113.5:3003

Uses: Home routers, Corporate networks, Internet sharing

Advantages

- Saves public IP addresses
- Cost-effective

Disadvantage: Incoming connections need port forwarding

Address Translation

- All of the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address.
- All incoming packets also pass through the NAT router, which replaces the destination address in the packet (the NAT router global address) with the appropriate private address.

Translation Table

- There may be tens or hundreds of private IP addresses, each belonging to one specific host.
- The problem arises when we want to translate the source address to an external address. This is solved if the NAT router has a translation table.

Network Layer Protocols: IP

- The IP (Internet Protocol) is a protocol that uses datagrams to communicate over a packet-switched network.
- The IP protocol operates at the network layer protocol of the OSI reference model and is a part of a suite of protocols known as TCP/IP.
- The IP network service transmits datagrams between intermediate nodes using IP routers.

Network Layer Protocols: IP

- The routers themselves are simple, since no information is stored concerning the datagrams which are forwarded on a link.
- The most complex part of an IP router is concerned with determining the optimum link to use to reach each destination in a network.
- This process is known as "routing". Although this process is computationally intensive, it is only performed at periodic intervals.

Internet Control Message Protocol (ICMP)

- Since IP does not have a inbuilt mechanism for sending error and control messages. It depends on Internet Control Message Protocol (ICMP) to provide an error control.
- It is used for reporting errors and management queries.
- It is a supporting protocol and used by networks devices like routers for sending the error messages and operations information. e.g. the requested service is not available or that a host or router could not be reached.

ICMP

- The Internet Control Message Protocol (ICMP) is a network layer protocol used by network devices to diagnose network communication issues.
- ICMP is mainly used to determine whether or not data is reaching its intended destination in a timely manner.
- Commonly, the ICMP protocol is used on network devices, such as routers.
- ICMP is crucial for error reporting and testing, but it can also be used in distributed denial-of-service (DDoS) attacks.

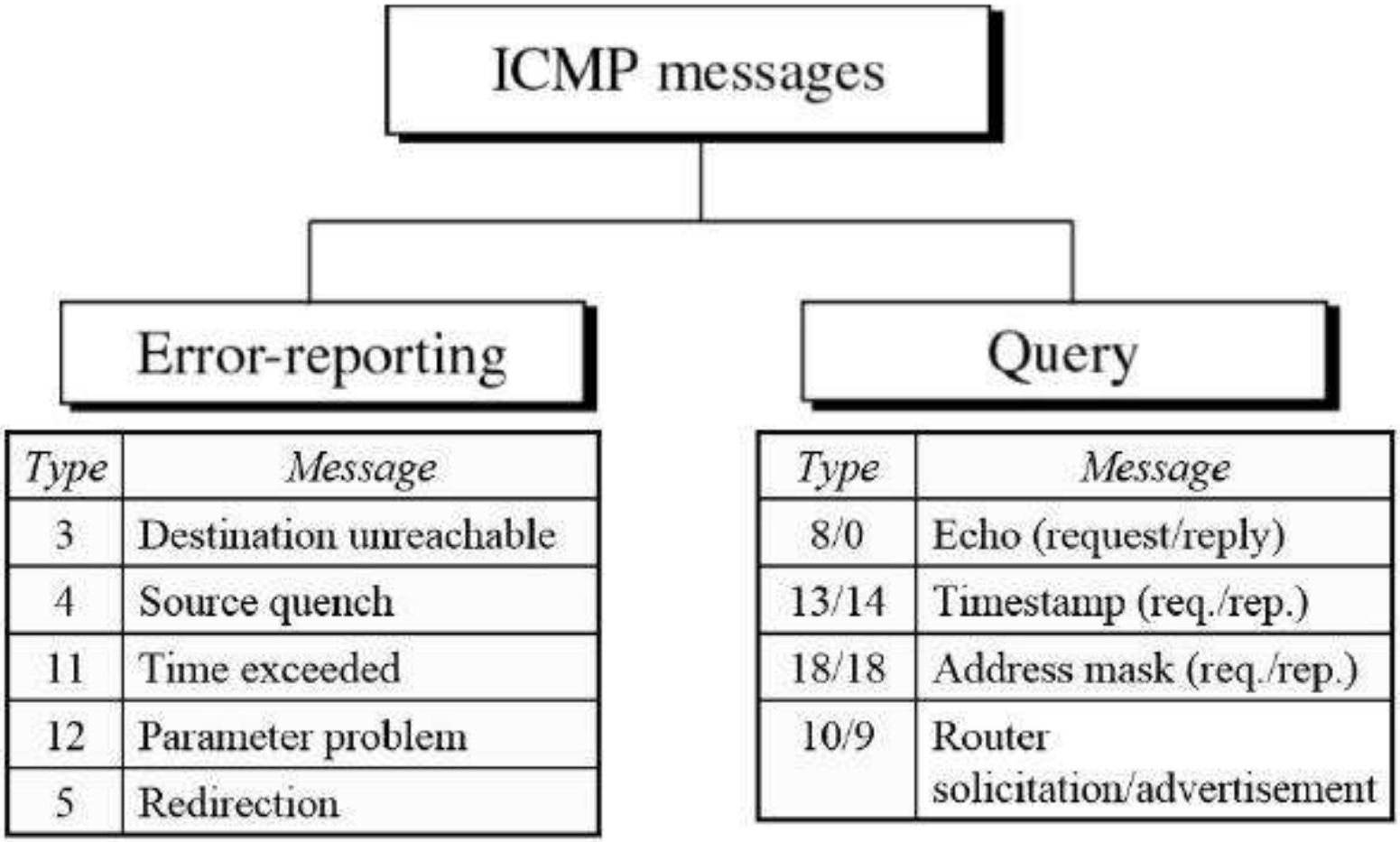
What is ICMP used for?

- The primary purpose of ICMP is for error reporting.
- When two devices connect over the Internet, the ICMP generates errors to share with the sending device in the event that any of the data did not get to its intended destination.
- For example, if a packet of data is too large for a router, the router will drop the packet and send an ICMP message back to the original source for the data.

- A secondary use of ICMP protocol is to perform network diagnostics; the commonly used terminal utilities traceroute and ping both operate using ICMP.
- The traceroute utility is used to display the routing path between two Internet devices.
- The routing path is the actual physical path of connected routers that a request must pass through before it reaches its destination.
- The journey between one router and another is known as a ‘hop,’ and a traceroute also reports the time required for each hop along the way. This can be useful for determining sources of network delay.

ICMP MESSAGE TYPES

- ICMP messages are divided into two broad categories: *error-reporting messages* and *query messages*.
- The error-reporting messages report problems that a router or a host (destination) may encounter when it processes an IP packet.
- The query messages help a host or a network manager get specific information from a router or another host.

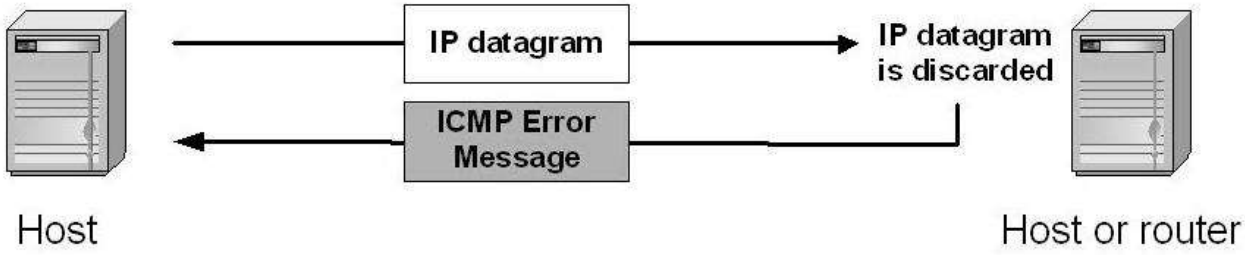


Type	*Message*
3	Destination unreachable
4	Source quench
11	Time exceeded
12	Parameter problem
5	Redirection

Type	*Message*
8/0	Echo (request/reply)
13/14	Timestamp (req./rep.)
18/18	Address mask (req./rep.)
10/9	Router solicitation/advertisement

ICMP Error – Reporting Messages

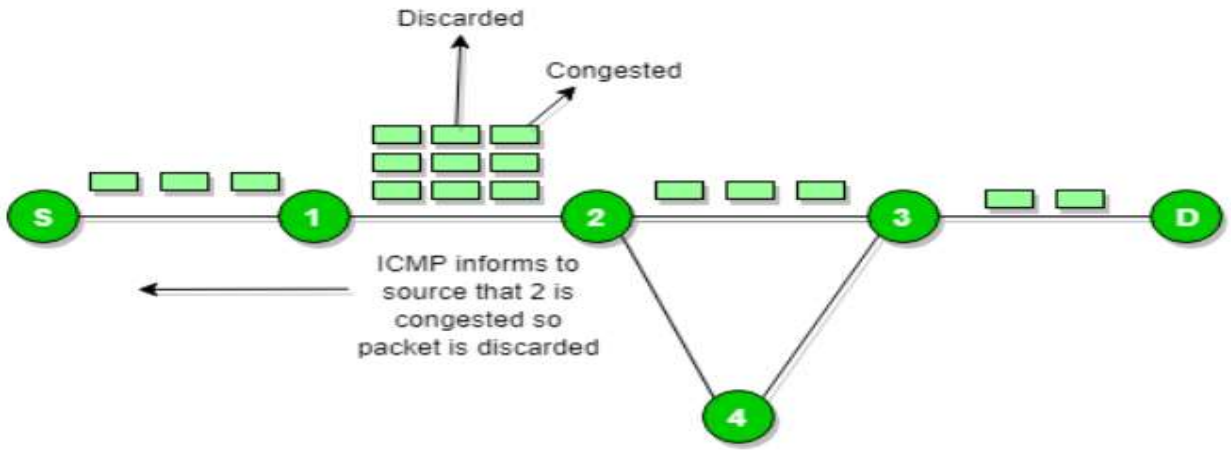
- ICMP error messages report error conditions
- Typically sent when a datagram is discarded
- Error message is often passed from ICMP to the application program



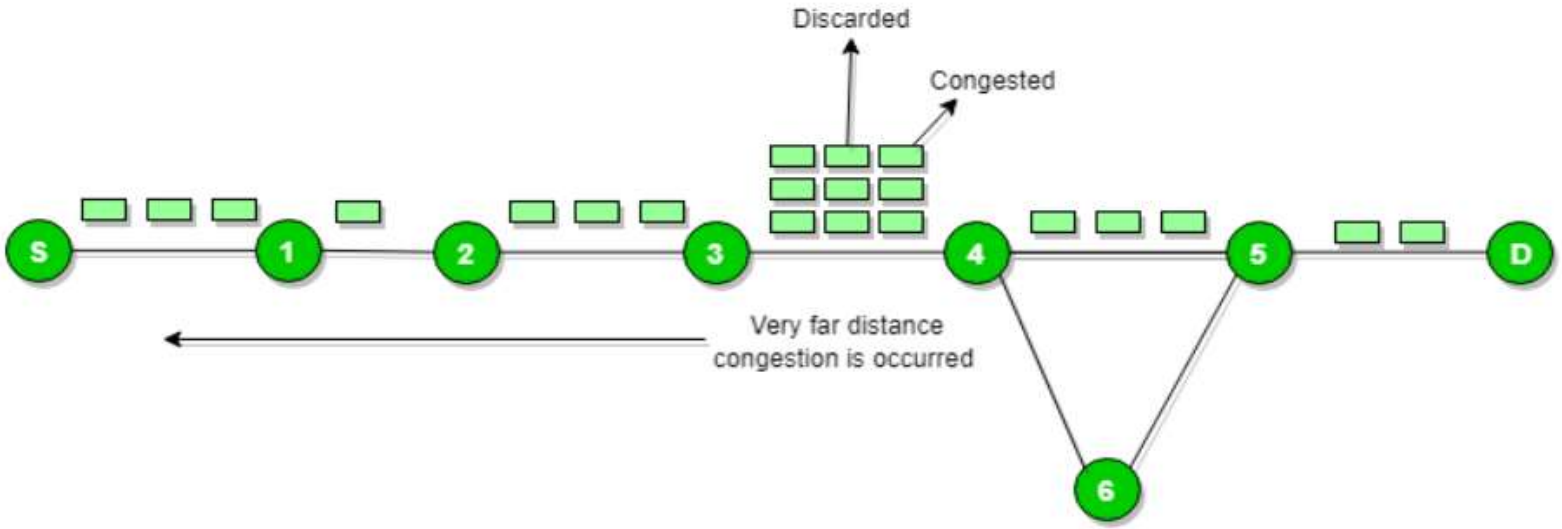
- ***Destination Unreachable***—When a router *cannot route* a datagram, the datagram is discarded and sends a destination unreachable message to source host.
- ***Source Quench***—When a router or host discards a datagram due to *congestion*, it sends a source-quench message to the source host. This message acts as flow control.
- ***Time Exceeded***—Router discards a datagram when TTL field becomes 0 and a time exceeded message is sent to the source host.
- ***Parameter Problem***—If a router discovers ambiguous or *missing* value in any field of the datagram, it discards the datagram and sends parameter problem message to source.
- ***Redirection***—Redirect messages are sent by the default router to inform the source host to *update* its forwarding table when the packet is routed on a wrong path.

Source quench message

- Source quench message is request to decrease traffic rate for messages sending to the host(destination).
- Or we can say, when receiving host detects that rate of sending packets (traffic rate) to it is too fast it sends the source quench message to the source to slow the pace down so that no packet can be lost.

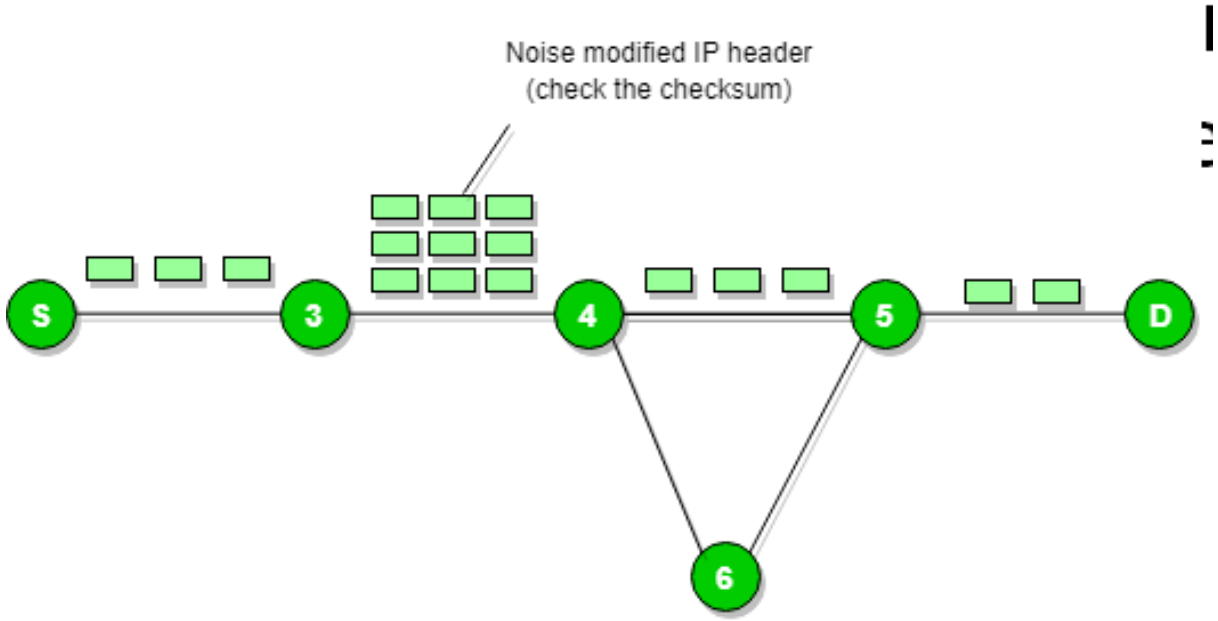


- ICMP will take source IP from the discarded packet and informs to source by sending source quench message.
- Then source will reduce the speed of transmission so that router will free for congestion.
- When the congestion router is far away from the source the ICMP will send hop by hop source quench message so that every router will reduce the speed of transmission.



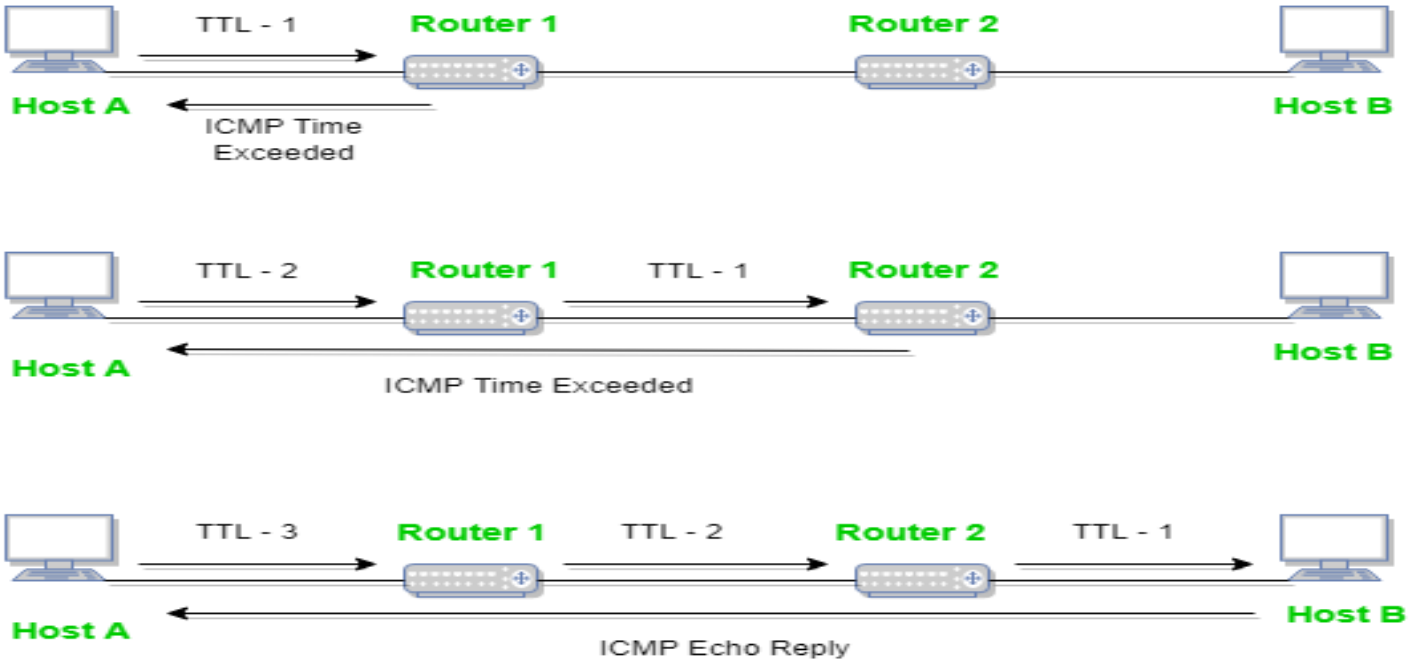
Parameter Problem

- Whenever packets come to the router then calculated header checksum should be equal to received header checksum then only packet is accepted by the router.
- If there is mismatch packet will be dropped by the router.
- ICMP will take the source IP from the discarded packet and informs to source by sending parameter problem message.



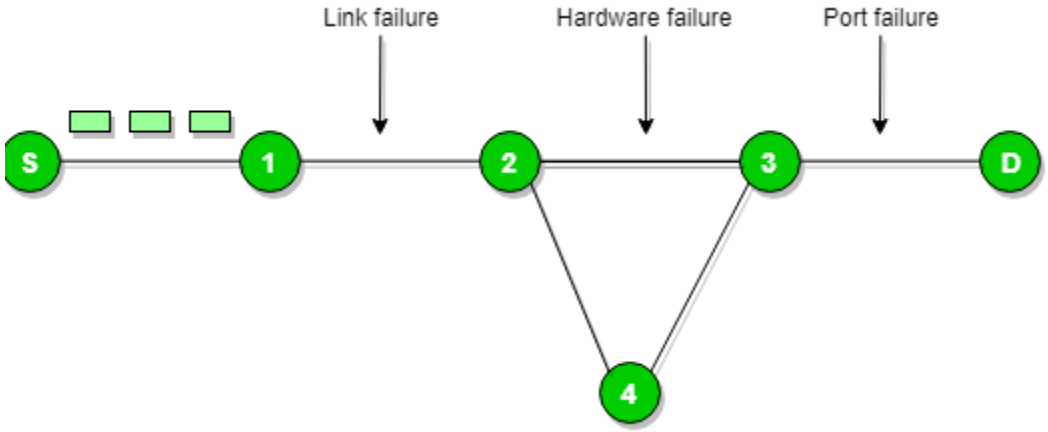
Time Exceeded Message

- When some fragments are lost in a network then the holding fragment by the router will be dropped then ICMP will take source IP from discarded packet and informs to the source, of discarded datagram due to time to live field reaches to zero, by sending time exceeded message.



Destination unreachable

- Destination unreachable is generated by the host or its inbound gateway to inform the client that the destination is unreachable for some reason.
- There is no necessary condition that only router give the ICMP error message some time destination host send ICMP error message when any type of failure (link failure,hardware failure,port failure etc) happen in the network.



Redirection message

- Redirect requests data packets be sent on an alternate route. The message informs to a host to update its routing information (to send packets on an alternate route).
- **Ex.** If host tries to send data through a router R1 and R1 sends data on a router R2 and there is a direct way from host to R2. Then R1 will send a redirect message to inform the host that there is a best way to the destination directly through R2 available. The host then sends data packets for the destination directly to R2.

Redirection message

- The router R2 will send the original datagram to the intended destination.
- But if datagram contains routing information then this message will not be sent even if a better route is available as redirects should only be sent by gateways and should not be sent by Internet hosts.

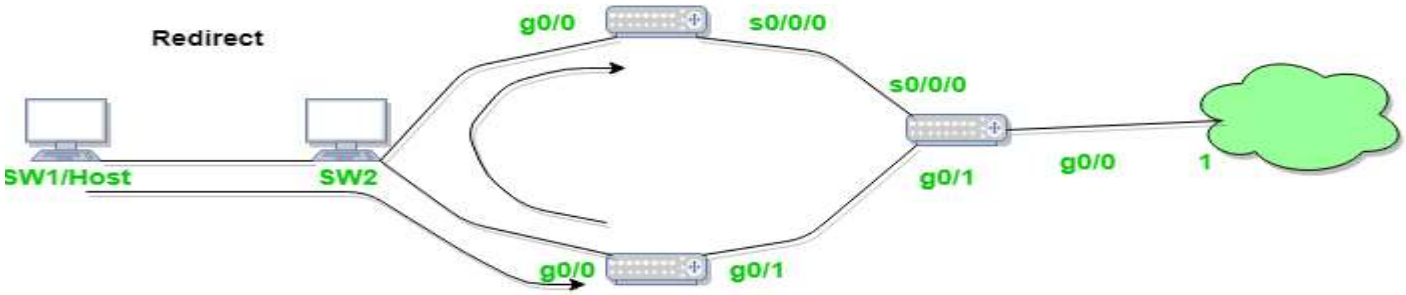
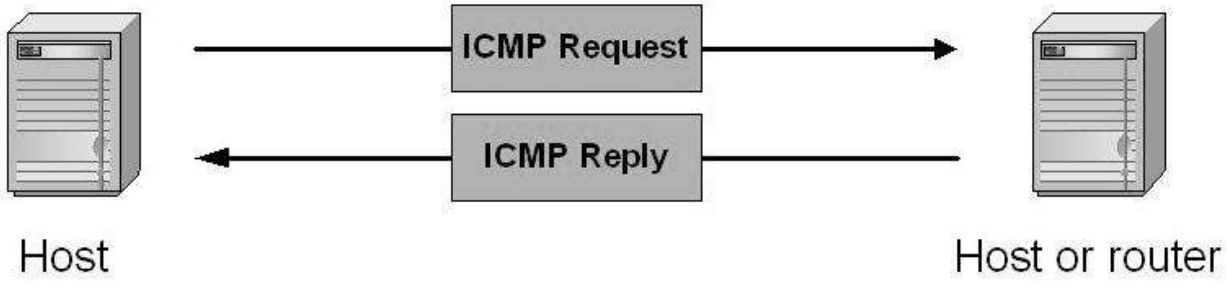


Figure - ICMP redirect Verification CCNP 2.0 100 - 101 (v - 71)

- ICMP Redirect
- ICMP Redirect for host
- ICMP Redirect for network
- How ICMP redirect work
- ICMP Redirect verification step by step

ICMP Query Messages

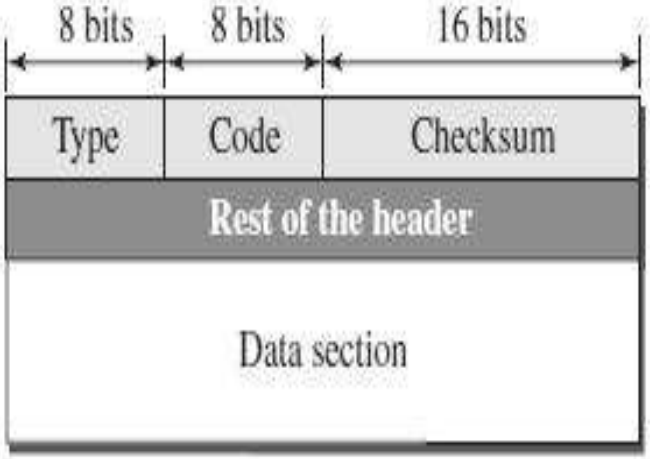
- Request sent by host to a router or host
- Reply sent back to querying host



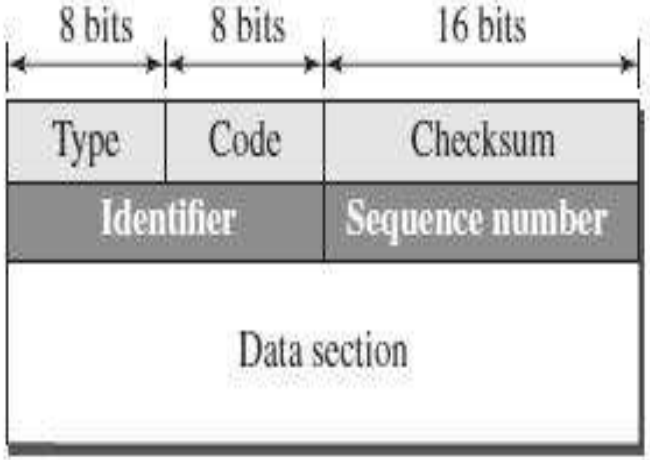
ICMP Query Messages

- ***Echo Request & Reply***—Combination of echo request and reply messages determines whether two systems communicate or not.
- ***Timestamp Request & Reply***—Two machines can use the timestamp request and reply messages to determine the round-trip time (RTT).
- ***Address Mask Request & Reply***—A host to obtain its subnet mask, sends an address mask request message to the router, which responds with an address mask reply message.
- ***Router Solicitation/Advertisement***—A host broadcasts a router solicitation message to know about the router. Router broadcasts its routing information with router advertisement message.

ICMP MESSAGE FORMAT



Error-reporting messages



Query messages

ICMP DEBUGGING TOOLS

- Two tools are used for debugging purpose. They are

(1) Ping

(2) Traceroute

Ping

- The *ping* program is used to find if a host is alive and responding.
- The source host sends ICMP echo-request messages; the destination, if alive, responds with ICMP echo-reply messages.
- The *ping* program sets the identifier field in the echo-request and echo-reply message and starts the sequence number from 0; this number is incremented by 1 each time a new message is sent.
- The *ping program* can calculate the round-trip time.
- It inserts the sending time in the data section of the message.
- When the packet arrives, it subtracts the arrival time from the departure time to get the round-trip time (RTT).

$ ping google.com

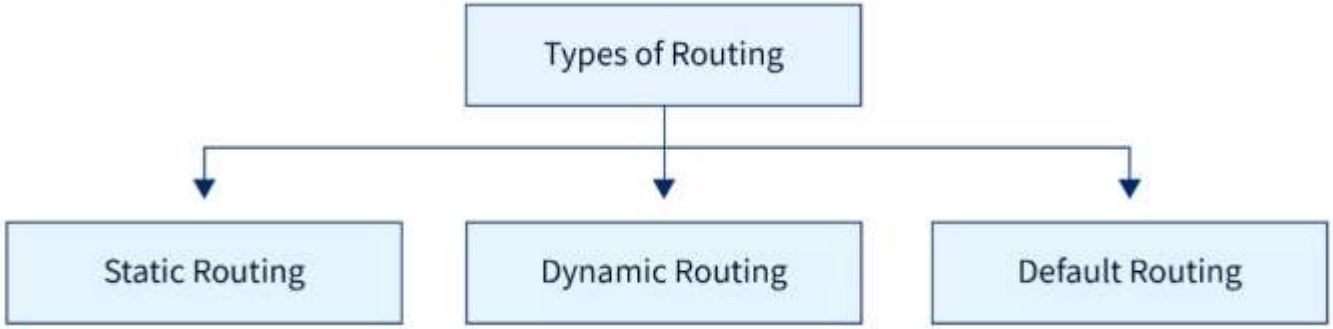
Traceroute or Tracert

- The *traceroute* program in UNIX or *tracert* in Windows can be used to trace the path of a packet from a source to the destination.
- It can find the IP addresses of all the routers that are visited along the path.
- The program is usually set to check for the maximum of 30 hops (routers) to be visited.
- The number of hops in the Internet is normally less than this.

$ traceroute google.com

Routing

- Routing is a process which is performed by layer 3 (or network layer) devices in order to deliver the packet by choosing an optimal path from one network to another.
- There are 3 types of routing:



Routing

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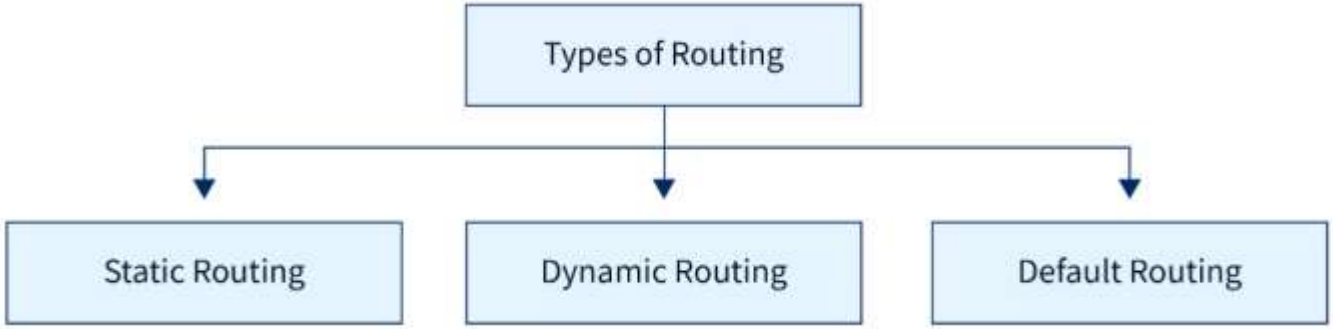
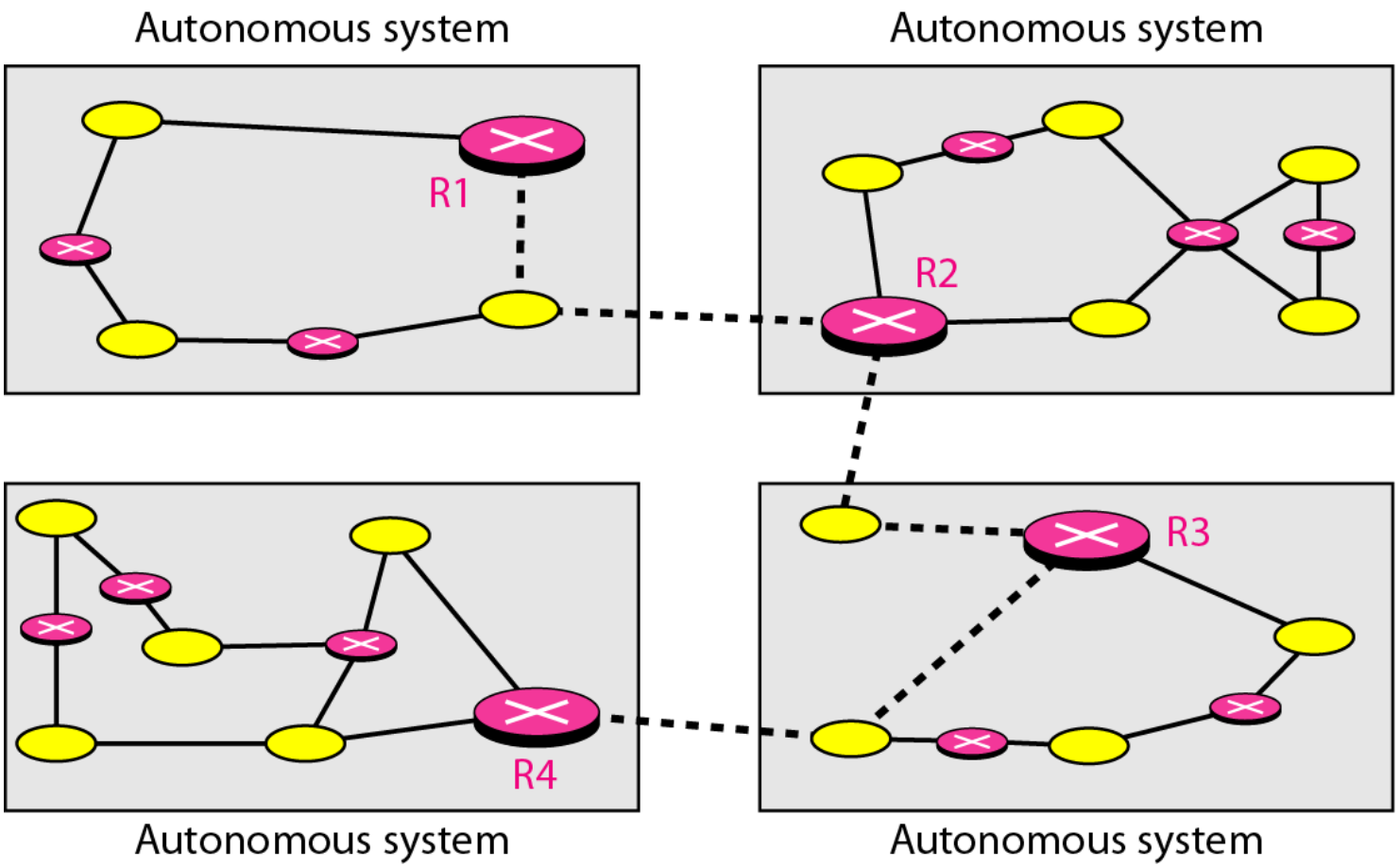


Figure *Autonomous systems*



- **1. Static routing :** Static routing is a process in which we have to manually add routes in routing table.

Advantages –

- No routing overhead for router CPU which means a cheaper router can be used to do routing.
- It adds security because only administrator can allow routing to particular networks only.
- No bandwidth usage between routers.

Disadvantage –

- For a large network, it is a hectic task for administrator to manually add each route for the network in the routing table on each router.
- The administrator should have good knowledge of the topology. If a new administrator comes, then he has to manually add each route so he should have very good knowledge of the routes of the topology.

2. Default Routing

- This is the method where the router is configured to send all packets towards a single router (next hop).
- It doesn’t matter to which network the packet belongs, it is forwarded out to router which is configured for default routing.
- It is generally used with stub routers. A stub router is a router which has only one route to reach all other networks.

3. Dynamic Routing

- Dynamic routing makes automatic adjustment of the routes according to the current state of the route in the routing table.
- Dynamic routing uses protocols to discover network destinations and the routes to reach it.
- RIP and OSPF are the best examples of dynamic routing protocol. Automatic adjustment will be made to reach the network destination if one route goes down.

A dynamic protocol have following features:

- The routers should have the same dynamic protocol running in order to exchange routes.
- When a router finds a change in the topology then router advertises it to all other routers.

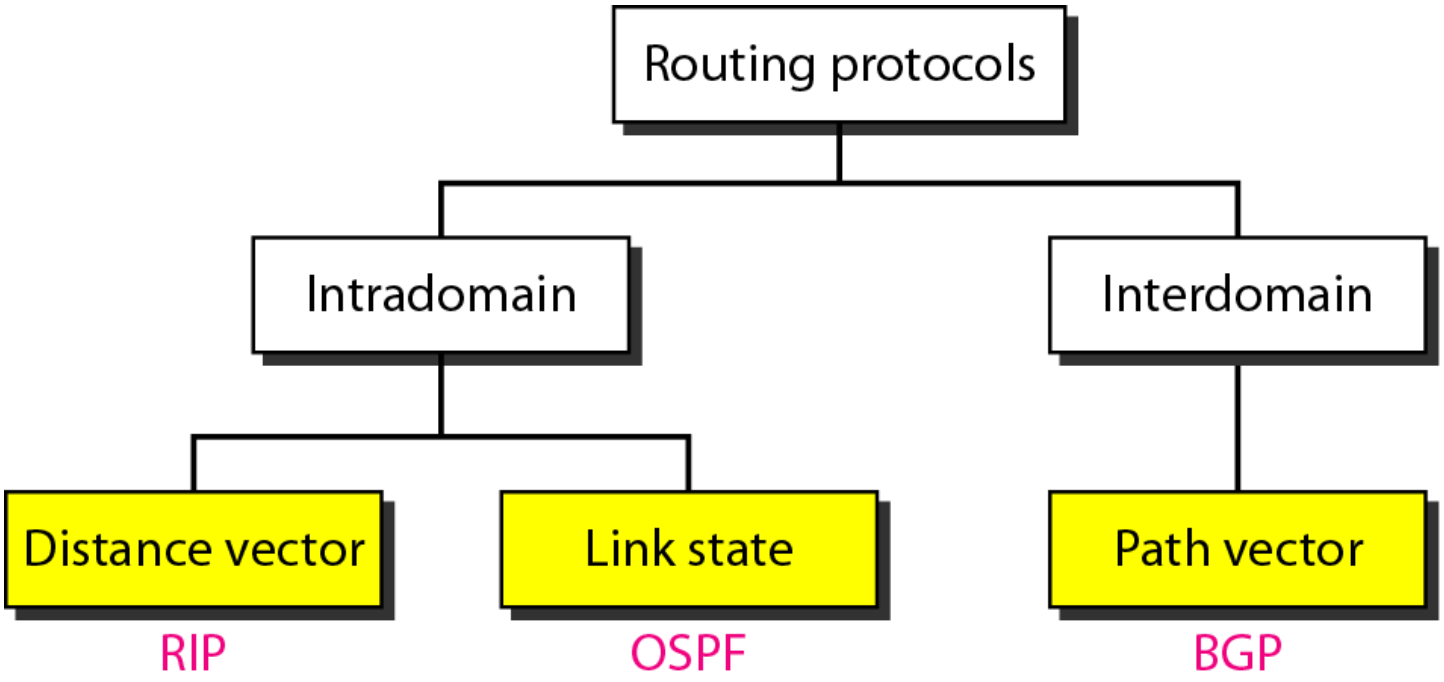
Advantages –

- Easy to configure.
- More effective at selecting the best route to a destination remote network and also for discovering remote network.

Disadvantage –

- Consumes more bandwidth for communicating with other neighbors.
- Less secure than static routing.

TYPES OF ROUTING PROTOCOLS



TYPES OF ROUTING PROTOCOLS

- Two types of Routing Protocols are used in the Internet:

1) Intradomain routing

- Routing within a single autonomous system
- Routing Information Protocol (RIP) - based on the distance-vector algorithm
- Open Shortest Path First (OSPF) - based on the link-state algorithm

2) Interdomain routing

- Routing between autonomous systems.
- Border Gateway Protocol (BGP) - based on the path-vector algorithm

Unicast Routing Protocols

- **Unicast –** Unicast means the transmission from a single sender to a single receiver. It is a point to point communication between sender and receiver. There are various unicast protocols such as TCP, HTTP, etc.
- TCP is the most commonly used unicast protocol. It is a connection oriented protocol that relay on acknowledgement from the receiver side.
- HTTP stands for Hyper Text Transfer Protocol. It is an object oriented protocol for communication.

Unicast Routing

- There are three main classes of routing protocols:

1) Distance Vector Routing Algorithm – Routing Information Protocol

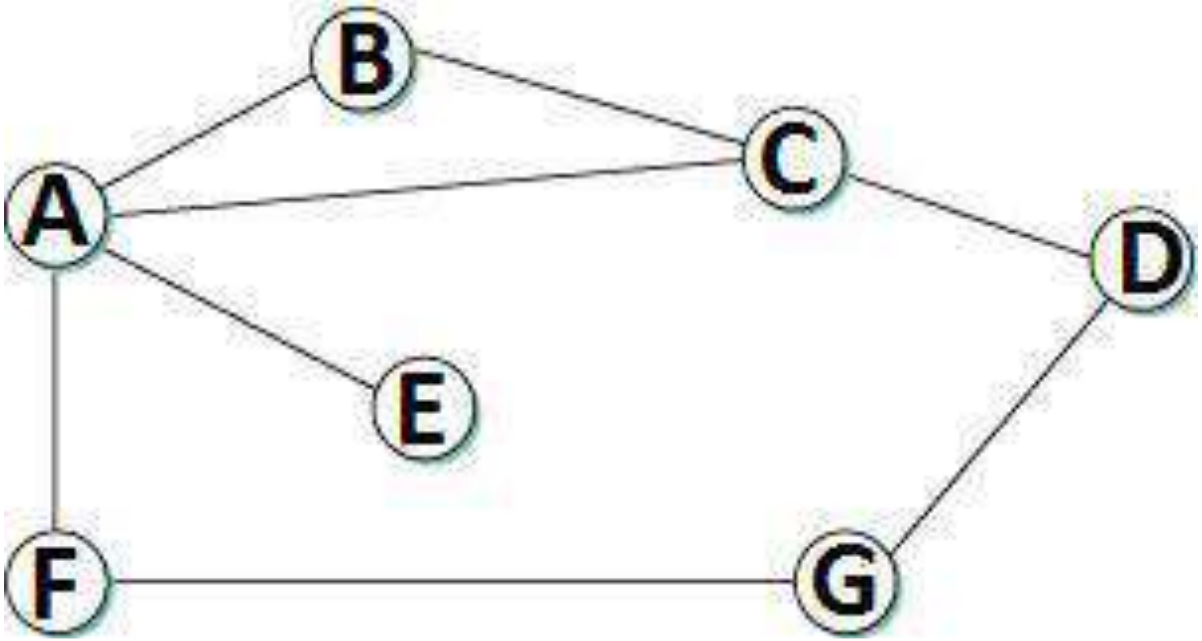
2) Link State Routing Algorithm – Open Shortest Path First Protocol

3) Path-Vector Routing Algorithm - Border Gateway Protocol

DISTANCE VECTOR ROUTING (DSR)

- Distance vector routing is *distributed*, i.e., algorithm is run on all nodes.
- Each node *knows* the distance (cost) to each of its directly connected neighbors.
- Nodes construct a *vector* (Destination, Cost, NextHop) and distributes to its neighbors.
- Nodes compute routing table of *minimum* distance to every other node via NextHop using information obtained from its neighbors.

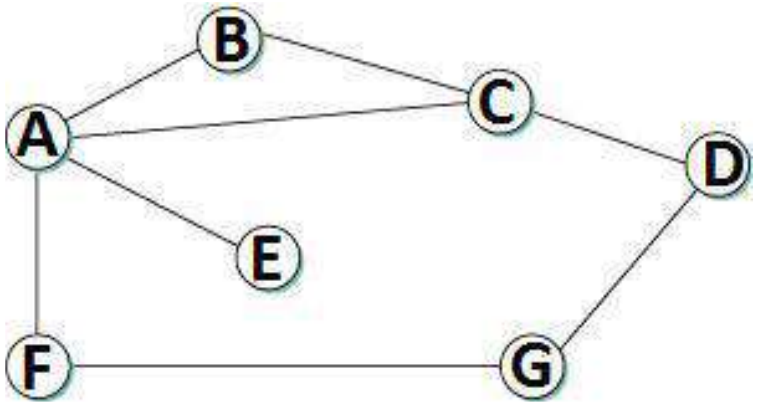
Initial State



Note: Cost of each link is 1 hop

Initial State

- In given network, *cost* of each link is 1 hop.
- Each node sets a distance of 1 (hop) to its *immediate* neighbor and cost to itself as 0.
- Distance for non-neighbors is marked as *unreachable* with value ∞ (infinity).
- For node *A*, nodes *B*, *C*, *E* and *F* are *reachable*, whereas nodes *D* and *G* are *unreachable*.



Destination	Cost	NextHop
A	0	A
B	1	B
C	1	C
D	∞	—
E	1	E
F	1	F
G	∞	—

Node A's initial table

Destination	Cost	NextHop
A	1	A
B	1	B
C	0	C
D	1	D
E	∞	—
F	∞	—
G	∞	—

Node C's initial table

Destination	Cost	NextHop
A	1	A
B	∞	—
C	∞	—
D	∞	—
E	∞	—
F	0	F
G	1	G

Node F's initial table

- The initial table for all the nodes are given below

Initial Distances Stored at Each Node (Global View)							
Information Stored at Node	Distance to Reach Node						
	A	B	C	D	E	F	G
A	0	1	1	∞	1	1	∞
B	1	0	1	∞	∞	∞	∞
C	1	1	0	1	∞	∞	∞
D	∞	∞	1	0	∞	∞	1
E	1	∞	∞	∞	0	∞	∞
F	1	∞	∞	∞	∞	0	1
G	∞	∞	∞	1	∞	1	0

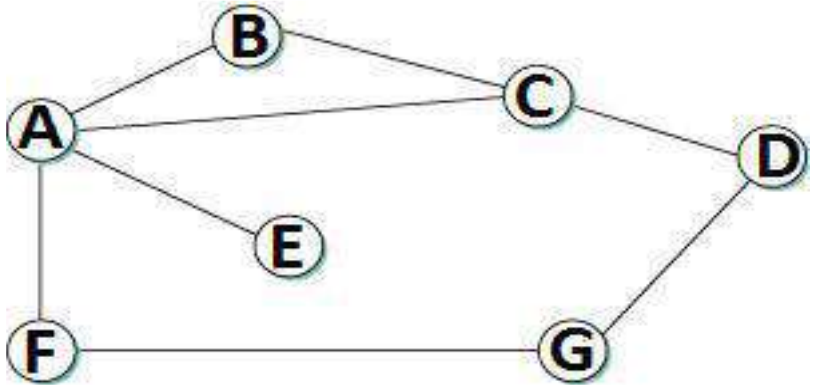
- Each node *sends* its initial table (distance vector) to neighbors and receives their estimate.
- Node *A* sends its table to nodes *B*, *C*, *E* & *F* and receives tables from nodes *B*, *C*, *E* & *F*.
- Each node *updates* its routing table by comparing with each of its neighbor's table
- For each destination, Total Cost is computed as:
  - ***Total Cost*** = Cost (*Node* to *Neighbor*) + Cost (*Neighbor* to *Destination*)
- If Total Cost < Cost then
  - ***Cost*** = Total Cost and NextHop = *Neighbor*
- Node *A learns* from *C*'s table to reach node *D* and from *F*'s table to reach node *G*.
- Total Cost to reach node *D* via *C* = Cost (*A* to *C*) + Cost(*C* to *D*)

  Cost = 1 + 1 = 2.
  - Since $2 < \infty$, entry for destination *D* in *A*'s table is changed to (*D*, 2, *C*)
  - Total Cost to reach node *G* via *F* = Cost(*A* to *F*) + Cost(*F* to *G*) = 1 + 1 = 2
  - Since $2 < \infty$, entry for destination *G* in *A*'s table is changed to (*G*, 2, *F*)
- Each node builds *complete* routing table after few exchanges amongst its neighbors.

Node A's final routing table

Destination	Cost	NextHop
A	0	A
B	1	B
C	1	C
D	2	C
E	1	E
F	1	F
G	2	F

- System stabilizes when all nodes have complete routing information, i.e., **convergence.**
- Routing tables are exchanged *periodically or* in case of *triggered update*.
- The final distances stored at each node is given below:



Final Distances Stored at Each Node (Global View)							
Information Stored at Node	Distance to Reach Node						
	A	B	C	D	E	F	G
A	0	1	1	2	1	1	2
B	1	0	1	2	2	2	3
C	1	1	0	1	2	2	2
D	2	2	1	0	3	2	1
E	1	2	2	3	0	2	3
F	1	2	2	2	2	0	1
G	2	3	2	1	3	1	0

Updation of Routing Tables

- There are two different circumstances under which a given node decides to send a routing update to its neighbors.

1. ***Periodic Update***
2. ***Triggered Update***

Periodic Update

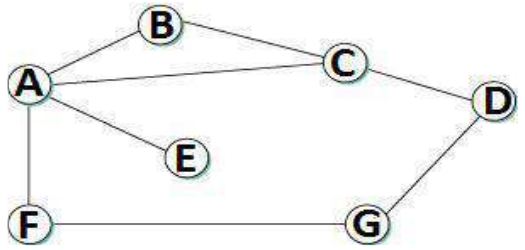
- In this case, each node automatically sends an update message every so often, even if nothing has changed.
- The frequency of these periodic updates varies from protocol to protocol, but it is typically on the order of several seconds to several minutes.

Triggered Update

- In this case, whenever a node notices a link failure or receives an update from one of its neighbors that causes it to change one of the routes in its routing table.
- Whenever a node's routing table changes, it sends an update to its neighbors, which may lead to a change in their tables, causing them to send an update to their neighbors.

Count-To-Infinity (or) Loop Instability Problem

- Suppose link from node *A* to *E* goes *down*.
  - Node *A* advertises a distance of ∞ to *E* to its neighbors
  - Node B receives periodic update from C before A's update reaches B
  - Node *B* updated by *C*, concludes that *E* can be reached in 3 hops via *C*
  - Node *B* advertises to *A* as 3 hops to reach *E*
  - Node *A* in turn updates *C* with a distance of 4 hops to *E* and so on



- Thus nodes update each other until cost to *E* reaches *infinity*, i.e., *no convergence*.
- Routing table does not stabilize.
- This problem is called *loop instability* or *count to infinity*

Why Count-to-Infinity Happens

- Distance vector routing works using **neighbor information only**. Nodes **do not know the entire topology**.

Step 1 — Immediately After Failure

- A updates: Distance(A → E) = ∞
- But neighbors still have old information:
- B thinks: E reachable via A (cost 2)
  C thinks: E reachable via A (cost 2)
  F thinks: E reachable via A (cost 2)

Step 2 — Wrong Assumption by A

- Before receiving A's update (or due to timing):
- B advertises to A:
- "I can reach E with distance = 2"
- A now assumes: "Oh! Maybe B has another path to E."
- So A updates: Distance(A → E) = 3 via B (WRONG — B's route actually goes back through A!)

Step 3 — Loop Formation

Now updates continue:

Exchange	A thinks	B thinks
Initial	∞	2
After update	3	2
Next update	3	4
Next	5	4
Next	5	6
...	...	...

Distance keeps increasing:

$3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow \ldots \rightarrow \infty$

This gradual increase is called: **Count-to-Infinity Problem**

Why It Happens

Because:

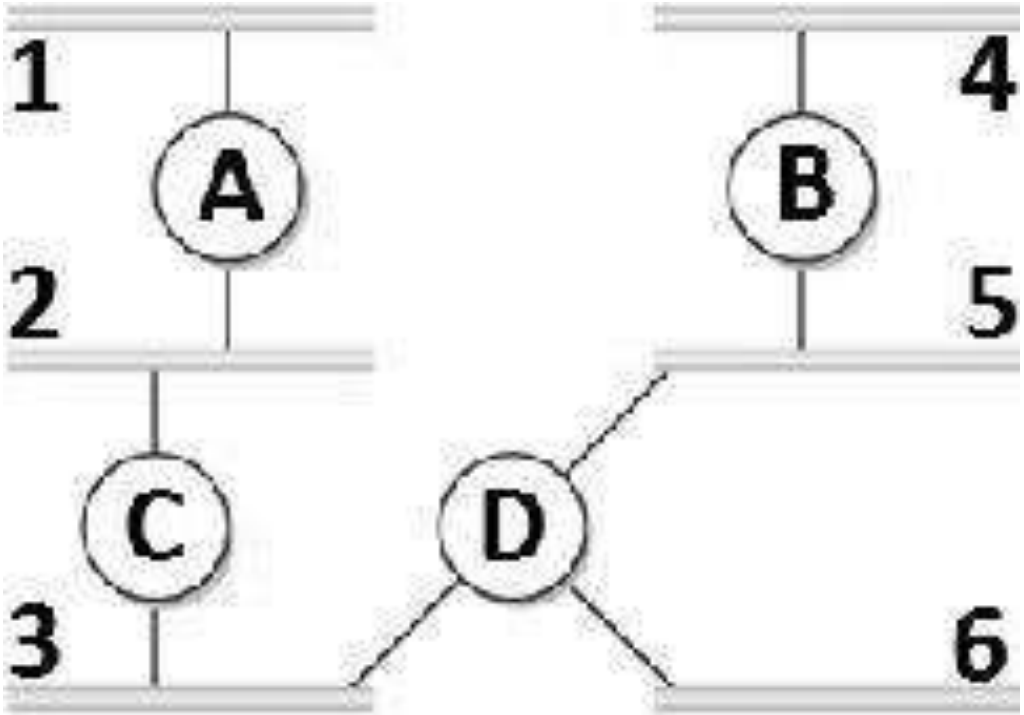
- Routers trust neighbors blindly
- No global network knowledge
- Old routing information circulates in loops
- In your topology: A ↔ B ↔ A ↔ B ...
- Packets keep believing a path exists even though **E is unreachable.**

Solution to Count-To-Infinity (or) Loop Instability Problem :

- *Infinity* is redefined to a small number, say 16.
- Distance between any two nodes can be 15 hops maximum. Thus distance vector routing *cannot be used* in large networks.
- When a node updates its neighbours, it does not send those routes it learned from each neighbour back to that neighbour. This is known as **split horizon**.
- **Split horizon with poison reverse** allows nodes to advertise routes it learnt from a node back to that node, but with a warning message.

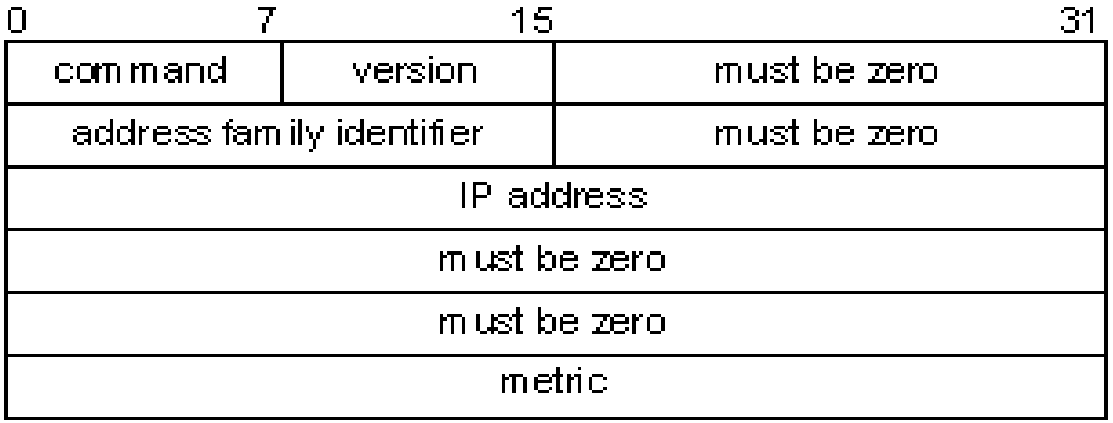
ROUTING INFORMATION PROTOCOL (RIP)

- RIP is an intra-domain routing protocol based on distance-vector algorithm.



ROUTING INFORMATION PROTOCOL (RIP)

- Routers *advertise* the cost of reaching networks. Cost of reaching each link is 1hop.
- For example, router *C* advertises to *A* that it can reach network *2*, *3* at cost 0 (directly connected), networks *5*, *6* at cost 1 and network *4* at cost 2.
- Each router *updates* cost and next hop for each network number.
- Infinity is defined as 16, i.e., any route cannot have more than 15 hops.
- Therefore RIP can be implemented on small-sized networks only.
- Advertisements are sent every 30 seconds or in case of triggered update.



- ***Command*** - It indicates the packet type.
- Value 1 represents a request packet. Value 2 represents a response packet.
- ***Version*** - It indicates the RIP version number. For RIPv1, the value is 0x01, 2 → RIP Version 2
- ***Address Family Identifier*** - When the value is 2, it represents the IP protocol.
- ***IP Address*** - It indicates the destination IP address of the route.
- ***Metric*** - It indicates the hop count of a route to its destination.

LINK STATE ROUTING (LSR)

- Each node knows *state* of link to its neighbors and *cost*.
- Nodes create an update packet called *link-state packet* (LSP) that contains:
  - ID of the node
  - List of neighbors for that node and associated cost
  - 64-bit Sequence number
  - Time to live
- Link-State routing protocols rely on two mechanisms:
  - ***Reliable flooding*** of link-state information to all other nodes
  - ***Route calculation*** from the accumulated link-state knowledge

Link State Routing

- Link state routing is a technique in which each router shares the knowledge of its neighborhood with every other router in the internetwork.

The three keys to understand the Link State Routing algorithm:

- **Knowledge about the neighborhood:** Instead of sending its routing table, a router sends the information about its neighborhood only. A router broadcast its identities and cost of the directly attached links to other routers.
- **Flooding:** Each router sends the information to every other router on the internetwork except its neighbors. This process is known as Flooding. Every router that receives the packet sends the copies to all its neighbors. Finally, each and every router receives a copy of the same information.
- **Information sharing:** A router sends the information to every other router only when the change occurs in the information.

Link state Routing

- Link state routing is the second family of routing protocols.
- While distance vector routers use a distributed algorithm to compute their routing tables, link-state routing uses link-state routers to exchange messages that allow each router to learn the entire network topology.
- Based on this learned topology, each router is then able to compute its routing table by using a shortest path computation.

Features of link state routing protocols –

- **Link state packet –** A small packet that contains routing information.
- **Link state database –** A collection information gathered from link state packet.
- **Shortest path first algorithm (Dijkstra algorithm) –** A calculation performed on the database results into shortest path
- **Routing table –** A list of known paths and interfaces.

Working Principle

- Discover its neighbors and learn their network addresses.
- Measure the delay or cost to each of its neighbours.
- Construct a packet telling all it has just learned.
- Send this packet to all other routers and Compute the shortest path to every other router.

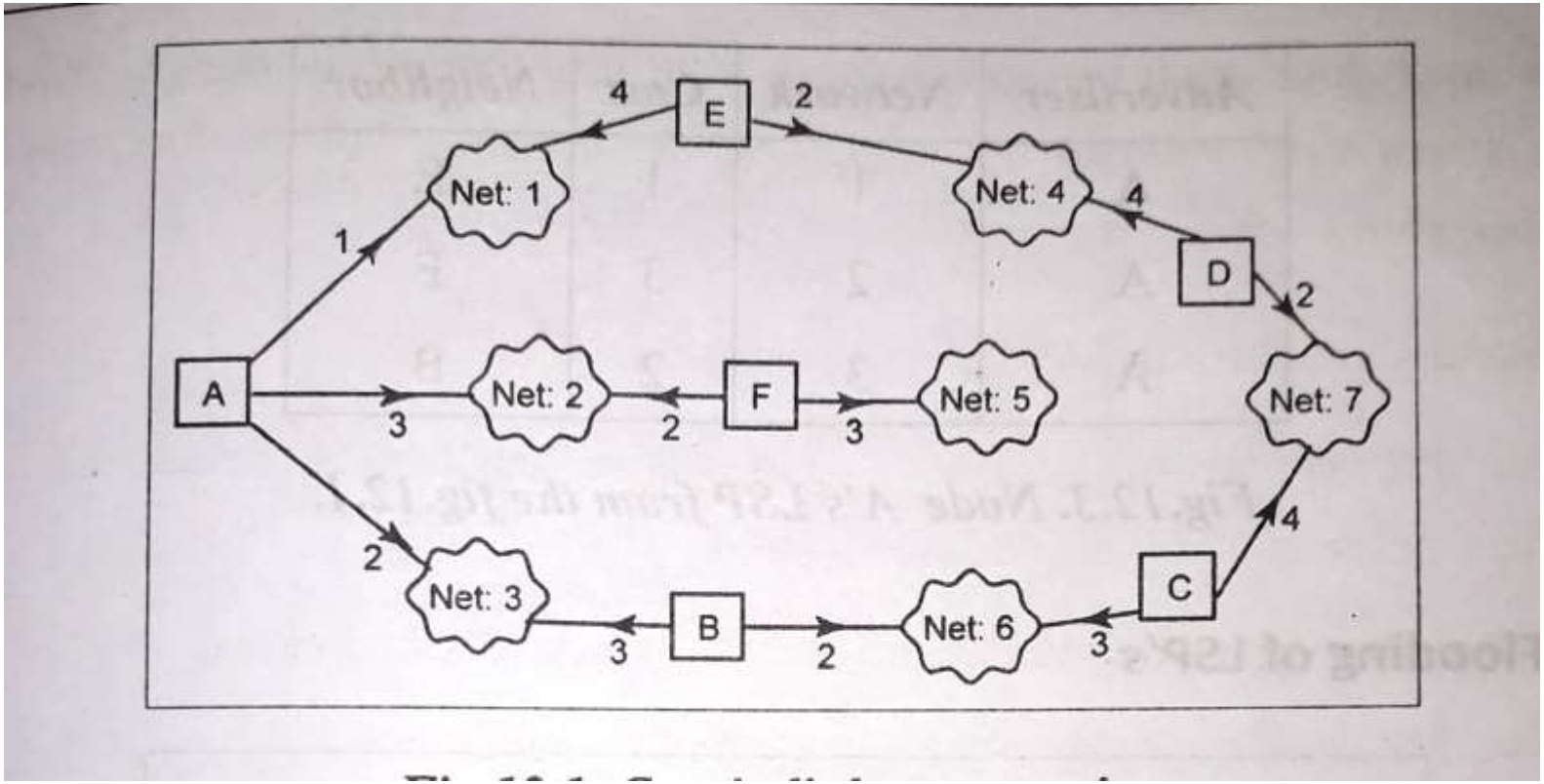
Step 1 Information Sharing

- The first step in link state routing is information sharing.
- Each router sends its knowledge about its neighborhood to every other router in the internetwork.

Step 2 Measuring Packet Cost

- Both distance vector and link state routing are lowest cost algorithms.
- In distance vector routing, cost refers to hop count.
- In link state routing, cost is a weighted value based on a variety of factors such as security levels, traffic or the state of the link.
- The cost from router A to network 1, therefore, might be different from the cost from A to network 2.
- Cost is applied only by routers and not by any other stations on a network.
- In determining a route, the cost of a hop is applied to each packet as it leaves a router and enters a network.

Cost in link state routing



Step 3 Building link state packet

- When a router floods the network with information about its neighbourhood, it is said to be advertising.
- In link state routing, a small packet containing routing information sent by the router to all other routers is referred as **Link-State Packet(LSP).**

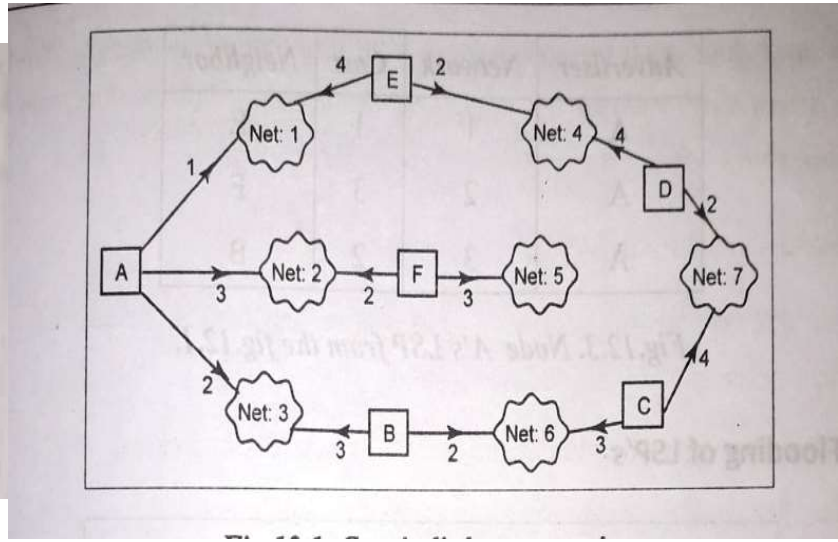
Link State Packet Format

- Link- State Packet contains 4 fields
  - The ID of the advertiser
  - The ID of the destination network
  - The cost
  - The ID of the neighbor router

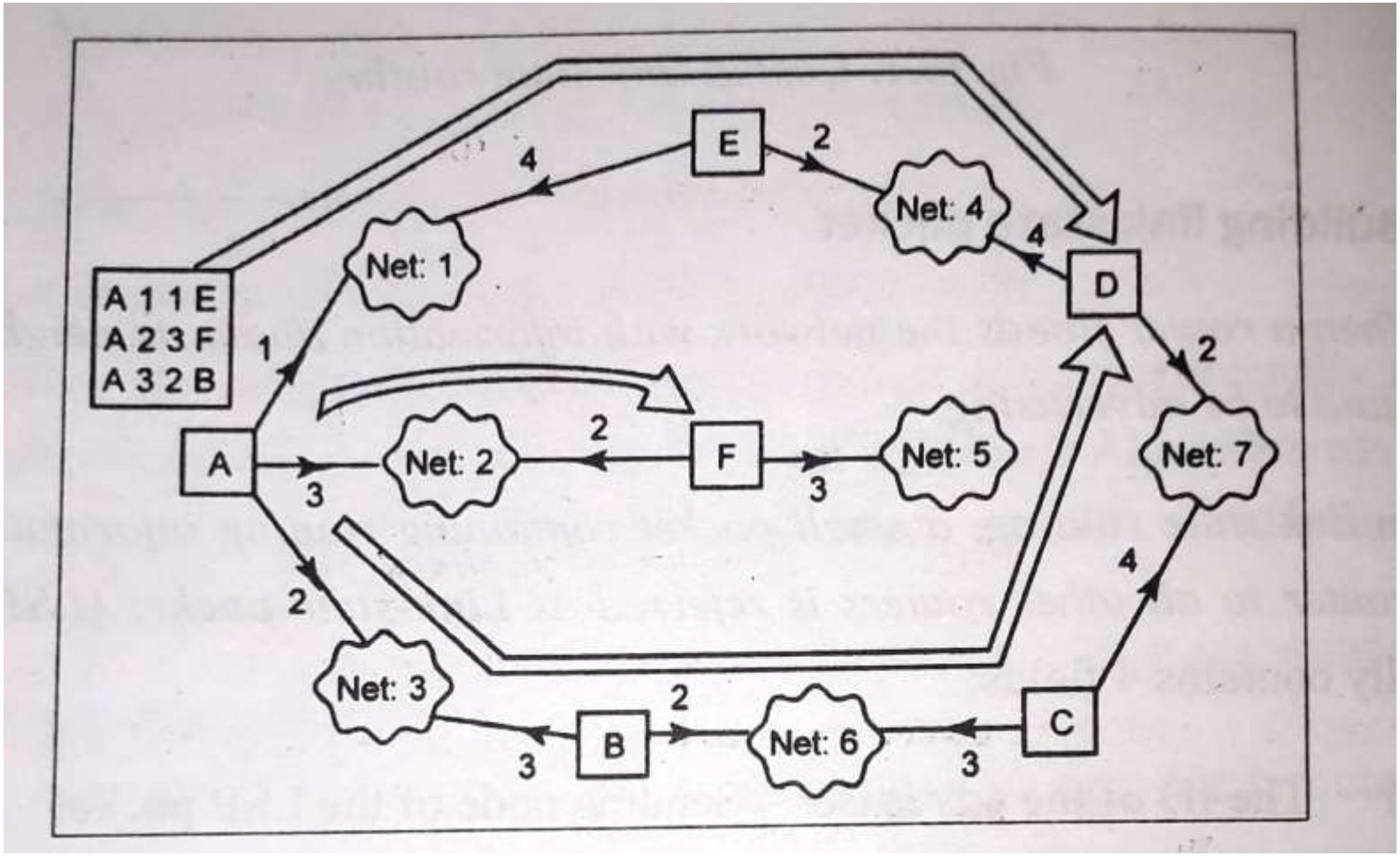
Link State Packet Format

Advertiser	Network	Cost	Neighbor
----------	--------	----	-----
----------	--------	----	-----
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Advertiser	Network	Cost	Neighbor
A	1	1	E
A	2	3	F
A	3	2	B



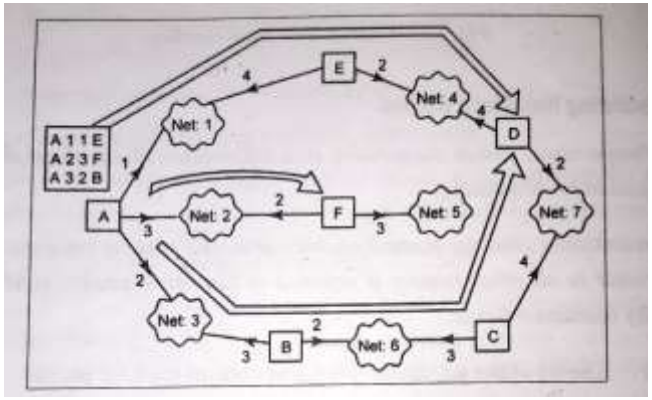
Step 4: Flooding of LSP



Step 4: Flooding of LSP

- After creating LSP, every router forwards it to each and every other router in the internet.
- Link State Database – With the help of the information in the every LSP packet, which is received by every router and it creates a common database to all routers is referred as Link State Database.
- Every router stores this database on its disk, and uses it for routing packets.

Link State Database



Advertiser	Network	Cost	Neighbor
A	1	1	E
A	2	3	F
A	3	2	B
B	3	3	A
B	6	2	C
C	6	3	B
C	7	4	D
D	4	4	E
D	7	2	C
E	1	4	A
E	4	2	D
F	2	2	A
F	5	3	--

Step 5: Computing shortest path tree

- Once the link state database has been created, each router applies an algorithm called Dijkstra algorithm to it, in order to calculate its routing table.

Shortest Path Routing

- 2 algorithms for computing the shortest path between two nodes of a graph are known.

1. Dijkstra's algorithm
2. Bellman-Ford algorithm

Calculation of Shortest Path

- To find shortest path, each node need to run the famous **Dijkstra algorithm**. This famous algorithm uses the following steps:
- **Step-1:** The node is taken and chosen as a root node of the tree, this creates the tree with a single node, and now set the total cost of each node to some value based on the information in Link State Database
- **Step-2:** Now the node selects one node, among all the nodes not in the tree like structure, which is nearest to the root, and adds this to the tree. The shape of the tree gets changed .
- **Step-3:** After this node is added to the tree, the cost of all the nodes not in the tree needs to be updated because the paths may have been changed.
- **Step-4:** The node repeats the Step 2. and Step 3. until all the nodes are added in the tree

Example

Example 3.10.1 *Find the shortest path between node A and node H for the following Fig. 3.10.1 by applying Dijkstra's algorithm. Show each steps output.*

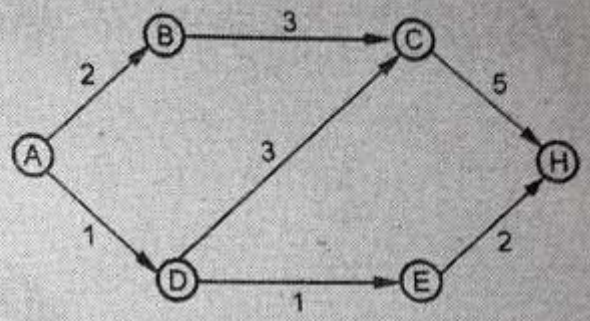


Fig. 3.10.1

Solution :

Step-I : Node A is initialized as source node.

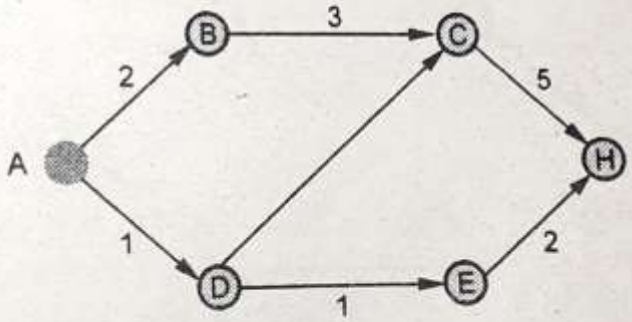


Fig. 3.10.1 (a)

Step-II : Link cost is computed for the adjacent node.

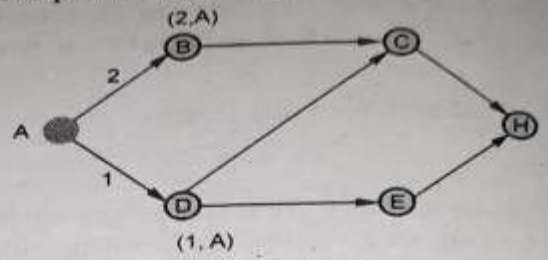


Fig. 3.10.1 (b)

Step-III : Since AD is smallest path, now D is working node.

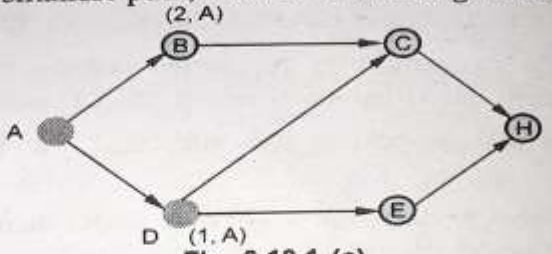


Fig. 3.10.1 (c)

Step-IV : Adjacent nodes to D are C and E.

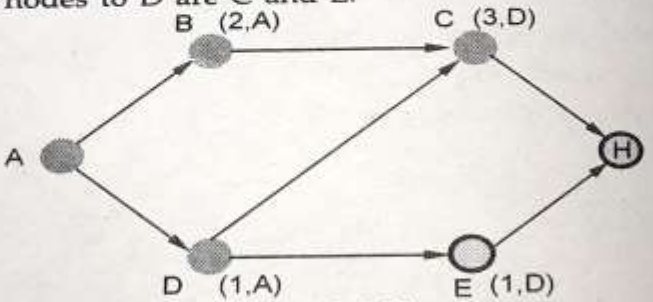


Fig. 3.10.1 (d)

Step-V : Since shortest is E, now E is working node.

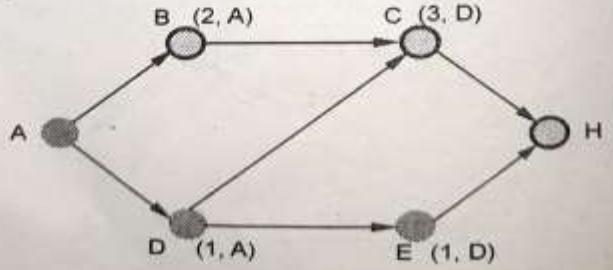


Fig.3.10.1 (e)

Step-VI :

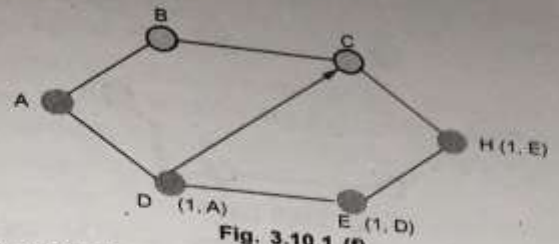


Fig. 3.10.1 (f)

Hence the shortest path between node A and node H is ADEH.

ii) Bellman-Ford algorithm :

Bellman-Ford algorithm is somewhat similar to Dijkstra's algorithm but here the shortest paths from a given source node is computed subject to the constraint that the path contain at most one link, i.e. from source node, at each step least-cost path with maximum number of links are found. Finally the least-cost path to each node and the cost of that path is computed. Bellman-Ford algorithm is illustrated in the following example.

Example 3.10.2 *Find the shortest path between node A and node H using Bellman-Ford algorithm, for the Fig. 3.10.2 shown in example 3.10.1.*

Solution : Step-1 :

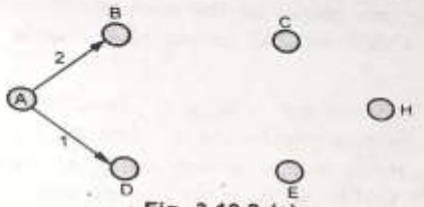


Fig. 3.10.2 (a)

Distance AD is shorter than AB. So route AD is chosen.

Step-2 :

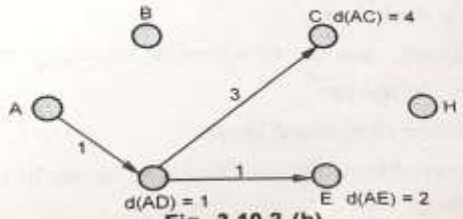


Fig. 3.10.2 (b)

$\because$ d(AE) < d(AC)

$\therefore$ d(AE) is chosen.

Step-3 :

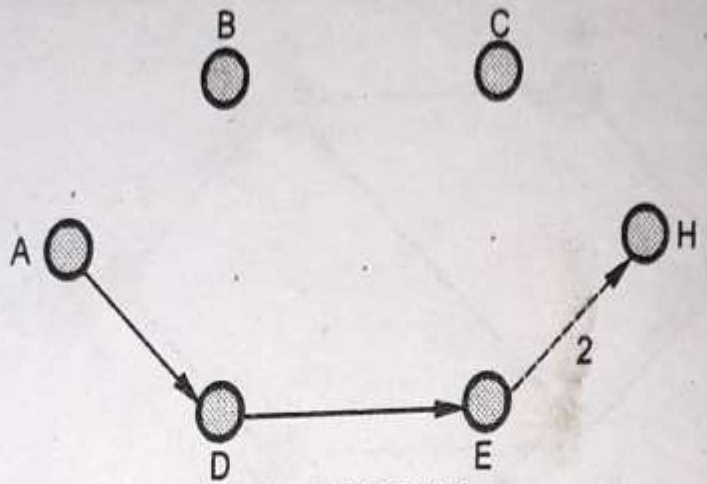


Fig. 3.10.2 (c)

So the shortest distance is ADEH, the result is same as in Dijkstra's algorithm.

Open Shortest Path First (OSPF) Protocol fundamentals

- Open shortest path first (OSPF) is a **link-state routing protocol** which is used to **find the best path between the source and the destination router using its own shortest path first (SPF) algorithm.**
- A link-state routing protocol is a protocol which uses the concept of **triggered updates**, i.e., if there is a change observed in the learned routing table then the updates are triggered only, not like the distance-vector routing protocol where the routing table are exchanged at a period of time.

- It is a **network layer protocol** which works on the protocol number 89 and uses AD value 110.
- OSPF uses multicast address 224.0.0.5 for normal communication and 224.0.0.6 for update to designated router(DR)/Backup Designated Router (BDR).
- **Criteria** – To form neighbourship in OSPF, there is a criteria for both the routers:
- It should be present in same area
- Router I'd must be unique
- Subnet mask should be same
- Hello and dead timer should be same
- Stub flag must match (This field used during **neighbor relationship formation** to indicate whether a router belongs to a **stub area** or not.)
- Authentication must match

OSPF messages –

- **OSPF messages –**
  OSPF uses certain messages for the communication between the routers operating OSPF.
- **Hello message –** These are keep alive messages used for neighbor discovery /recovery. These are exchanged in every 10 seconds. This include following information : Router I'd, Hello/dead interval, Area I'd, Router priority, DR and BDR IP address, authentication data.
- **Database Description (DBD) –** It is the OSPF routes of the router. This contains topology of an AS or an area (routing domain).

OSPF messages –

- **Link state request (LSR) –** When a router receive DBD, it compares it with its own DBD. If the DBD received has some more updates than its own DBD then LSR is being sent to its neighbor.
- **Link state update (LSU) –** When a router receives LSR, it responds with LSU message containing the details requested.
- **Link state acknowledgement –** This provides reliability to the link state exchange process. It is sent as the acknowledgement of LSU.
- **Link state advertisement (LSA) –** It is an OSPF data packet that contains link-state routing information, shared only with the routers to which adjacency has been formed.

Timers –

- **Hello timer –** The interval in which OSPF router sends a hello message on an interface. It is 10 seconds by default.
- **Dead timer –** The interval in which the neighbor will be declared dead if it is not able to send the hello packet . It is 40 seconds by default.It is usually 4 times the hello interval but can be configured manually according to need.

OSPF supports/provides/advantages –

- Both IPv4 and IPv6 routed protocols
- Load balancing with equal cost routes for same destination
- VLSM and route summarization
- Unlimited hop counts
- Trigger updates for fast convergence
- A loop free topology using SPF algorithm
- Run on most routers
- Classless protocol

Open shortest path first (OSPF) router roles –

- An area is a group of contiguous network and routers.
- Routers belonging to same area shares a common topology table and area I'd. The area I'd is associated with router's interface as a router can belong to more than one area. There are some roles of router in OSPF:

OSPF

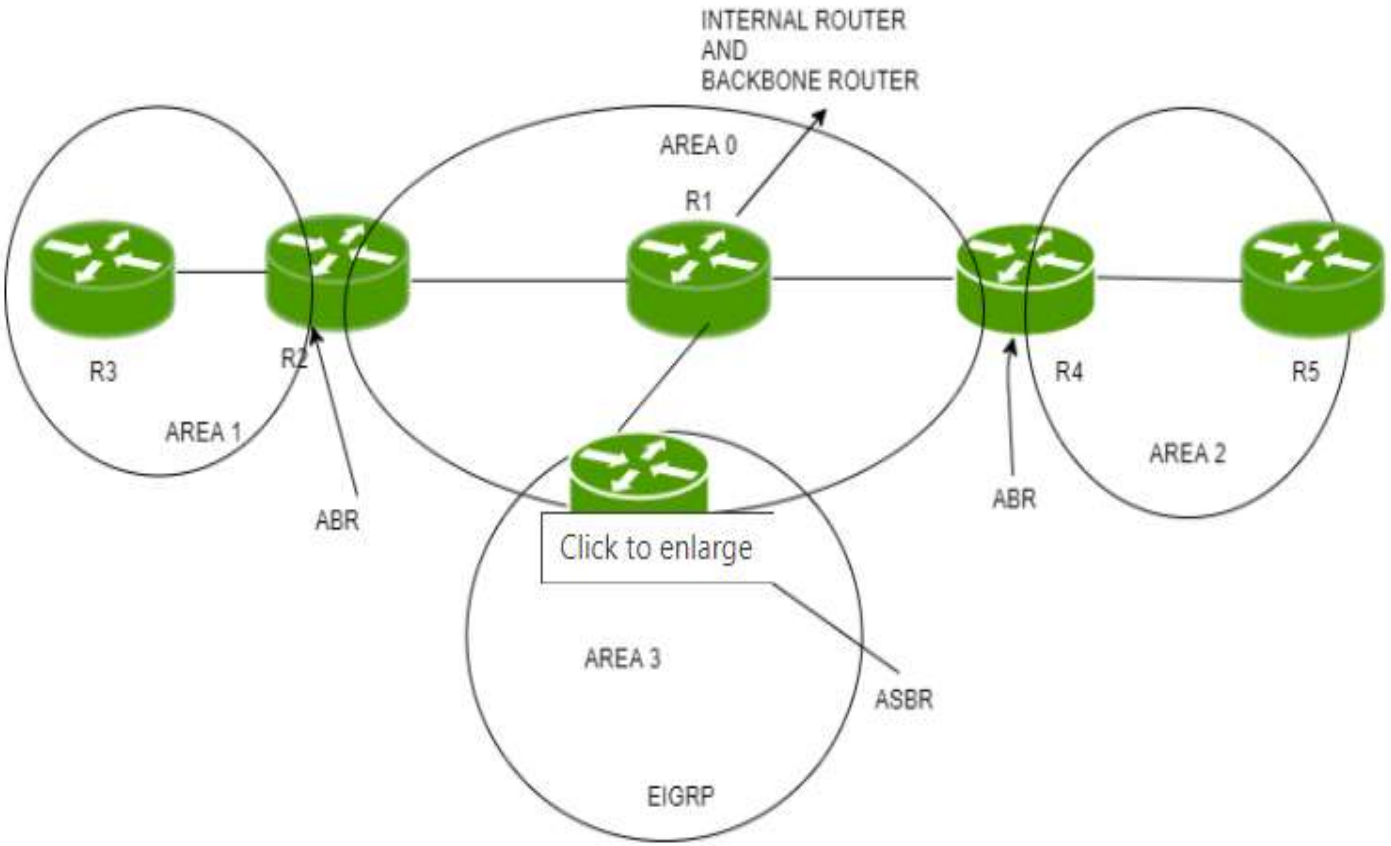

INTERNAL ROUTER
AND
BACKBONE ROUTER
AREA 0
R1
R3
R2
AREA 1
ABR
R4
R5
AREA 2
ABR
Click to enlarge
AREA 3
ASBR
EIGRP

Figure ***Areas in an autonomous system***

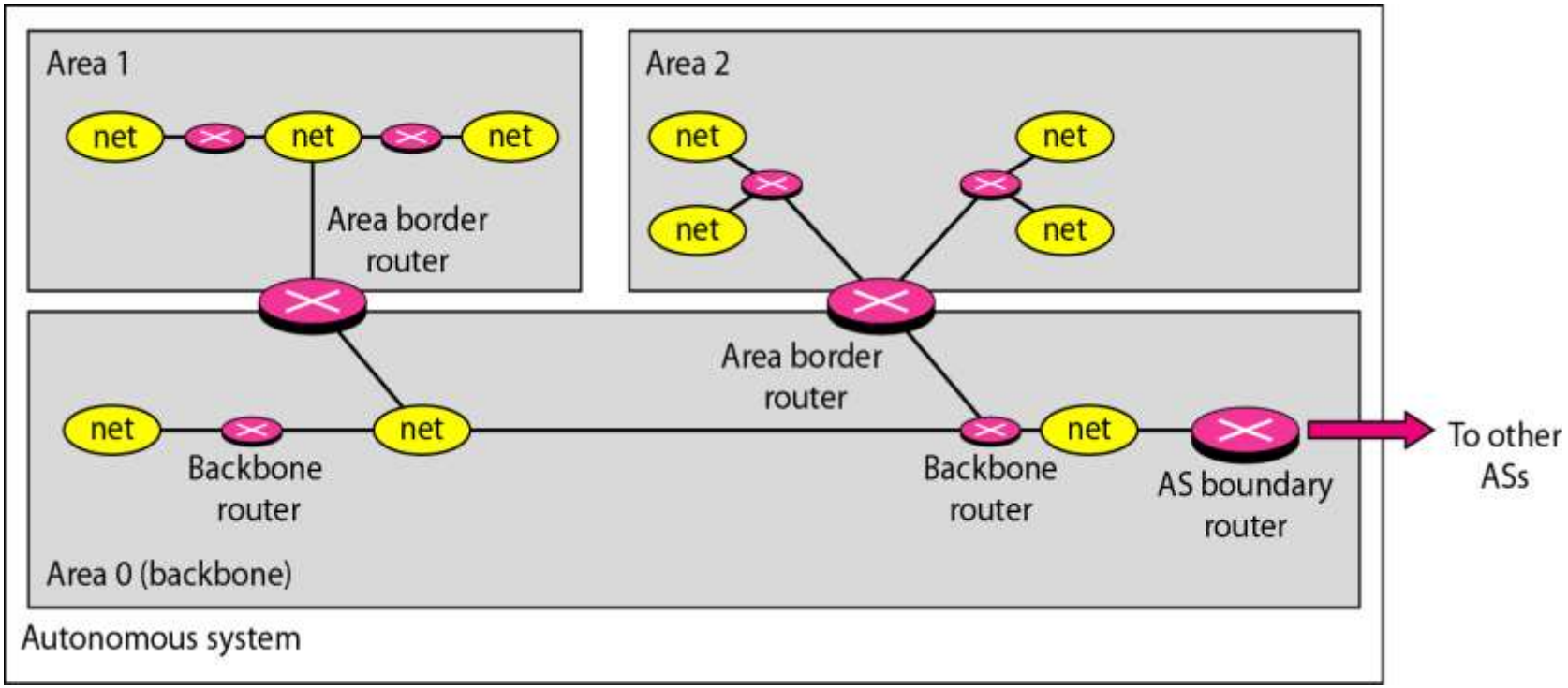
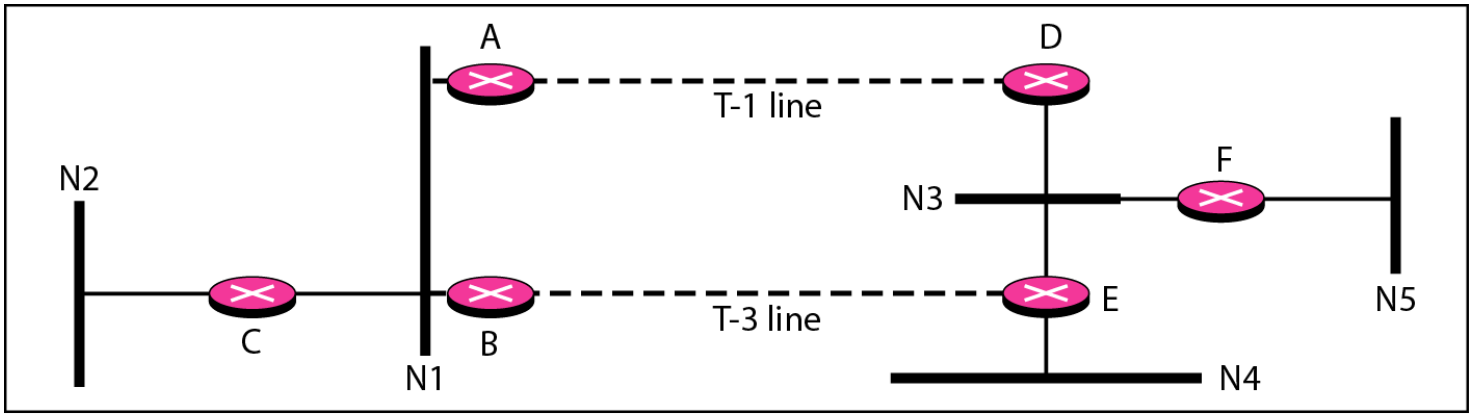
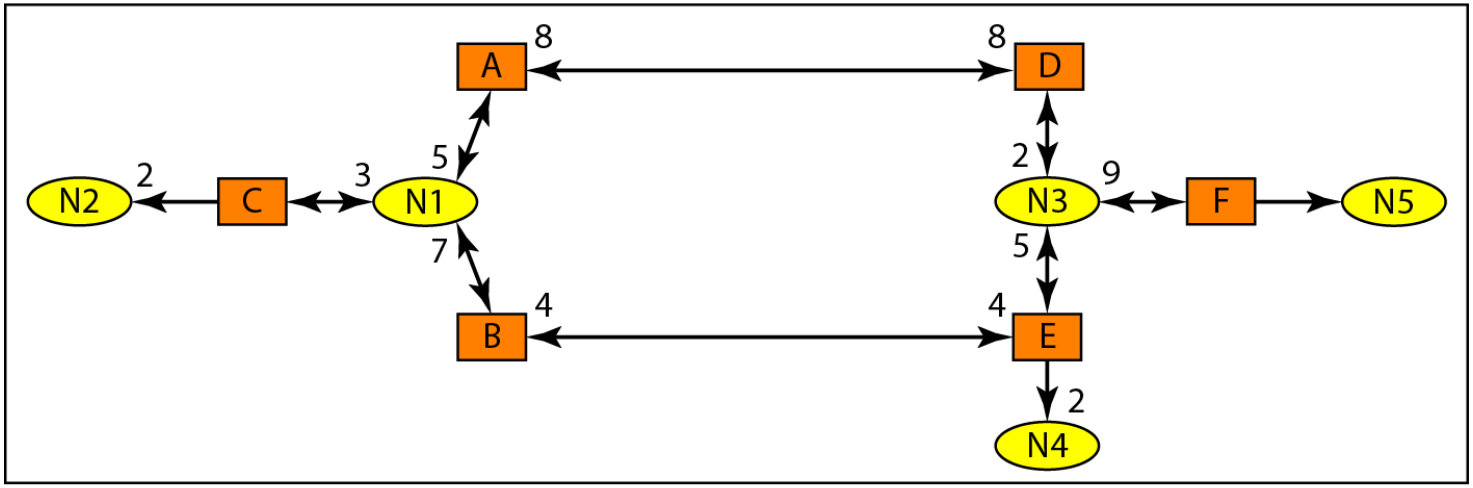


Figure *Example of an AS and its graphical representation in OSPF*



a. Autonomous system



b. Graphical representation

Area Border Router

- At the borders of an area, special routers are present called as area border routers, it is used to summarize the information about the area and send it to the other areas.
- These routers belong to both an area and the backbone.
- These routers are responsible for routing packets outside the area through backbone router.

Backbone Router

- Among the areas inside an autonomous system, is a special area called the backbone and the routers inside the backbone is called backbone routers.
- The primary role of the backbone area is to route traffic between the other areas in an AS.
- This router constructs a complete topological map of the entire autonomous system.

Boundary Routers

- A boundary router exchanges routing information with routers belonging to others autonomous systems

Internal Routers

- These routers are in non backbone areas and perform only intra AS routing in the area.

- **Area Summary Border Router (ASBR) –** When an OSPF router is connected to a different protocol like EIGRP, or Border Gateway Protocol, or any other routing protocol then it is known as AS.
- The router which connects two different AS (in which one of the interface is operating OSPF) is known as Area Summary Border Router.
- These routers perform redistribution. ASBRs run both OSPF and another routing protocol, such as RIP or BGP. ASBRs advertise the exchanged external routing information throughout their AS.

OSPF terms –

- **Router I'd –** It is the highest active IP address present on the router. First, highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.
- **Router priority –** It is a 8 bit value assigned to a router operating OSPF, used to elect DR and BDR in a broadcast network.

OSPF terms –

- **Designated Router (DR) –** It is elected to minimize the number of adjacency formed. DR distributes the LSAs to all the other routers. DR is elected in a broadcast network to which all the other routers shares their DBD. In a broadcast network, router requests for an update to DR and DR will respond to that request with an update.
- **Backup Designated Router (BDR) –** BDR is backup to DR in a broadcast network. When DR goes down, BDR becomes DR and performs its functions.

OSPF terms –

- **DR and BDR election** – DR and BDR election takes place in broadcast network or multi-access network. Here are the criteria for the election:
- Router having the highest router priority will be declared as DR.
- If there is a tie in router priority then highest router I'd will be considered. First, the highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.

OSPF states –

The device operating OSPF goes through certain states. These states are:

- **Down –** In this state, no hello packet have been received on the interface.

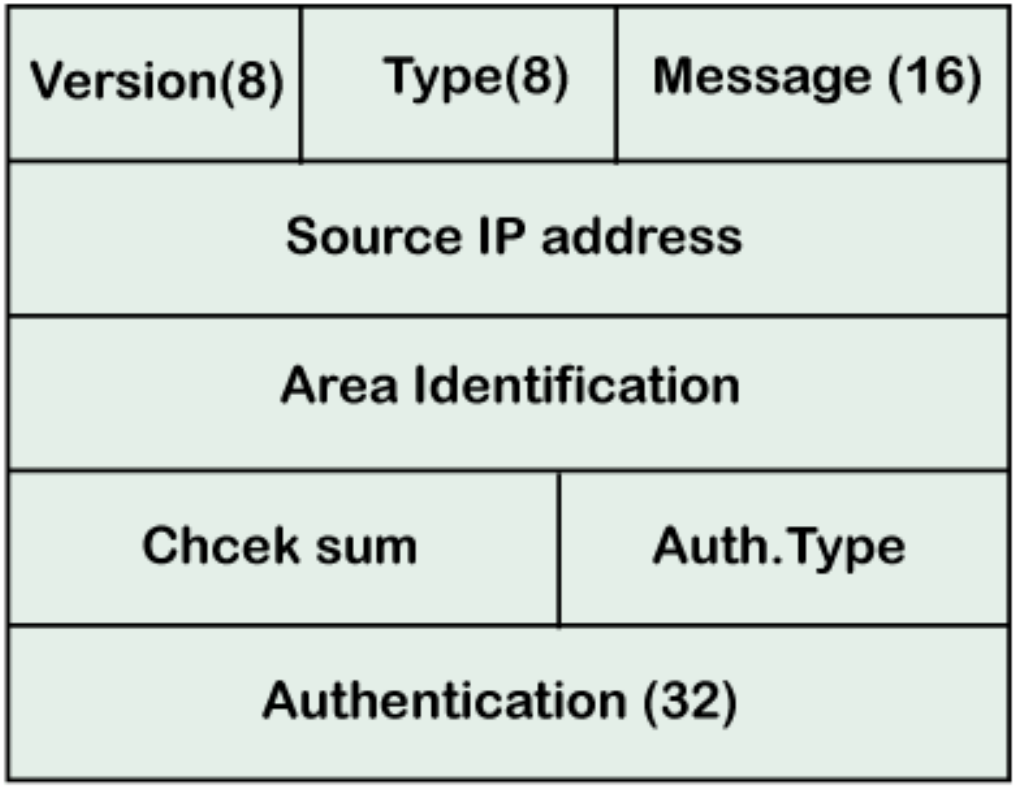
  Note – The Down state doesn't mean that the interface is physically down. Here, it means that OSPF adjacency process has not started yet.
- **INIT –** In this state, hello packet have been received from the other router.
- **2WAY –** In the 2WAY state, both the routers have received the hello packets from other routers. Bidirectional connectivity has been established.

  Note – In between the 2WAY state and Exstart state, the DR and BDR election takes place.

- **Exstart –** In this state, NULL DBD are exchanged.In this state, master and slave election take place. The router having the higher router I’d becomes the master while other becomes the slave. This election decides Which router will send it’s DBD first (routers who have formed neighbourship will take part in this election).
- **Exchange –** In this state, the actual DBDs are exchanged.

- **Loading –** In this sate, LSR, LSU and LSA (Link State Acknowledgement) are exchanged.
- **Full –** In this state, synchronization of all the information takes place. OSPF routing can begin only after the Full state.

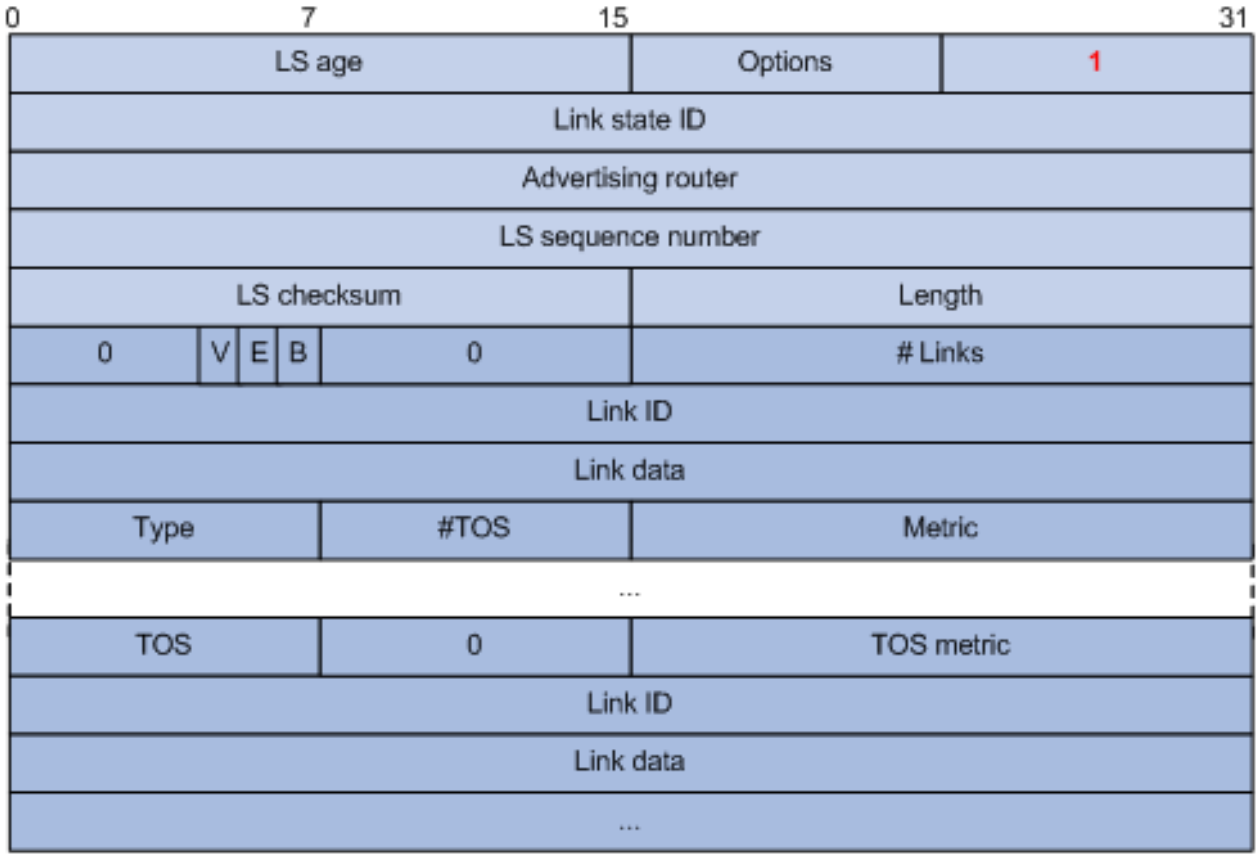
OSPF Message Format



OSPF Message Format

- **Version:** It is an 8-bit field that specifies the OSPF protocol version.
- **Type:** It is an 8-bit field. It specifies the type of the OSPF packet.
- **Message:** It is a 16-bit field that defines the total length of the message, including the header. Therefore, the total length is equal to the sum of the length of the message and header.
- **Source IP address:** It defines the address from which the packets are sent. It is a sending routing IP address.
- **Area identification:** It defines the area within which the routing takes place.
- **Checksum:** It is used for error correction and error detection.
- **Authentication type:** There are two types of authentication, i.e., 0 and 1. Here, 0 means for none that specifies no authentication is available and 1 means for pwd that specifies the password-based authentication.
- **Authentication:** It is a 32-bit field that contains the actual value of the authentication data.

Link State Advertisement (LSA)



- **Link state ID**—ID of the router that originated the LSA.
- **V (Virtual Link)**—Set to 1 if the router that originated the LSA is a virtual link endpoint.
- **E (External)**—Set to 1 if the router that originated the LSA is an ASBR.
- **B (Border)**—Set to 1 if the router that originated the LSA is an ABR.
- **# Links**—Number of router links (interfaces) to the area, as described in the LSA.
- **Link ID**—Determined by link type.
- **Link data**—Determined by link type.
- **Type**—Link type. A value of 1 indicates a point-to-point link to a remote router; a value of 2 indicates a link to a transit network; a value of 3 indicates a link to a stub network; and a value of 4 indicates a virtual link.
- **#TOS**—Number of different TOS metrics given for this link. If no TOS metric is given for the link, this field is set to 0. TOS is not supported in RFC 2328. The #TOS field is reserved for early versions of OSPF.
- **Metric**—Cost of using this router link.
- **TOS**—IP Type of Service that this metric refers to.
- **TOS metric**—TOS-specific metric information.

PATH VECTOR ROUTING (PVR)

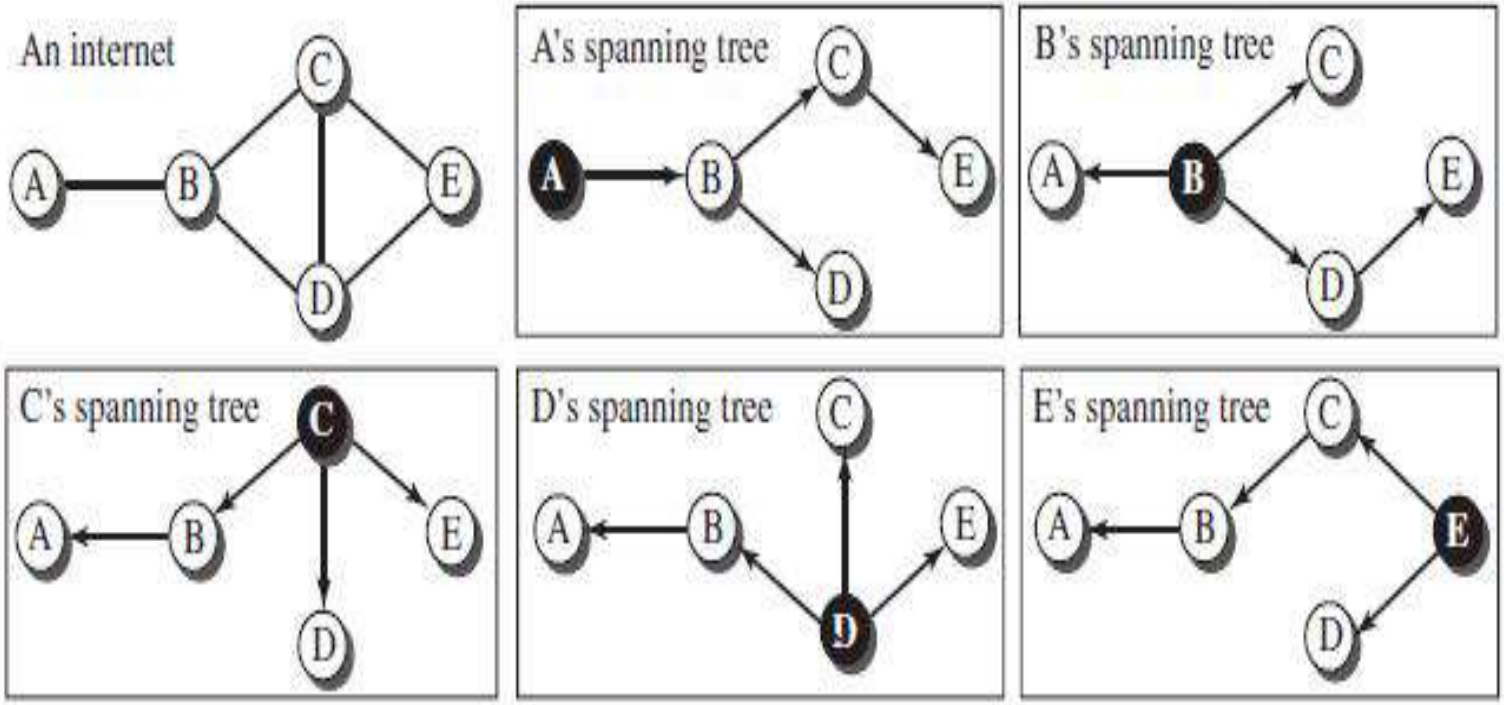
- Path-vector routing is an asynchronous (operations do **not occur at the same time** or in fixed intervals.) and distributed routing algorithm.
- The Path-vector routing is not based on least-cost routing.
- Routing decisions are based on the **path information**, not only distance.
- The best route is determined by the source using the policy it imposes on the route.
- In other words, the source can control the path.
- Path-vector routing is not actually used in an internet, and is mostly designed to route a packet between ISPs.

Spanning Trees

- In path-vector routing, the path from a source to all destinations is determined by the *best* spanning tree.
- The best spanning tree is not the least-cost tree.
- It is the tree determined by the source when it imposes its own policy.
- If there is more than one route to a destination, the source can choose the route that meets its policy best.
- A source may apply several policies at the same time.
- One of the common policies uses the minimum number of nodes to be visited.
- Another common policy is to avoid some nodes as the middle node in a route.

- The spanning trees are made, gradually and asynchronously, by each node.
- When a node is booted, it creates a *path vector* based on the information it can obtain about its immediate neighbor.
- A node sends greeting messages to its immediate neighbors to collect these pieces of information.
- Each node, after the creation of the initial path vector, sends it to all its immediate neighbors.
- Each node, when it receives a path vector from a neighbor, updates its path vector using the formula
- The policy is defined by selecting the *best* of multiple paths.
- Path-vector routing also imposes one more condition on this equation.
- If Path (**v**, **y**) includes ***x***, that path is discarded to avoid a loop in the path.
- In other words, ***x*** does not want to visit itself when it selects a path to **y**.

Example:

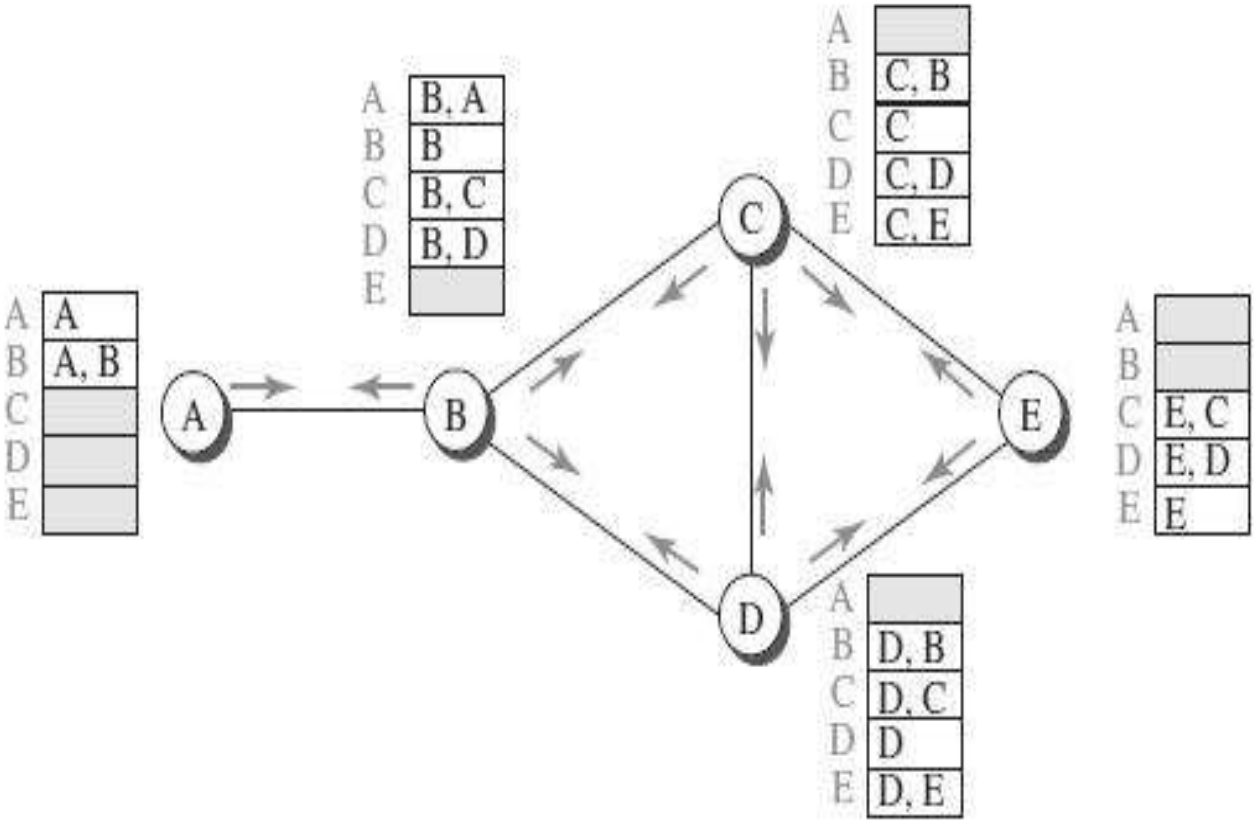

An internet
A
B
C
D
E
A's spanning tree
B's spanning tree
C's spanning tree
D's spanning tree
E's spanning tree

- Each source has created its own spanning tree that meets its policy.
- The policy imposed by all sources is to use the minimum number of nodes to reach a destination.
- The spanning tree selected by A and E is such that the communication does not pass through D as a middle node.
- Similarly, the spanning tree selected by B is such that the communication does not pass through C as a middle node.

Path Vectors made at booting time

- The Figure below shows all of these path vectors for the example.
- Not all of these tables are created simultaneously.
- They are created when each node is booted.
- The figure also shows how these path vectors are sent to immediate neighbors after they have been created.

Path Vectors made at booting time



Updating path vector for node C

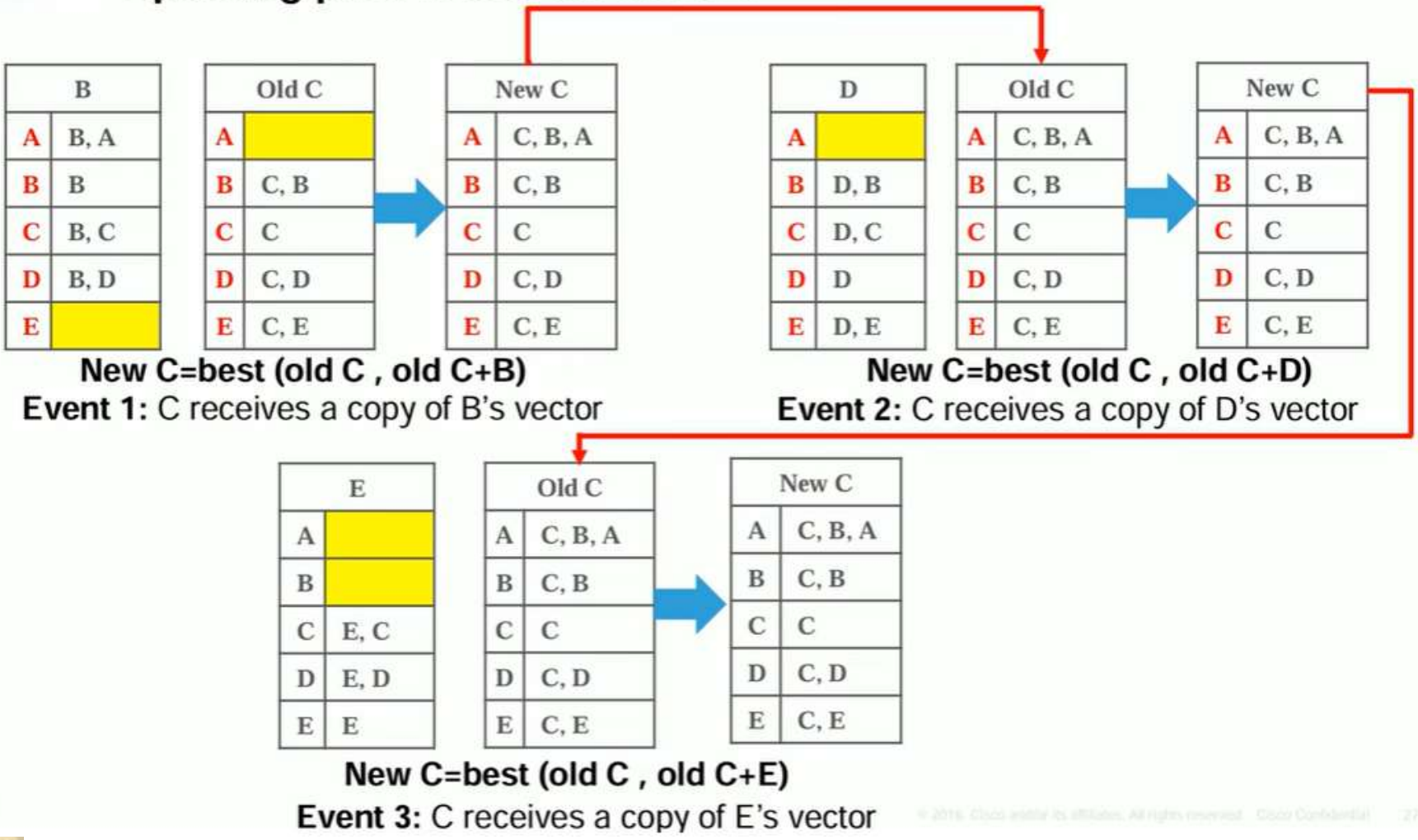


Figure ***Initial routing tables in path vector routing***

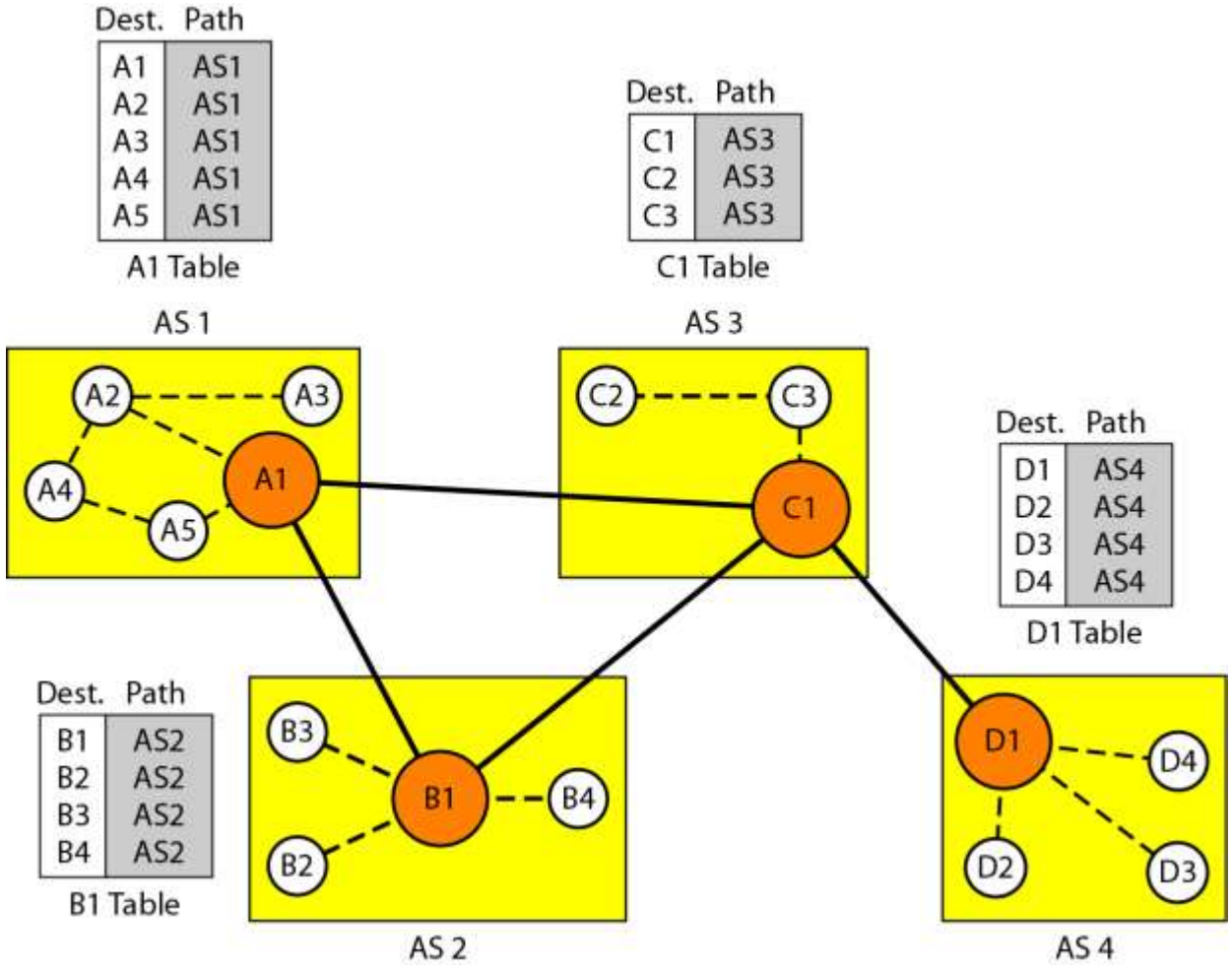
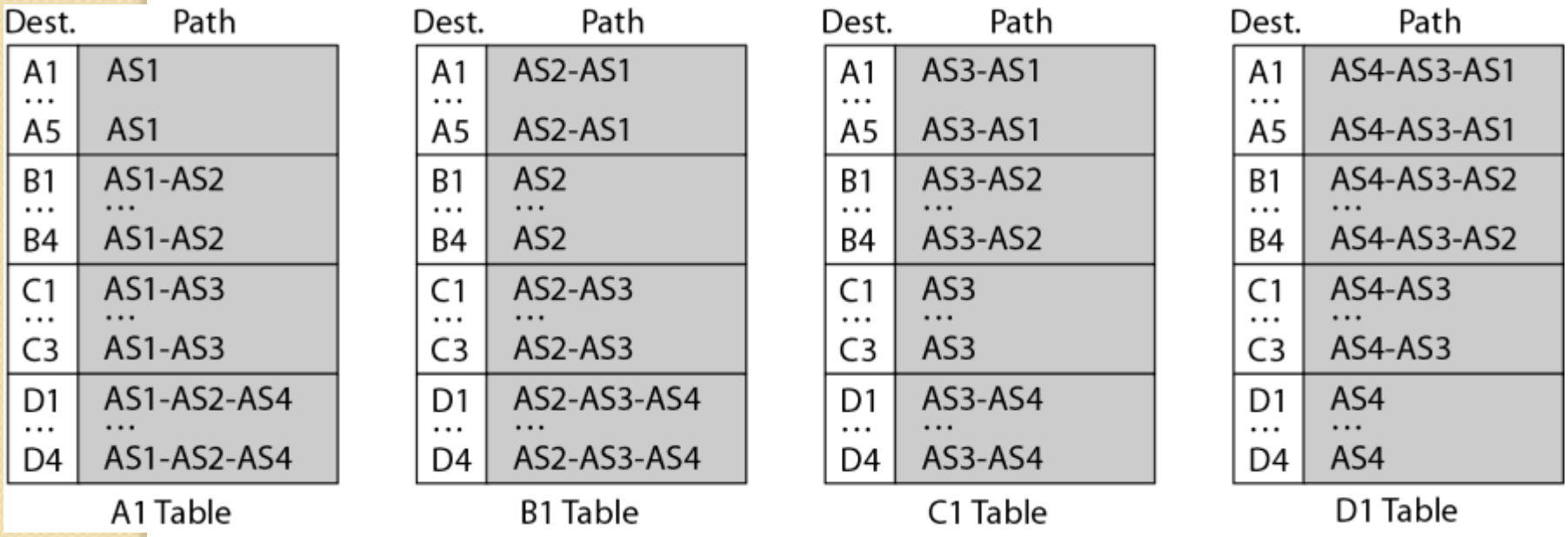


Figure ***Stabilized tables for three autonomous systems***



Dest.	Path
A1 … A5	AS1
B1 … B4	AS1-AS2 … AS1-AS2
C1 … C3	AS1-AS3 … AS1-AS3
D1 … D4	AS1-AS2-AS4 … AS1-AS2-AS4

A1 Table

Dest.	Path
A1 … A5	AS2-AS1
B1 … B4	AS2 … AS2
C1 … C3	AS2-AS3 … AS2-AS3
D1 … D4	AS2-AS3-AS4 … AS2-AS3-AS4

B1 Table

Dest.	Path
A1 … A5	AS3-AS1
B1 … B4	AS3-AS2 … AS3-AS2
C1 … C3	AS3 … AS3
D1 … D4	AS3-AS4 … AS3-AS4

C1 Table

Dest.	Path
A1 … A5	AS4-AS3-AS1
B1 … B4	AS4-AS3-AS2 … AS4-AS3-AS2
C1 … C3	AS4-AS3 … AS4-AS3
D1 … D4	AS4 … AS4

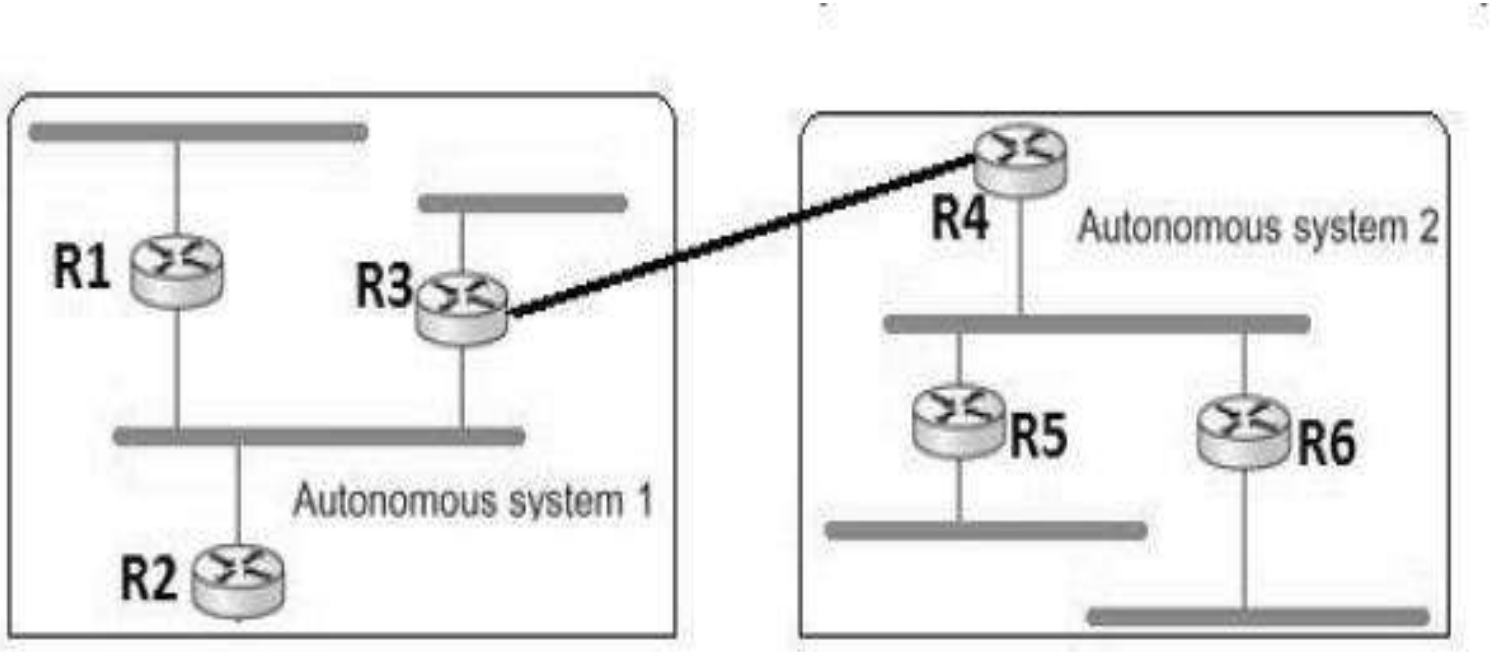
D1 Table

BORDER GATEWAY PROTOCOL (BGP)

- The Border Gateway Protocol version (BGP) is the only interdomain routing protocol used in the Internet today.
- BGP4 is based on the path-vector algorithm.
- It provides information about the reachability of networks in the Internet.
- BGP views internet as a set of autonomous systems interconnected arbitrarily.

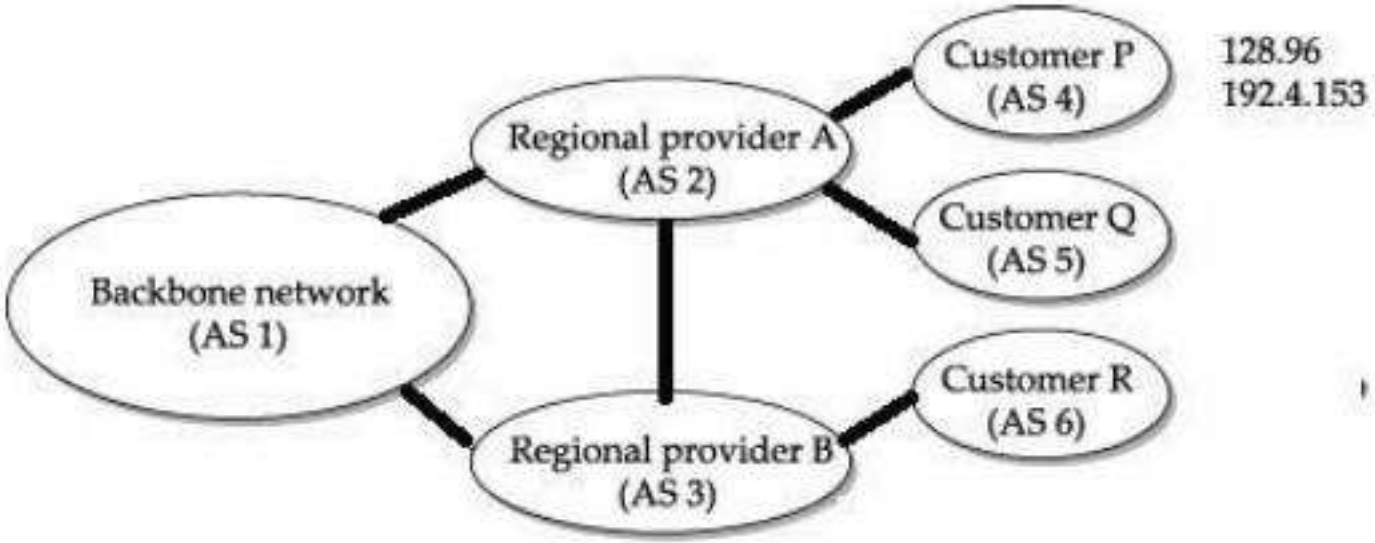
BORDER GATEWAY PROTOCOL (BGP)

Border Routers



- Each AS have a *border router* (gateway), by which packets enter and leave that AS. In above figure, *R3* and *R4* are border routers.
- One of the router in each autonomous system is designated as BGP *speaker*.
- BGP Speaker *exchange* reachability information with other BGP speakers, known as *external* BGP session.
- BGP advertises complete *path* as enumerated list of AS (path vector) to reach a particular network.
- Paths must be without any *loop*, i.e., AS list is unique.

- For *example*, backbone network advertises that networks 128.96 and 192.4.153 can be reached along the path *<AS1, AS2, AS4>*.

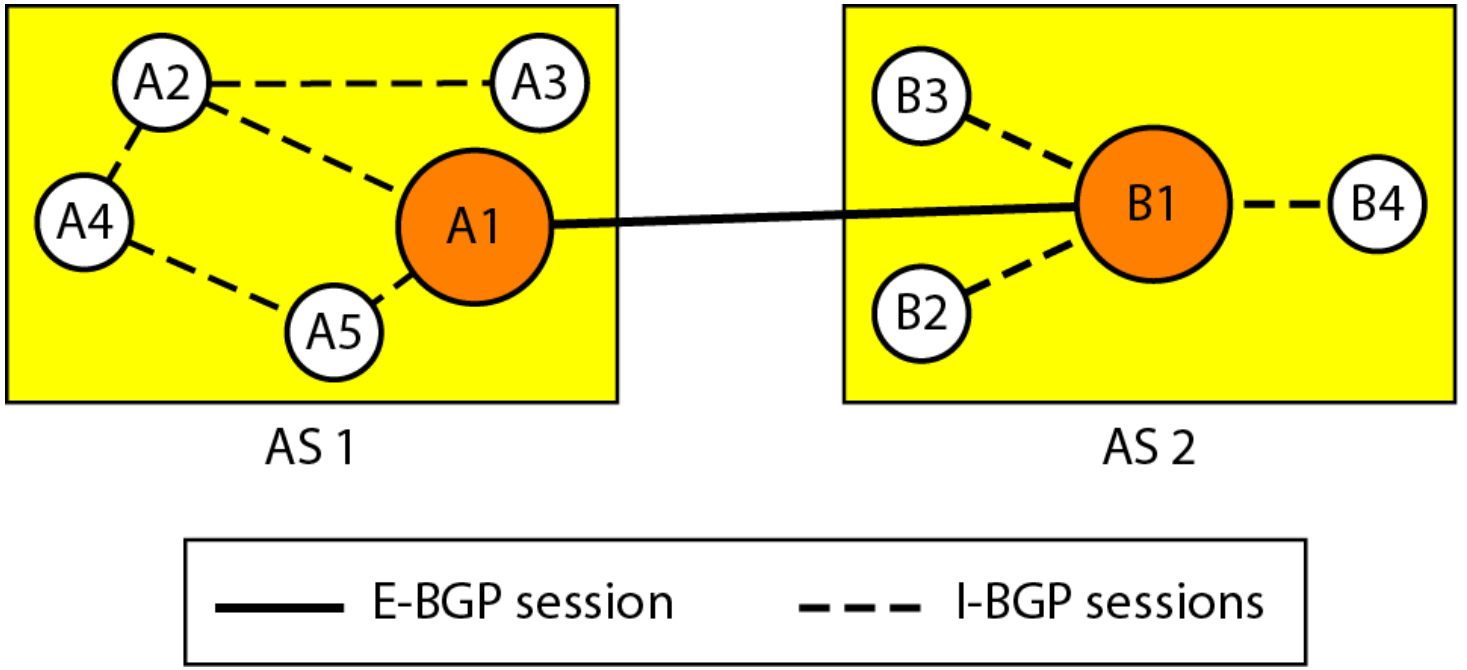


- If there are multiple routes to a destination, BGP speaker chooses one based on policy.
- Speakers need not advertise any route to a destination, even if one exists.
- Advertised paths can be cancelled, if a link/node on the path goes down. This negative advertisement is known as withdrawn route.
- Routes are not repeatedly sent. If there is no change, keep alive messages are sent.

iBGP - interior BGP

- A Variant of BGP
- Used by routers to update routing information learnt from other speakers to routers inside the autonomous system.
- Each router in the AS is able to determine the appropriate next hop for all prefixes.

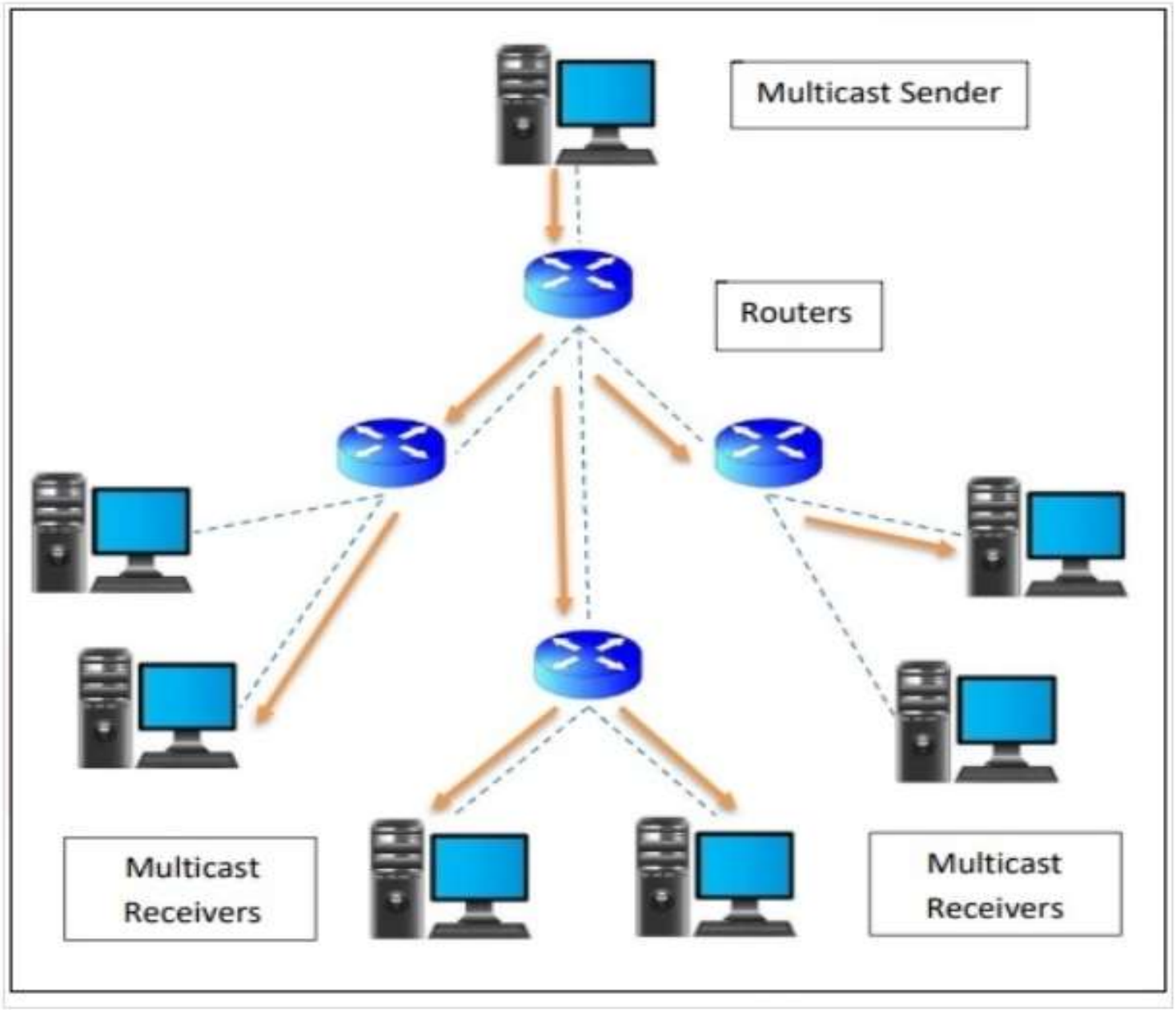
Figure ***Internal and external BGP sessions***



MULTICASTING

MULTICASTING Basic

- In multicasting, there is one source and a group of destinations.
- Multicast supports efficient delivery to multiple destinations.
- The relationship is one to many or many-to-many.
- **One-to-Many (Source Specific Multicast)**
    - o Radio station broadcast
    - o Transmitting news, stock-price
    - o Software updates to multiple hosts
- **Many-to-Many (Any Source Multicast)**
    - o Multimedia teleconferencing
    - o Online multi-player games
    - o Distributed simulations
- In this type of communication, the source address is a unicast address, but the destination address is a group address.
- The group address defines the members of the group.


Multicast Sender
Routers
Multicast Receivers
Multicast Receivers

Multicasting

- **Multicasting** works in **similar to Broadcasting**, but in Multicasting, the information is sent to the targeted or specific members of the network.
- This task can be accomplished by **transmitting individual copies to each user or node present in the network**, but sending individual copies to each user is **inefficient and might increase the network latency**.
- To overcome these shortcomings, **multicasting allows a single transmission that can be split up among the multiple users**, consequently, this **reduces the bandwidth of the signal.**

Applications :

Multicasting is used in many areas like:

- **Bulk data transfer** – for example, the transfer of the software upgrade from the software developer to users needing the upgrade.
- Internet protocol (IP)
- **Streaming Media** – for example, the transfer of the audio, video and text of the live lecture to a set of distributed lecture participants.
- It also supports video conferencing applications and webcasts.
- Data feeds
- Interactive gaming

MULTICAST ROUTING

- There are two types of Multicast Distribution Trees used in multicast routing.
- They are

Source-Based Tree: (DVMRP)

- For each combination of (source , group), there is a shortest path spanning tree.

Flood and prune

- Send multicast traffic everywhere
- Prune edges that are not actively subscribed to group

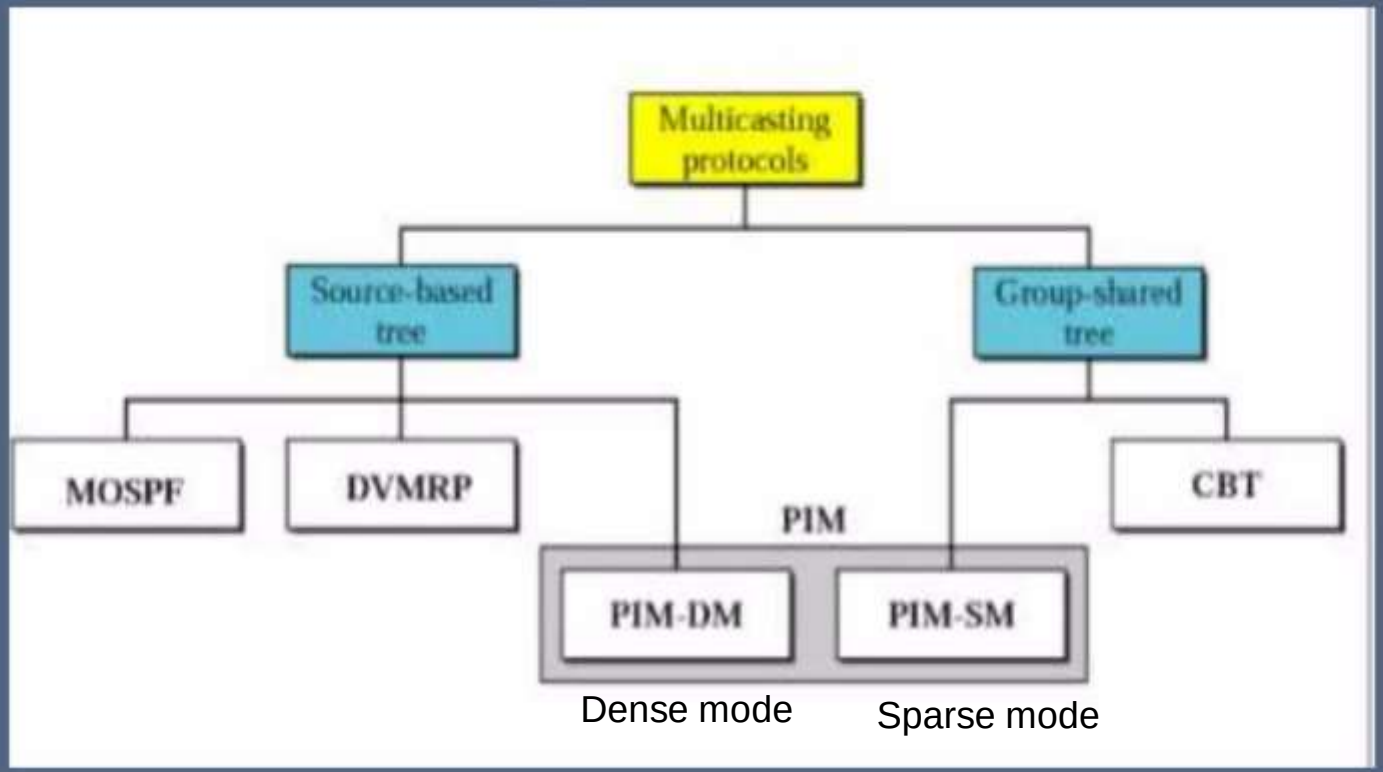
Link-state

- Routers flood groups they would like to receive
- Compute shortest-path trees on demand

Group Shared Tree (PIM (Protocol Independent Multicast))

- Single distributed tree shared among all sources
- Specify **rendezvous point (RP)** for group
- Senders send packets to RP, receivers join at RP

Multicasting routing protocol



MULTICAST ROUTING PROTOCOLS

- Internet multicast is implemented on physical networks that support broadcasting by ***extending forwarding functions.***
- Major multicast routing protocols are:

1. Distance-Vector Multicast Routing Protocol (DVMRP)

2. Protocol Independent Multicast (PIM)

1. Distance Vector Multicast Routing Protocol

- The DVMRP, is a routing protocol used to share information between routers to facilitate the transportation of IP multicast packets among networks.
- It formed the basis of the Internet's historic multicast backbone.
- Distance vector routing for unicast is extended to support multicast routing.
- Each router maintains a routing table for all destination through exchange of distance vectors.
- DVMRP is also known as ***flood-and-prune protocol***.
- DVMRP consists of two major components:
- A conventional distance-vector routing protocol, like RIP
- A protocol for determining how to forward multicast packets, based on the routing table
- DVMRP router forwards a packet if
- The packet arrived from the link used to reach the source of the packet
- If downstream links have not pruned the tree

DVMRP protocol uses the **basic packet types** as follows:

- DVMRP Probes
  - for DVMRP Neighbor Discovery
- DVMRP Reports
  - for Multicast Route Exchange
- DVMRP Prunes
  - for pruning multicast delivery trees
- DVMRP Grafts
  - for grafting multicast delivery trees
- DVMRP Graft Ack's
  - for acknowledging graft msgs

Multicasting is added to distance-vector routing in four stages.

- Flooding
- Reverse Path Forwarding (RPF)
- Reverse Path Broadcasting (RPB)
- Reverse Path Multicast (RPM)

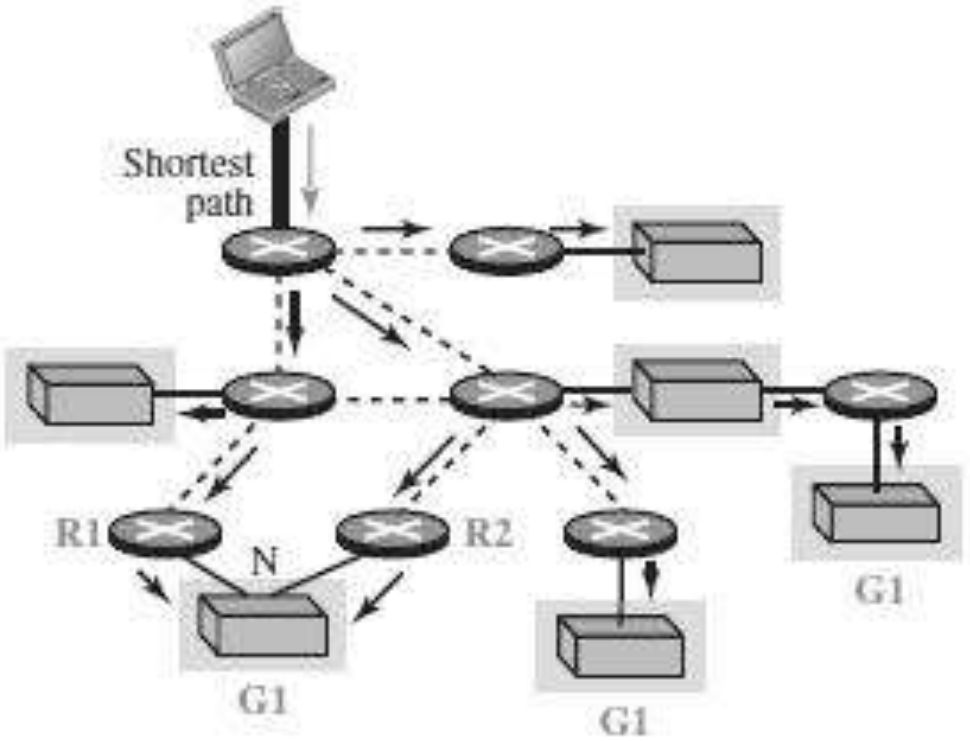
Flooding

- Router on receiving a multicast packet from source *S* to a Destination from **NextHop, *forwards* the packet on all out-going links.**
- Packet is flooded and looped back to *S*.
- The drawbacks are:
    1. It floods a network, even if it has *no members* for that group.
    2. Packets are forwarded by each router connected to a LAN, i.e., *duplicate flooding*

Reverse Path Forwarding (RPF)

- RPF eliminates the looping problem in the flooding process.
- **Only one copy is forwarded and the other copies are discarded**.
- RPF forces the router to forward a multicast packet from one specific interface: the one which has come through the shortest path from the source to the router.
- Packet is flooded but not looped back to S.

Reverse Path Forwarding (RPF)

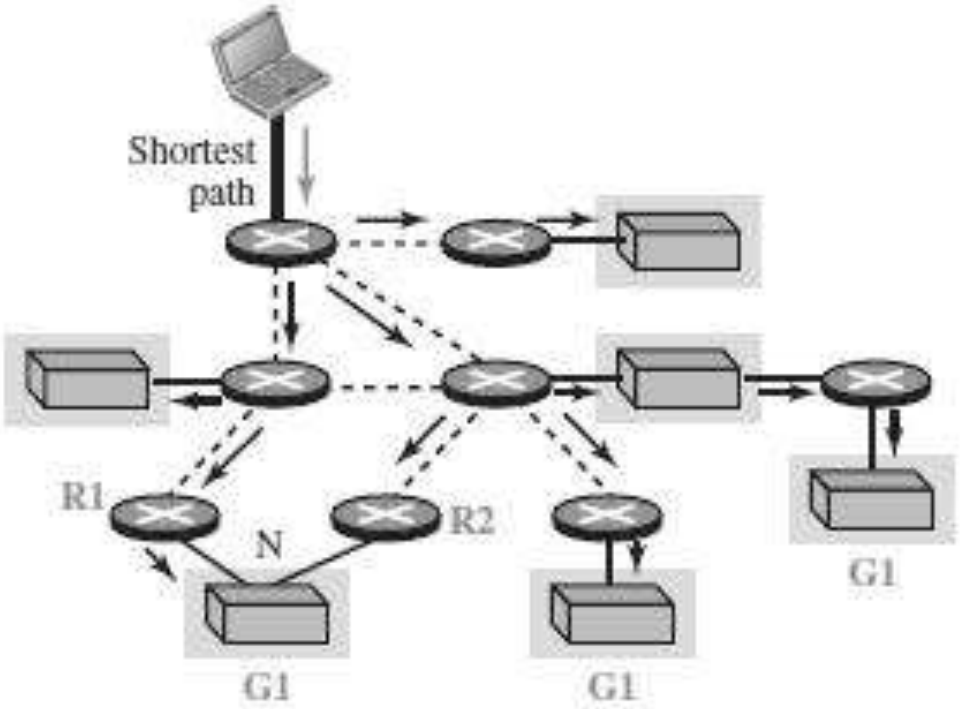


Using RPF, N receives two copies.

Reverse-Path Broadcasting (RPB)

- RPB does not multicast the packet, it broadcasts it.
- RPB creates a **shortest path broadcast tree from the source to each destination.**
- It guarantees that **each destination receives one and only one copy of the packet.**
- We need to **prevent each network from receiving more than one copy of the packet.**
- If a network is connected to more than one router, it may receive a copy of the packet from each router.
- One router identified as parent called **designated Router (DR).**
- Only parent router *forwards* multicast packets from source *S to the attached network.*
- When a router that is not the parent of the attached network receives a multicast packet, it simply drops the packet.

Reverse-Path Broadcasting (RPB)



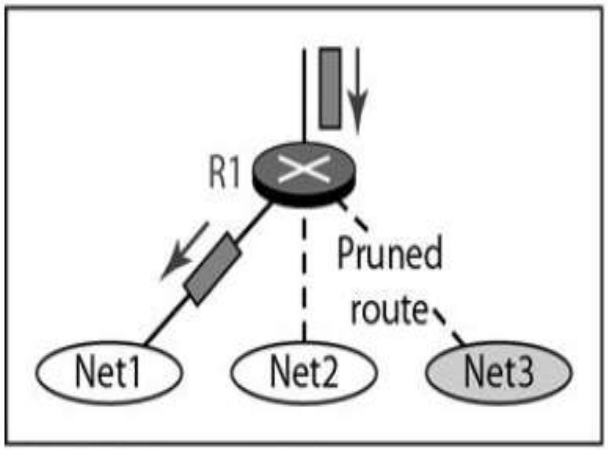
Using RPB, N receives only one copy.

Reverse-Path Multicasting (RPM)

- To **increase efficiency**, the multicast packet must reach only those networks that have active members for that particular group.
- RPM adds **pruning and grafting** to RPB to create a multicast shortest path tree that supports dynamic membership changes.

Pruning:

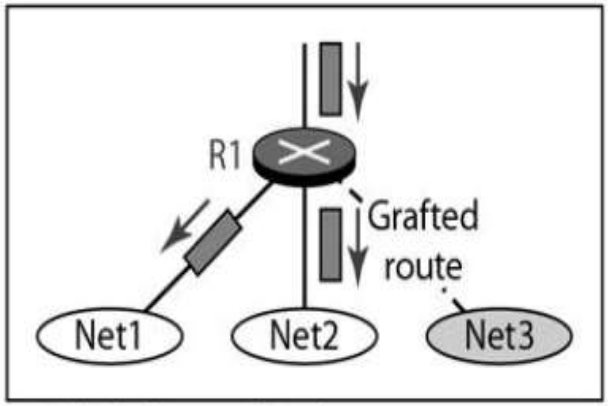
- Sent from routers receiving multicast traffic for which they have no active group members
- "Prunes" the tree created by DVMRP
- Stops needless data from being sent



RPM (after pruning)

Grafting:

- Used after a branch has been pruned back
- Sent by a router that has a host that joins a multicast group
- Goes from router to router until a router active on the multicast group is reached
- Sent for the following cases
  - A new host member joins a group
  - A new dependent router joins a pruned branch
  - A dependent router restarts on a pruned branch



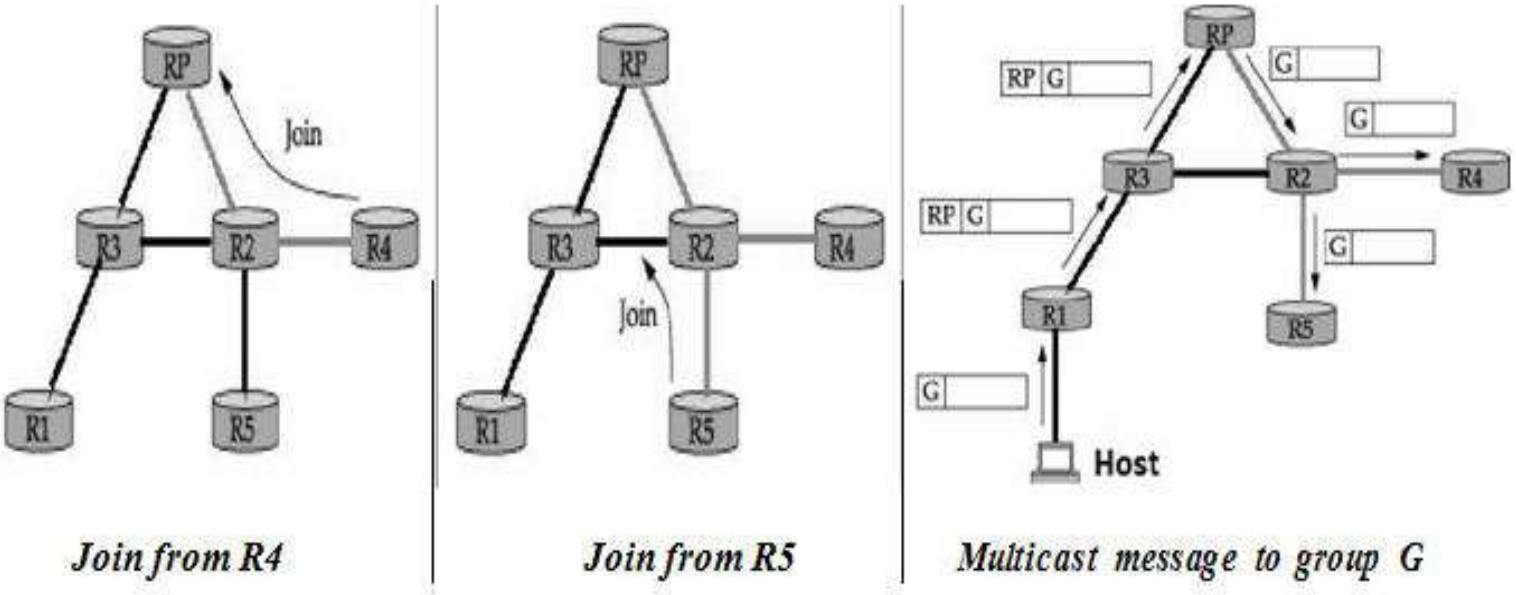
RPM (after grafting)

2. Protocol Independent Multicast (PIM)

- PIM divides multicast routing problem into *sparse* and *dense* mode.
- PIM sparse mode (PIM-SM) is widely used.
- PIM does not rely on any type of unicast routing protocol, hence protocol independent.
- Routers explicitly join and leave multicast group using ***Join and Prune messages***.
- One of the router is designated as *rendezvous point* (RP) for each group in a domain to receive PIM messages.
- Multicast forwarding *tree* is built as a result of routers sending Join messages to RP.
- Two types of trees to be constructed:
  - ***Shared tree*** - used by all senders
  - ***Source-specific* tree** - used only by a specific sending host
- The normal mode of operation creates the shared tree first, followed by one or more source-specific trees

Example

- Router *R4* sends Join message for group *G* to rendezvous router RP.
- Join message is received by router *R2*. It makes an entry (*, *G*) in its table and forwards the message to *RP*.
- When *R5* sends Join message for group *G*, *R2* does not forwards the Join. It *adds* an outgoing interface to the forwarding table created for that group.

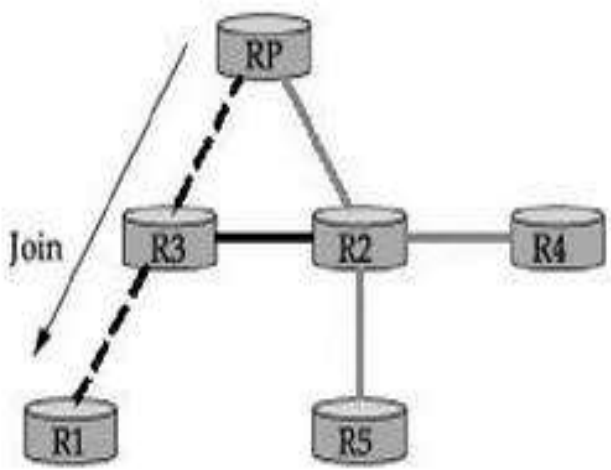


- As routers send Join message for a group, branches are *added* to the tree, i.e., shared.
- Multicast packets sent from hosts are forwarded to *designated* router RP.
- Suppose router *R1*, receives a message to group *G*.
    - *R1* has no state for group *G*.
    - Encapsulates the multicast packet in a Register message.
    - Multicast packet is tunneled along the way to RP.
- RP decapsulates the packet and sends multicast packet onto the shared tree, towards *R2*.
- *R2* forwards the multicast packet to routers *R4* and *R5* that have members for group *G*.

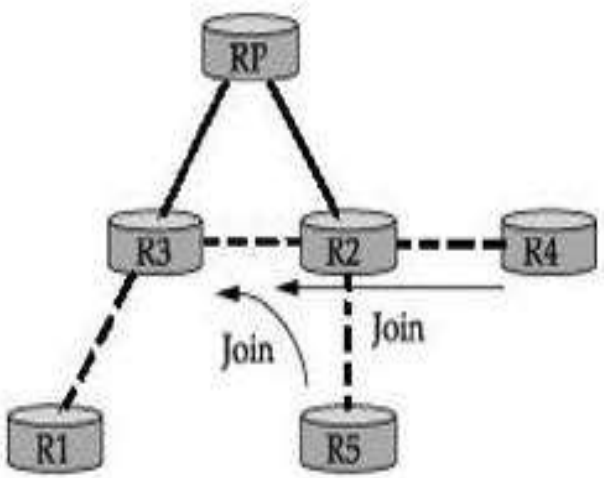
Source-Specific Tree

- RP can force routers to know about group *G*, by sending Join message to the sending host, so that tunneling can be avoided.
- Intermediary routers create *sender-specific* entry (*S*, *G*) in their tables. Thus a source-specific route from *R1* to RP is formed.
- If there is high rate of packets sent from a sender to a group *G*, then shared tree is *replaced* by source-specific tree with sender as root.

Example



Source-specific Join from RP



Routers switch to Source tree

- Rendezvous router RP sends a Join message to the host router *R1*.
- Router *R3* learns about group *G* through the message sent by RP.
- Router *R4* send a source-specific Join due to high rate of packets from sender.
- Router *R2* learns about group *G* through the message sent by *R4*.
- Eventually a source-specific tree is formed with *R1* as root.

PIM-DM
(Dense Mode)

- PIM-DM is used when there is a possibility that each router is involved in multicasting (dense mode)
- In this environment, the use of a protocol that broadcasts the packet is justified because almost all routers are involved in the process
- PIM-DM is a source-based tree routing protocol that uses RPF and pruning/grafting strategies for multicasting
- Its operation is like DVMRP; however, it does not depend on a specific unicating protocl

PIM (Protocol Independent Multicasting) Sparse Mode

- **PIM Sparse Mode** is a multicast routing protocol used when **multicast receivers are sparsely distributed** across a network.
- PIM Sparse Mode is a multicast routing protocol designed for networks where receivers are sparsely located. It forwards multicast traffic only after receivers explicitly request it using Join messages. A Rendezvous Point (RP) is used to connect sources and receivers, reducing unnecessary traffic and improving scalability. It is called *protocol independent* because it works with **any unicast routing protocol** (OSPF, RIP, IS-IS, BGP, etc.).
- Assumes **few receivers** exist.
- Data is forwarded **on demand**.
- Uses a special router called **Rendezvous Point (RP)**.

Working of PIM Sparse Mode

- **Receiver joins multicast group**
  - Sends **Join message** toward RP.
- **Shared Tree Creation**
  - Routers build a tree toward RP.
  - Called **Shared Tree (RPT – RP Tree)**.
- **Source Registration**
  - Source sends data to RP.
- **Data Forwarding**
  - RP forwards multicast packets to receivers.
- **Shortest Path Tree (Optional)**
  - Routers may switch to direct shortest path from source.

IPV6 - NEXT GENERATION IP

- IPv6 was evolved to solve address space problem and offers rich set of services.
- Some hosts and routers will run IPv4 only, some will run IPv4 and IPv6 and some will run IPv6 only.

DRAWBACKS OF IPV4

- Despite subnetting and CIDR, address depletion is still a long-term problem.
- Internet must accommodate real-time audio and video transmission that requires minimum delay strategies and reservation of resources.
- Internet must provide encryption and authentication of data for some applications

FEATURES OF IPV6

- *Better header format*
- *New options*
- *Allowance for extension*
- *Support for resource allocation*

Additional Features

1. Need to accommodate scalable routing and addressing
2. Support for real-time services
3. Security support
4. Auto configuration - The ability of hosts to automatically configure themselves with such information as their own IP address and domain name.
5. Enhanced routing functionality, including support for mobile hosts
6. Transition from ipv4 to ipv6

ADDRESS SPACE ALLOCATION OF IPV6

- IPv6 provides a 128-bit address space to handle up to 3.4×10^{38} nodes.
- IPv6 uses *classless* addressing, but classification is based on MSBs.
- The address space is subdivided in various ways based on the leading bits.
- The current assignment of prefixes is listed in Table

Prefix	Use
00...0 (128 bits)	Unspecified
00...1 (128 bits)	Loopback
1111 1111	Multicast addresses
1111 1110 10	Link-local unicast
Everything else	Global Unicast Addresses

- A node may be assigned an "IPv4-compatible IPv6 address" by zero-extending a 32-bit IPv4 addressto128 bits.

ADDRESS NOTATION OF IPV6

- Standard representation of IPv6 address is ***x : x : x : x : x : x : x : x*** where *x* is a 16-bit hexadecimal address separated by colon (:).
  For example,
  47CD : 1234 : 4422 : ACO2 : 0022 : 1234 : A456 : 0124

- IPv6 address with contiguous 0 bytes can be written compactly.
  For example,
  47CD : 0000 : 0000 : 0000 : 0000 : 0000 : A456 : 0124 → 47CD : : A456 : 0124

- IPv4 address is mapped to IPv6 address by prefixing the 32-bit IPv4 address with 2 bytes of 1s and then zero-extending the result to 128 bits.
  For example,
  128.96.33.81 → : : FFFF : 128.96.33.81
  This notation is called as CIDR notation or slash notation.

ADDRESS AGGREGATION OF IPV6

- IPv6 provides *aggregation* of routing information to reduce the burden on routers.
- Aggregation is done by assigning prefixes at *continental* level.
- For *example*, if all addresses in Europe have a common prefix, then routers in other continents would need one routing table entry for all networks in Europe.

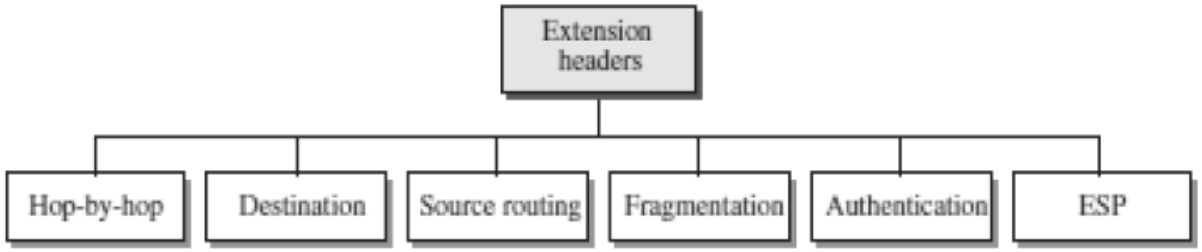


- ***Prefix*** - All addresses in the same continent have a common prefix
- ***RegistryID*** — identifies the continent
- ***ProviderID*** — identifies the provider for Internet access, i.e., ISP.
- ***SubscriberID*** — specifies the subscriber identifier

- ***SubnetID*** — contains subnet of the subscriber.
- ***InterfaceID*** —contains link level or physical address.

Extension Headers

- Extension header provides greater functionality to IPv6.
- Base header may be followed by six extension headers.
- Each extension header contains a NextHeader field to identify the header following it.



- ***Hop-by-Hop*** — source host passes information to all routers visited by the packet
- ***Destination*** — source host information is passed to the destination only.
- ***Source Routing*** — routing information provided by the source host.
- ***Fragmentation*** — In IPv6, only the source host can fragment. Source uses a path MTU discovery technique to find smallest MTU on the path.
- ***Authentication*** — used to validate the sender and ensures data integrity.
- ***ESP (Encrypted Security Payload)*** — provides confidentiality against eavesdropping.

ADVANCED CAPABILITIES OF IPV6

- Auto Configuration
- Advanced Routing
- Additional Functions
- Security
- Resource allocation

ADVANTAGES OF IPV6

- ***Address space*** — IPv6 uses 128-bit address whereas IPv4 uses 32-bit address.
- Hence IPv6 has huge address space whereas IPv4 faces address shortage problem.
- ***Header format*** — Unlike IPv4, optional headers are separated from base header in IPv6. Each router thus need not process unwanted addition information.
- ***Extensible*** — Unassigned IPv6 addresses can accommodate needs of future technologies.

- In IPv6 representation, we have three addressing methods :
- Unicast
- Multicast
- Anycast
- **Unicast Address:** Unicast Address identifies a single network interface. A packet sent to unicast address is delivered to the interface identified by that address

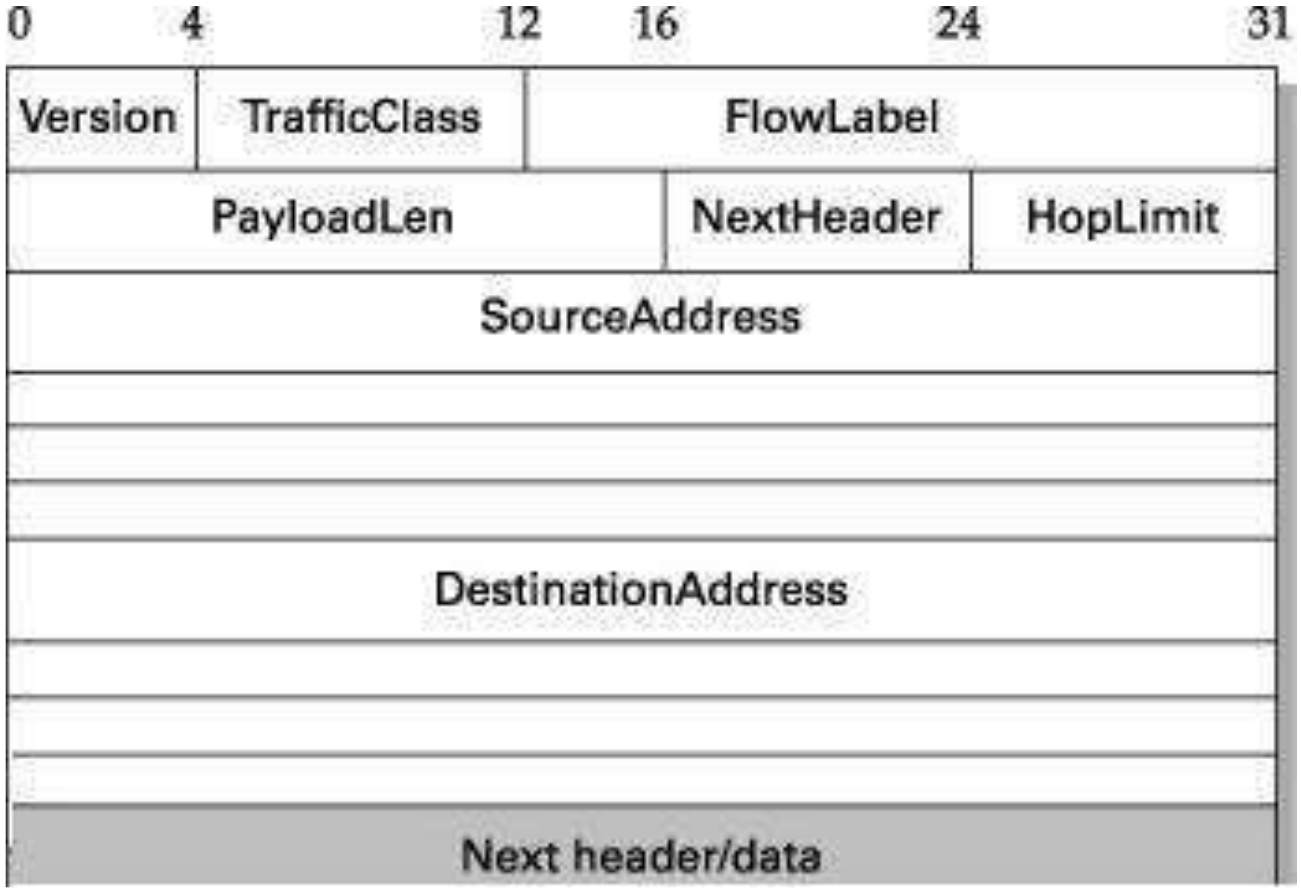
Multicast Address:

Multicast Address is used by multiple hosts, called as Group, acquires a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address.

Anycast Address:

Anycast Address is assigned to a group of interfaces. Any packet sent to anycast address will be delivered to only one member interface (mostly nearest host possible). Ex: One address → Multiple servers → Client reaches **closest server**.

PACKET FORMAT OF IPV6



PACKET FORMAT OF IPV6

- ***Version*** — specifies the IP version, i.e., 6.
- ***Traffic Class*** — defines priority of the packet with respect to traffic congestion. It is either congestion-controlled or non-congestion controlled
- ***Flow Label*** — provides special handling for a particular flow of data. Router handles different flows with the help of a flow table.
- ***Payload Len*** — gives length of the packet, excluding IPv6 header.
- ***Next Header*** — Options are specified as a header following IP header. NextHeader contains a pointer to optional headers.
- ***Hop Limit*** — Gives the TTL value of a packet.
- ***Source Address / Destination Address*** — 16-byte addresses of source and destination host

THANK YOU